Market Regimes and Hidden Markov Models

What a regime model actually estimates, how to fit a hidden Markov model honestly, and where its output belongs

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Abstract

A market regime is an output of a model the analyst chooses, not a property of the market. The standard regime model, the hidden Markov model, separates calm periods from turbulent ones well. It does not separate rising markets from falling ones. A century of US stock returns confirms this. The model's value is in setting risk, not in choosing direction.

A regime model returns state probabilities, not a single label. Collapsing the probabilities to the most likely state discards the model's uncertainty and is the most common implementation error. The model can predict only because states persist, and the decay of that persistence limits how far ahead it can help. Clustering, jump models and mixture models are special cases of the hidden Markov model with a restriction added or removed. One statistical fact governs the field. More observations sharpen the estimate of each state's volatility, but each state's mean return improves only with more calendar time. With 10 years of daily data and a correctly specified model, the estimated return gap between two states has the wrong sign about one time in six. The volatility gap from the same fit is 16 times more precise.

Fitted in real time on US stocks, a two-state model finds a calm state with about 10% annual volatility and a turbulent state with about 28%. In hindsight the turbulent state seems to lose about 40 points a year relative to the calm state. Out of sample that gap disappears. Days flagged as turbulent in advance were more than twice as volatile and earned no less. Several software packages report smoothed, look-ahead probabilities by default. A good fit is not evidence of regimes. Data with smoothly drifting volatility and no regimes fits regime models well, and a fat-tailed running volatility estimate forecast the return distribution better than the regime model did.

Over 80 years of US data, cutting exposure when the model expected turbulence raised the Sharpe ratio from about 0.52 to about 0.6, within the margin of error. It reduced the maximum drawdown from 59% to about 40%. The overlay breaks even at about 13 basis points of one-way cost. In forecasting models, dividing the target by trailing volatility matters more than any regime feature. The chapter recommends using regimes for risk rather than direction, keeping filtered probabilities, and normalizing forecast targets by trailing volatility.

What this is. "Regime" is the most overloaded word in systematic trading. It is used for a latent Markov state, for a VIX threshold, for a business-cycle phase, for a correlation cluster, and for the vague feeling that this year is not like last year. This chapter takes the concept apart: what a regime model *is* mathematically, what sixty years of literature on it has actually established, which parts of that literature survive contact with out-of-sample data, how to fit the central model of the field — the hidden Markov model — honestly, and how to use its output for sizing, risk, and forecasting without fooling yourself or wasting the effort.

How to read it. Part I (§1–§8) is the model layer: what a regime is, why one might exist, how the field arrived here, the formal model zoo, the hidden Markov model in practice, the taxonomy that collapses the zoo, and an honest scorecard of what works. Part II (§9–§13) is the use layer: the statistical framing that tells you *which* scheme your problem calls for when a forecasting model consumes the regime, the schemes themselves, the research and evaluation protocol, and what to do in live trading.

Appendix A defines every concept the main text uses without explaining, ordered so that it reads as a build-up rather than a glossary; §5, §6 and §10 lean on it most.

If you read three things, read **§8.1** (why regime models identify volatility and not returns, with the simulation), **§6.11** (the same result on a century of US equity returns, fitted walk-forward with Student-t emissions), and **§11.2** (the eight ways regime research leaks). If you are implementing, §6, §10 and §11 are the working sections and the rest is reference.

What you will be able to do. After this chapter you should be able to:

1. Say what a regime model estimates, and compute from a fitted transition matrix the horizon beyond which it cannot help.
2. Fit a hidden Markov model by EM with heavy-tailed emissions, a persistence prior and walk-forward parameters, and check it with pseudo-residuals.

3. Explain why state volatilities are identified and state means are not, and design around it.
4. Choose among threshold rules, continuous indices, jump models and HMMs for a given decision.
5. Feed a regime estimate into a position rule, a risk model or a forecasting model without look-ahead, and evaluate the result with episode-counted standard errors.

Relationship to the other notes. Trend-Following in Financial Markets derives what a trend rule earns; several of its results reappear here, because a trend filter is a two-state regime classifier with the states named "up" and "down". Momentum in Financial Markets covers signal measurement, and Simple and Log Returns covers the return conventions used throughout. Each stands alone.

Epistemic tags. Claims are flagged by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, derivations, or arithmetic. Several results come from simulations run for this chapter; those are labelled **[Simulated]**. The worked example of §6.11 uses real data. The generating code for both is committed alongside this document in `figures/`, so you can change the parameters and rerun.

Notation. r_t is the return of the traded asset over bar t (log returns unless noted), A the number of bars per year (252 for daily data), and $\mathcal{F}_t$ the information set available at the *end* of bar t.

$s_t \in \{1, \dots, K\}$ is the **regime** — a latent discrete state — and K the number of regimes. (S_t, the asset price elsewhere in these notes, does not appear in this chapter.) $\mathbf{P}$ is the $K \times K$ transition matrix with entries $p_{jk} = \Pr(s_{t+1} = k \mid s_t = j)$, π its stationary distribution, and λ_2 its second-largest eigenvalue in modulus. ψ collects the parameters of the state process; in §6.6 the vectors ψ_{jk} are the coefficients that tie transitions to lagged covariates z_{t-1}. θ_k collects the parameters of the observation (emission) distribution in state k: for Gaussian emissions the mean μ_k and standard deviation σ_k; for Student-t emissions (§6.3) the location μ_k, the scale σ_k and the degrees of freedom ν_k. D_k is the length of a visit to state k (§6.5).

Three different state beliefs recur throughout and are never interchangeable:

$$\underbrace{\xi_{t|t-1}}_{\text{predicted}} = \Pr(s_t \mid \mathcal{F}_{t-1}), \qquad \underbrace{\xi_{t|t}}_{\text{filtered}} = \Pr(s_t \mid \mathcal{F}_t), \qquad \underbrace{\xi_{t|T}}_{\text{smoothed}} = \Pr(s_t \mid \mathcal{F}_T).$$

Each is a K-vector of probabilities; $\xi_{t|t}^{(k)}$ denotes its k-th component. T is the last observation in the sample. The distinction between these three objects is the single largest source of false results in this field, and §5.1 is devoted to it. The pairwise smoothed probability of a transition, $\xi_{t-1,t|T}^{(jk)} = \Pr(s_{t-1} = j, s_t = k \mid \mathcal{F}_T)$, appears in estimation (§6.1).

On the forecasting side, $x_t \in \mathbb{R}^d$ is the feature vector known at t, y_t the prediction target realised over $(t, t+h]$, and h the forecast horizon in bars. In §5 and §6, where the HMM's observation is not a forecasting target, y_t is simply the observation at t (usually r_t). $f(\cdot)$ is the fitted model and $m(x) = \mathbb{E}[y \mid x]$ the true conditional mean. When f carries a parameter argument — $f(\cdot \mid \theta_k)$ — it means the state-k observation density instead; the argument is what distinguishes the two. $p(x)$, $p(y \mid x)$ and $p(x, y)$ denote the feature, conditional, and joint distributions; subscripting them by k (as in $p_k(y \mid x)$) means "conditional on regime k". w_t is a training sample weight, $g(\cdot)$ a gating function, and $\hat{\sigma}_t$ an $\mathcal{F}_t$-measurable estimate of return volatility.

Two symbols carry a standing qualification: σ is always a return volatility or scale (never a significance level or a sigmoid), and K is always the number of regimes (never a kernel or a Kalman gain).

Matrices and constant vectors are bold: $\mathbf{P}$, $\boldsymbol{\pi}$, and $\mathbf{1}$, the vector of ones. Every other vector (ξ, x_t, z_t, θ_k) is plain, and the context says which it is. The one bold exception is the cross-asset return vector $\mathbf{r}_t$ of §5.9. Symbols that belong to a single model (β, ω, c, γ and similar) are defined where the model appears, and a few letters change meaning between models. The one to watch is λ: it is the second eigenvalue (λ_2), the jump penalty (§5.7, §7.2) or an exponential decay factor (§6.8, §6.11, §10.3), and each use says which. In §12 and A.45–A.46 a plain π_t is a position size, not a component of the stationary distribution (which is always π or π_k, indexed by state).

Table of contents

- Abstract

Part I — What a regime is, and what is known about it

1. What a market regime is
2. Why regimes could exist
3. How the field evolved
4. Foundational references
5. The formal models
6. Hidden Markov models in practice
7. Taxonomy and equivalences
8. What is known to work, and what is not

Part II — Using a regime model

9. Which shift do you have?
10. Conditioning a forecasting model on the regime
11. Research and evaluation
12. Trading a regime-aware model
13. Synthesis

Appendix

- A. Concepts and prerequisites — every advanced idea the main text leans on, built from first principles and ordered by dependency. Start here if a term in §5, §6 or §10 is unfamiliar.

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1. What a market regime is

1.1 The wrong intuition

Ask a portfolio manager what regime the market is in and you will get an answer: risk-on, late-cycle, a high-inflation regime, a dispersion regime. The answer is delivered as a statement about the world, in the same grammatical register as "the S&P is at 5,400". That grammar is the problem, because it presupposes that the regime is a fact about the market which you could in principle observe and might currently be getting wrong.

It isn't. What you observe is a sequence of returns. Any such sequence can be represented as a discrete-state process modulating a family of distributions, in infinitely many ways. Fit a two-state Gaussian mixture and you get a calm state and a turbulent one. Fit four and you get something like crash, slow growth, bull, and recovery — which is exactly what Guidolin and Timmermann (2007) report for joint stock and bond returns, because four is what they fitted. Fit twelve and the likelihood will find work for all twelve. None of these is more correct than the others in any sense the data can adjudicate, because a K-component mixture is a universal approximator for densities: with enough components you can fit any distribution to any tolerance, and with enough states you can fit any path.

So "this market has regimes" is not, on its own, a claim about the world. It becomes one only when you fix K, fix the state dynamics, and commit to out-of-sample evaluation. Everything hard about this field descends from that sentence, and the fact that the standard test for "how many regimes are there" does not have a standard distribution (§8.2) is not a technical footnote — it is the central difficulty wearing a technical disguise.

1.2 The right starting point: a conditional density with a discrete bottleneck

Here is the object, stated once and used throughout. A regime model asserts that the one-step-ahead predictive density of returns is a finite mixture:

$$p(r_t \mid \mathcal{F}_{t-1}) = \sum_{k=1}^{K} \underbrace{\xi_{t|t-1}^{(k)}}_{\text{weight}} \cdot \underbrace{f(r_t \mid \theta_k)}_{\text{component}},$$

where $\xi_{t|t-1}^{(k)} = \Pr(s_t = k \mid \mathcal{F}_{t-1})$ is your predicted probability of being in state k and $f(\cdot \mid \theta_k)$ is the return distribution that state implies. Read what this says:

- **The regime is never observed.** The mixture is what generates data; the state is a latent index. What you can compute is a *posterior over states*, never the state itself.
- **The state label is not the object; the predictive density is.** Collapsing $\xi_{t|t-1}$ to its argmax and treating that label as data throws away the model's own statement about how confident it is. This is the most common practitioner error in the field, and §10.1 says what to feed instead.
- **The model earns its keep only by beating the alternatives**, which are a single distribution ($K = 1$) and a continuously-varying parameter (stochastic volatility, GARCH). It frequently does not. §8.7.

The phrase to hold on to is **discrete bottleneck**. Asserting K regimes is asserting that everything relevant about the conditional distribution of returns can be compressed into which of K boxes you are in. That compression buys statistical efficiency — every observation assigned to box k contributes to estimating the single parameter vector θ_k — and costs fidelity, because the world is not boxes. Whether the trade is worth making is an empirical question with a different answer for volatility (usually yes) than for expected returns (usually no), and §8 is about why.

1.3 Persistence is the entire forecasting content

One consequence of §1.2 deserves its own subsection because it is quantitative, it is routinely violated in practice, and it takes three lines to derive.

Suppose s_t were independent across t — a mixture with no memory. Then $\Pr(s_t = k \mid \mathcal{F}_{t-1}) = \pi_k$ for every history, the predictive density collapses to the *unconditional* mixture $\sum_k \pi_k f(\cdot \mid \theta_k)$, and the model forecasts nothing. It has told you that returns are fat-tailed and heteroskedastic on average. That is a real fact, but it is not a signal.

All forecasting content therefore lives in the gap between the transition matrix $\mathbf{P}$ and the no-memory matrix $\mathbf{1}\pi^\top$, and that gap is governed by the second-largest eigenvalue of $\mathbf{P}$. Write the eigendecomposition of an ergodic chain: $\lambda_1 = 1$ with left eigenvector π, and the remaining eigenvalues $|\lambda_2| \geq \ldots \geq |\lambda_K|$ control the transient. Then

$$\mathbf{P}^h = \mathbf{1}\pi^\top + O(|\lambda_2|^h),$$

so an h-step-ahead state forecast reverts to the unconditional distribution geometrically at rate $|\lambda_2|$. For a two-state chain there is a closed form:

$$\lambda_2 = p_{11} + p_{22} - 1, \qquad \text{half-life} = \frac{\ln 2}{-\ln |\lambda_2|} \text{ bars.}$$

The absolute value matters only in the anti-persistent case $p_{11} + p_{22} < 1$, where $\lambda_2 < 0$ and the state belief alternates as it decays; fitted financial chains are sticky, with λ_2 close to one.

This is a hard ceiling on the horizon at which a regime model can help you, and it is computable from the fitted transition matrix before you run a single backtest. Take the calibration used throughout this document — $p_{11} = 0.99$ for a calm state with expected duration 100 days, $p_{22} = 0.96$ for a turbulent state with expected duration 25 days. Then $\lambda_2 = 0.95$, the half-life is 13.5 days, and at a 60-day horizon $\lambda_2^{60} = 0.046$: **more than 95% of the state information is gone.** A model fitted on daily data with these dynamics cannot inform a quarterly asset-allocation decision, no matter how convincing its in-sample regime plot looks. If you intend to rebalance quarterly, you need a state process whose half-life is measured in quarters — which is a different model with different data requirements, not the same model sampled less often.

> **Diagnostic.** Compute $|\lambda_2|$ and its half-life the moment you fit a transition matrix, and compare it to your rebalancing horizon. If the half-life is shorter than the horizon, the regime model is decoration. **[Practice]**

1.4 The master form

Generalising §1.2 to cover everything in §5, a regime model is a pair of equations — a state process and a state-conditional observation model:

$$\begin{aligned} s_t &\sim q(s_t \mid s_{t-1},\, z_{t-1};\, \psi) && \text{(state dynamics)} \\ y_t \mid s_t = k &\sim f(\cdot \mid x_t;\, \theta_k) && \text{(regime-conditional observation model)} \end{aligned}$$

Here ψ collects the parameters of the state process, z_{t-1} is any observable that may drive transitions, and x_t any observable that may drive the observation. Four slots vary, and every model in §5 is a choice of the four:

Slot	Question	Range of choices
What varies	Which parameters depend on k?	mean only; variance only; mean and variance; full covariance; the whole regression function; the noise distribution
State dynamics	How does s_t move?	Markov; semi-Markov with explicit durations; threshold on observable z; i.i.d. mixture; single permanent break
State driver	What determines the transition?	endogenous (past y); exogenous (z); free (latent only)
Inference and use	What do you condition on, and what consumes it?	predicted / filtered / smoothed; hard label or posterior; feeds parameters, weights, features, or a gate

Two things are worth noticing immediately. First, the *inference and use* slot is not part of the model — it is part of the research protocol — yet it swamps the other three in its effect on measured performance. Second, most of the literature varies slot 1 and slot 2 while holding slots 3 and 4 fixed, which is why the model zoo of §5 looks larger than the design space of §7 turns out to be.

1.5 What a regime is not

A great deal of confusion is dissolved by putting the near neighbours in a table and saying how each differs.

Object	What it is	How it differs from a regime
Trend	a signed conditional mean, $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$	continuous-valued and *ordered*; regimes are categorical and unordered
Structural break	a one-time permanent parameter change	non-recurrent — you never come back, so there is nothing to estimate from history about the new state
Stochastic volatility	a continuous latent scale σ_t	no discrete bottleneck; σ_t takes a continuum of values
GARCH	σ_t^2 as a deterministic function of past returns	no latent state at all — σ_t is $\mathcal{F}_{t-1}$-measurable, so there is nothing to filter
Conditioning variable	an observable such as the dividend yield or the VIX	observable and continuous; you can regress on it directly
Business-cycle phase	an economic classification (e.g. NBER dates)	published with a 6–18 month lag and revised; not $\mathcal{F}_t$-measurable, so unusable as a live signal †
Microstructure state	order-book condition (wide/tight, toxic/benign)	different time scale by five orders of magnitude, and a different object
Cross-sectional cluster	a partition of *assets*	a regime is a partition of *time*; the two are often confused because both use k-means

† The NBER Business Cycle Dating Committee's announcements are the canonical example of a variable that is informative, widely cited, and completely untradable at the dates it labels. Any backtest that conditions on NBER recession dates without lagging them by the actual announcement delay is reporting a fantasy. **[Fact]**

The GARCH row is the one to sit with. **The most common alternative to a regime model is not "no model" — it is a continuous-state model, and the continuous model usually wins.** A regime model buys you interpretability and a natural place to hang discrete decisions; it pays for that with a coarse approximation to a state variable that is, on the evidence, continuous. §8.7 puts numbers on the trade.

§1 Key takeaways

1. A regime is a property of a model, not of the market. The question "what regime are we in" is only well posed after you have fixed K and the state dynamics.
2. The object a regime model produces is a predictive *density* $\sum_k \xi_{t|t-1}^{(k)} f(\cdot \mid \theta_k)$, not a label. Collapsing to the argmax label discards the model's own uncertainty and is the most common implementation error in the field.
3. All forecasting content lives in the persistence of the state. An i.i.d. mixture predicts nothing beyond the unconditional distribution.
4. The second eigenvalue of the transition matrix sets a hard ceiling on the useful forecast horizon. Compute its half-life before you backtest; if it is shorter than your rebalancing interval, stop.
5. Regimes are categorical partitions of *time*. Trends, stochastic volatility, GARCH, conditioning variables, and cross-sectional clusters are all different objects, and three of them are usually better tools.
6. NBER-style economic regime labels are published with a lag and revised. Using them unlagged is a look-ahead, not a modelling choice.

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2. Why regimes could exist

Before fitting anything it is worth asking what would have to be true of the world for a *discrete* state model to be the right description. This matters more than it sounds, because most proposed mechanisms predict *continuous* variation in conditional moments, and a continuous mechanism observed through a discrete model produces exactly the kind of unstable, sample-dependent state estimates that §8 documents.

There is one mechanism family that genuinely predicts discreteness, and it is not the one usually cited.

2.1 Family A — occasionally binding constraints

The strongest microfoundation for discreteness is an inequality constraint that sometimes binds and sometimes does not. Intermediaries face leverage limits, margin requirements, VaR budgets, and redemption thresholds. Each is a constraint of the form $c(\text{state}) \leq \bar{c}$, and the associated Lagrange multiplier is exactly zero when it is slack and strictly positive when it binds. That is not an approximation to a discrete state — **it is a discrete state**, arising from the complementary-slackness condition of an optimisation problem.

The consequences are asymmetric and fast. Brunnermeier and Pedersen (2009) model the feedback between market liquidity and funding liquidity, in which a shock that tightens funding forces deleveraging, which widens spreads, which tightens funding further. He and Krishnamurthy (2013) and Adrian, Etula and Muir (2014) develop the asset-pricing consequences: intermediary net worth is a priced state variable, and its effect on risk premia is strongly nonlinear near the constraint. **[Contested]** — the mechanism is well formalised and the intermediary-capital factor prices assets in-sample, but whether the constraint-binding indicator is *measurable in real time* well enough to trade is a different and much weaker claim.

This family predicts things the others do not, which makes it the most falsifiable: state transitions should be **fast in one direction and slow in the other** (constraints bind abruptly and relax gradually), correlations should *rise* in the constrained state as everyone liquidates the same assets, and the state should be visible in funding-market observables (repo spreads, cross- currency basis, dealer balance sheets) before it is visible in returns.

2.2 Family B — policy and institutional regimes

Central-bank reaction functions really do change, discretely, when the committee or the mandate changes. Sims and Zha (2006) fit a multivariate regime-switching model to U.S. monetary policy and find three coefficient regimes roughly matching the periods that narrative accounts identify. Their headline result is more interesting than the existence of the regimes, though, and it recurs throughout this document: **the best-fitting specification allowed time variation in disturbance variances only.** When they allowed coefficients to switch as well, the improvement was modest and the regime differences were too small to explain the inflation of the 1970s–80s. **[Fact]**

Regulatory changes (Reg NMS, Basel III, the Volcker Rule), index-methodology changes, and market-structure changes (decimalisation, the rise of ETFs) are the same kind of event: genuinely discrete, dated, and externally observable. Note what that last property implies — if the regime is externally observable, you should condition on the observable directly rather than inferring a latent state from returns (§5.5, §7.4).

2.3 Family C — the leverage cycle and multiple equilibria

Geanakoplos's leverage cycle and the limits-to-arbitrage literature (Shleifer and Vishny, 1997) describe positive feedback: falling prices reduce collateral value, which forces sales, which lowers prices. Positive feedback loops admit multiple equilibria, and multiple equilibria are discrete by construction. **[Hypothesis]** — the theory is well developed, but distinguishing "the system jumped between equilibria" from "a large shock hit a continuous system" using return data alone has not, on the published evidence, been done convincingly.

2.4 Family D — investor composition and beliefs

Barberis, Shleifer and Vishny (1998) model investor sentiment *as* a regime-switching process: the representative investor believes earnings follow either a mean-reverting or a trending regime, and Bayesian updating between the two generates under- and over-reaction. This is a rare case where the regime model is the economic theory rather than a statistical convenience. Flow-driven versions — retail participation, systematic-strategy crowding, dealer gamma positioning — make the same argument about who is trading rather than what they believe. **[Contested]**

2.5 Family E — the business cycle

Expected returns vary with business conditions: Fama and French (1989) show that dividend yields and term spreads forecast returns with cyclical patterns, and Cochrane's (2011) survey makes the case that essentially all of the variation in asset prices is discount-rate variation, which is cyclical. Henkel, Martin and Nardari (2011) sharpen this into a regime statement: **short-horizon return predictability is concentrated in recessions and close to zero in expansions.** Farmer, Schmidt and Timmermann (2023) find the same shape without imposing a cycle — predictability arrives in short, localised "pockets".

This is the most economically respectable case for regime-conditional *return* models, and it comes with two caveats. First, business-cycle phases are long: a model with cycle-length persistence has perhaps 10–15 independent observations in a post-war sample, which is the identification problem of §8.1 in its most acute form. Second, the pockets result has been challenged — Cakici and co-authors published a replication in the *Journal of Finance* in 2025 disputing the out-of-sample economic significance. **[Contested]**

2.6 Family F — there are no regimes

The null deserves to be stated as forcefully as the alternatives, because it is the hypothesis most consistent with the evidence and the one most rarely tested.

A single process with persistent stochastic volatility and heavy tails — no discrete states anywhere — produces sample paths that a regime model will happily describe with two to four states, with convincing-looking state plots and a significant-looking likelihood improvement. Diebold and Inoue (2001) prove a version of this analytically: **stochastic regime switching and long memory are easily confused, even asymptotically**, and a process with occasional breaks generates estimated long-memory parameters, while a long-memory process generates apparent breaks. Granger and Hyung (2004) demonstrate the same confusion empirically on S&P 500 absolute returns. The precedent is Perron (1989), who showed that a trend-stationary series with one break is nearly indistinguishable from a unit root.

The practical statement: **a likelihood improvement from adding regimes is not evidence that regimes exist.** It is evidence that your single-state model was misspecified, which you already knew, and a mixture is a flexible way to absorb misspecification. **[Fact]**

2.7 Telling the families apart

Most of the time you cannot, and pretending otherwise is how research programmes die. But the families do make different predictions, and a few of them are testable on data you already have.

Prediction	Constraints (A)	Policy (B)	Leverage cycle (C)	Business cycle (E)	No regimes (F)
Transition asymmetry	fast in, slow out	either	fast in, slow out	symmetric	none expected
Correlations rise in bad state	strongly yes	no	yes	mildly	mildly †
State visible in non-return data	yes (funding)	yes (dated)	partly	yes (macro)	no

Prediction	Constraints (A)	Policy (B)	Leverage cycle (C)	Business cycle (E)	No regimes (F)
Optimal K stable across samples	yes	yes	—	—	**no**
Beaten by a continuous-vol model	no	no	—	—	**yes**

† A stochastic-volatility null does generate correlation increases in high-vol periods, purely mechanically, because correlations estimated over high-variance windows are biased upward when the common factor's variance rises faster than the idiosyncratic. Forbes and Rigobon (2002) made this point about "contagion" and it applies verbatim here: **an increase in measured correlation during a crisis is partly an artefact of the volatility increase, not independent evidence of a regime.**

The last two rows are the ones to act on, and they yield two concrete experiments you can run in an afternoon:

1. **Stability of K.** Fit $K = 2, 3, 4, 5$ on disjoint halves of your sample and on bootstrapped resamples. If the selected K and the fitted θ_k move around, you are fitting misspecification, not structure.
2. **The continuous-model horse race.** Fit a GARCH or stochastic-volatility model and a regime model, and compare *out-of-sample predictive log-likelihood*, not in-sample fit. In most published comparisons, and in the worked example of §6.11, the continuous model wins on volatility and the two are hard to separate on everything else. **[Practice]**

And one more that is worth more than both: **make the state predict something it was not fitted on.** A state extracted from equity returns that also lines up with credit spreads, funding costs, or realised dispersion is more likely to be real than one that only explains the series it was estimated from.

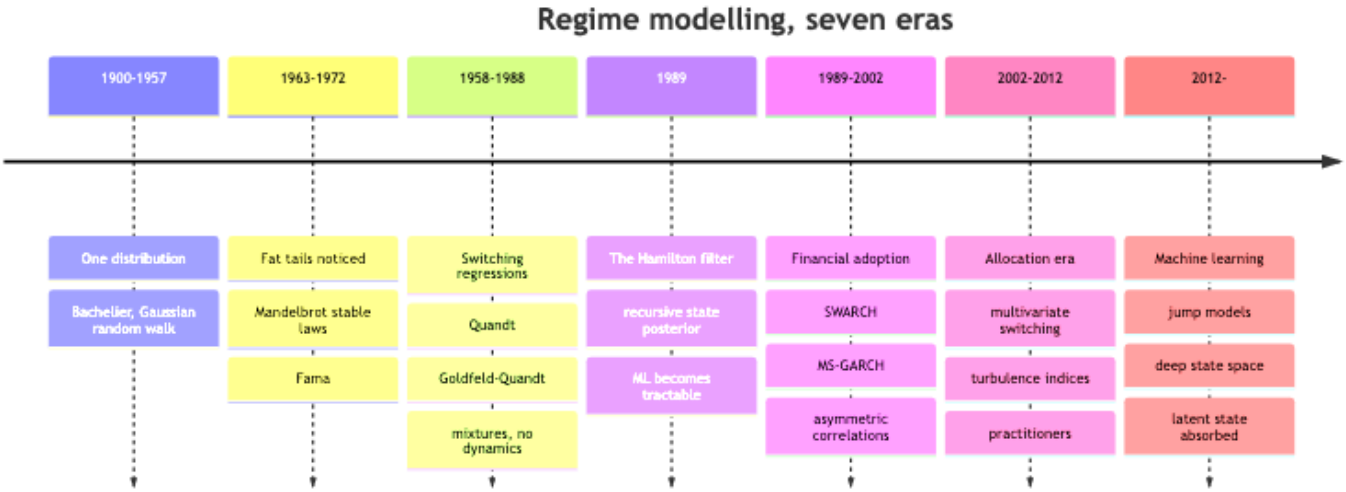
§2 Key takeaways

1. Occasionally binding constraints are the only mechanism family that predicts genuine discreteness rather than continuous variation, because a complementary-slackness condition literally *is* a two-state system.
2. Policy regimes are real, dated, and externally observable — which argues for conditioning on the observable rather than filtering a latent state.
3. Sims and Zha's result generalises: when you let both means and variances switch, the variance switching does nearly all the work. This recurs at every level of this document.
4. Return predictability is concentrated in bad times — the best economic case for regime-conditional alpha — but bad times are rare, so the sample is tiny and the finding is contested.
5. A stochastic-volatility process with no regimes anywhere produces data that regime models fit beautifully. Likelihood improvement from adding states is not evidence that states exist.
6. Two cheap discriminating tests: is the selected K stable across subsamples, and does a continuous-state model beat the discrete one out of sample? Run both before building anything.
7. Rising correlations in crises are partly a mechanical artefact of rising volatility (Forbes and Rigobon, 2002), not independent evidence of a regime.

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3. How the field evolved

The history explains the shape of the current literature — in particular why the econometrics is deep and the trading evidence is thin, and why the speech-recognition and machine-learning communities developed the same objects under different names.



3.1 Era I — One distribution (1900–1957)

Contribution. Bachelier's 1900 thesis modelled prices as Brownian motion; the mid-century finance literature adopted the Gaussian random walk as the baseline.

What changed. The idea that price changes are draws from a *fixed* distribution, which is the null every subsequent era has been arguing with.

Limitations. Empirically wrong in the second and fourth moments, as the next era established.

Lasting influence. Enormous, and mostly as a foil. Every regime paper's introduction is a variation on "returns are not i.i.d. Gaussian".

3.2 Era II — Fat tails, and the first alternative to regimes (1963–1972)

Contribution. Mandelbrot (1963) documented that speculative price changes have far heavier tails than the Gaussian and proposed stable Paretian distributions; Fama (1965) confirmed and extended the evidence for U.S. stocks.

What changed. The field accepted non-normality. Critically, **the first explanation offered was not regimes — it was a single heavy-tailed distribution.** Mandelbrot's alternative was one law with infinite variance, not two laws that alternate.

Limitations. Stable laws with $\alpha < 2$ imply infinite variance, which is awkward for portfolio theory and, on the evidence of later work, not what the data show; the tails are heavy but the variance appears finite.

Lasting influence. The unresolved question is the important legacy. "Is this one fat-tailed process or two thin-tailed ones taking turns?" was never settled; the profession simply moved on to models where the question does not have to be answered. §2.6 and §8.7 are that question, still open.

3.3 Era III — Switching regressions (1958–1988)

Contribution. Quandt (1958) estimated a linear regression obeying two separate regimes with an unknown switch point; Goldfeld and Quandt (1973) introduced a Markov model for switching regressions.

What changed. Regimes became estimable objects rather than narrative descriptions.

Limitations. The estimation was awkward and, in the mixture versions, the state carried no persistence, so the models had no forecasting content (§1.3). There was no efficient recursion for the state posterior, which made likelihood evaluation expensive and multivariate extensions impractical.

Lasting influence. Direct — Hamilton's paper is explicitly a solution to the computational problem this era posed.

3.4 Era IV — The Hamilton filter (1989)

Contribution. Hamilton (1989) gave a recursive algorithm — one forward pass — for the state posterior in a Markov-switching autoregression, making maximum likelihood estimation tractable and, at the same time, making the distinction between filtered and smoothed inference explicit and computable. Kim (1994) supplied the efficient backward smoother that completes the pair.

What changed. Almost everything. This is the founding moment of the modern field, and the filter is still what every implementation runs. Its structure — predict, then update on the new observation — is identical to the Kalman filter with a discrete state space, which is why the machine-learning literature's forward-backward algorithm for HMMs (Baum et al., 1970; Rabiner, 1989) is the same algorithm discovered a generation earlier in a different community.

Limitations. Hamilton's own application was to U.S. GNP growth, a low-frequency, low-noise series. Financial returns are the opposite, and the transplant was less successful than the enthusiasm of the following decade suggested.

Lasting influence. Total. If you fit a regime model today, you are running Hamilton's recursion.

3.5 Era V — Financial adoption, and the discovery that it is about variance (1989–2002)

Contribution. Turner, Startz and Nelson (1989) applied switching to stock returns; Hamilton and Susmel (1994) combined switching with ARCH (SWARCH); Gray (1996) and Haas, Mittnik and Paolella (2004) solved the path-dependence problem that makes naive MS-GARCH inestimable; Ang and Bekaert (2002) built the international asset-allocation application; Longin and Solnik (2001) and Ang and Chen (2002) documented that correlations rise in down markets.

What changed. Two things, in opposite directions. Positively, the field established that **regime models describe second moments well** — volatility states are large, persistent, and estimable. Negatively, the critiques arrived and were not answered. Hansen (1992) showed that the likelihood-ratio test for the number of regimes does not have a standard distribution, because the transition probabilities are unidentified under the null; applying his bound to Hamilton's own GNP model, **the switching specification could not reject a plain AR(4)**. Dacco and Satchell (1999) showed that regime-switching models fit exchange rates beautifully in sample and lose to a random walk out of sample, and — this is the part usually forgotten — explained *why*: the loss function is asymmetric in the state-classification error, so even a model with correctly estimated parameters forecasts worse than a constant if it misclassifies the state often enough.

Limitations. The era's applied work systematically reported smoothed state probabilities and in-sample fit.

Lasting influence. The second-moment result is the durable one. The first-moment results from this era have mostly not replicated.

3.6 Era VI — Multivariate models and the allocation application (2002–2012)

Contribution. Guidolin and Timmermann (2007) fitted four-state models to joint stock and bond returns and solved the resulting dynamic portfolio problem; Pelletier (2006) built regime-switching dynamic correlations; Kritzman and Li (2010) introduced the Mahalanobis "turbulence" index and Kritzman, Page and Turkington (2012) the practitioner framework built on it; Bulla and co-authors (2011) ran the honest out-of-sample test of a Markov-switching allocation rule including transaction costs.

What changed. The application shifted from *predicting returns* to *managing risk*, which is where the evidence actually supports regime models. Kritzman, Page and Turkington's framing — the value is in avoiding large losses, not in picking direction — is the correct reading of the previous era's results.

Limitations. Much of the practitioner work in this era comes from firms that sell regime-based products. That does not make it wrong — these authors have data and implementation experience academics lack — but the backtests are selection-prone and the reported improvements are typically not adjusted for the number of specifications tried.

Lasting influence. High. This is the era whose framing most institutional allocators still use.

3.7 Era VII — Machine learning, and the disappearing regime (2012–present)

Contribution. Several strands at once. Unsupervised methods — Gaussian mixtures, k-means on engineered features, Wasserstein clustering of correlation matrices — replaced likelihood-based HMMs where the emission model was the weak link. Statistical jump models (Bemporad et al., 2018; Nystrup, Kolm and Lindström, 2020) replaced the probabilistic state process with an explicit jump *penalty*, giving an objective whose state assignment is solved exactly by dynamic programming (§5.7), a tunable persistence knob, and — by the authors' simulations — better classification accuracy than a correctly

specified maximum-likelihood HMM. Deep latent-variable models (recurrent switching linear dynamical systems, deep state-space models) brought neural-network emissions to the same graphical structure.

What changed. The most consequential shift is a negative one. **The dominant machine-learning approach to asset pricing does not use an explicit regime model at all.** Gu, Kelly and Xiu (2020) handle time variation by interacting stock characteristics with a small set of observable macro predictors and letting the function approximator find whatever conditional structure exists; Chen, Pelger and Zhu (2024) learn a latent macro state with a recurrent network as part of an end-to-end asset-pricing objective rather than fitting it separately. The latent state did not disappear — it got absorbed into the approximator, where it is estimated jointly with the thing you actually care about instead of being handed over from a separately fitted model with its own errors.

Meanwhile the likelihood-based HMM did not stand still. Student-t emissions, semi-Markov durations, covariate-driven transitions, Bayesian estimation with persistence priors and adaptive estimation with forgetting were refined and brought to financial returns in this period (§6; several have older roots, §4), and each addresses a specific failure of the 1989 model. The practical question the era leaves is: **should the regime be a separate model whose output you use directly or feed in, or a structure a forecasting model discovers for itself?** §6.10 covers direct use, and §9 and §10 answer the second half, which is "it depends on which distribution is shifting"; §10.9 ranks the options.

Limitations. The ML era has largely not engaged with the identification critiques of Era V. A deep switching model has all of Hamilton's identification problems plus its own, and the papers rarely report the diagnostics of §2.7 and §8.2.

§3 Key takeaways

1. Hamilton (1989) is the founding technical contribution; every implementation since runs his recursion, which is the discrete-state Kalman filter and the same object as the HMM forward algorithm from speech recognition.
2. The first proposed explanation for non-normal returns was one heavy-tailed distribution, not two alternating ones. That debate was never settled — it was bypassed — and it is still the live null.
3. The critiques landed early and were not answered. Hansen (1992) showed the number-of-regimes test is non-standard and that Hamilton's own model could not reject an AR(4); Dacco and Satchell (1999) showed out-of-sample failure and explained the mechanism.
4. The durable empirical finding across every era is that regimes describe second moments well and first moments badly.
5. The 2002–2012 practitioner era got the framing right — use regimes for risk, not direction — but its evidence base is dominated by interested parties.
6. The modern machine-learning answer is usually to skip the explicit regime model and let the approximator learn the conditioning. Whether that is right depends on which distribution is shifting (§9). The HMM meanwhile gained the refinements — heavy tails, durations, covariates, priors — that address its original failures (§6).

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4. Foundational references

Grouped by kind, because kinds are read differently. Each entry says why it matters. Where a free copy exists it is linked from the title; paywalled-only entries are marked.

4.1 The founding econometrics

- **Quandt, R. E. (1958).** "The Estimation of the Parameters of a Linear Regression System Obeying Two Separate Regimes." *Journal of the American Statistical Association* 53(284), 873–880. — The origin. Two regimes, one unknown switch point, no state process.
- **Goldfeld, S. M. & Quandt, R. E. (1973).** "A Markov Model for Switching Regressions." *Journal of Econometrics* 1(1), 3–15. — Adds the Markov chain. The model exists; the estimation is still impractical.
- **Hamilton, J. D. (1989).** "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica* 57(2), 357–384. — **The** paper. The recursive filter that makes everything else computable. Read §2–3 of it even if you read nothing else in this list.

- **Hamilton, J. D. (1990).** "Analysis of Time Series Subject to Changes in Regime." *Journal of Econometrics* 45(1–2), 39–70. [publisher — paywalled] — The EM algorithm for Markov-switching models, in the notation econometricians use (§6.1).
- **Hamilton, J. D. (1994).** *Time Series Analysis.* Princeton University Press, chapter 22. — The textbook treatment of the filter, the smoother, and the EM algorithm, worked in full. This is the reference you will actually implement from.
- **Kim, C.-J. (1994).** "Dynamic Linear Models with Markov-Switching." *Journal of Econometrics* 60(1–2), 1–22. — The efficient backward smoother. Necessary for estimation; dangerous if you use its output as a signal (§5.1).
- **Kim, C.-J. & Nelson, C. R. (1999).** *State-Space Models with Regime Switching.* MIT Press. — The systematic treatment, including the Bayesian versions and the approximations that make multivariate models tractable.
- **Diebold, F. X., Lee, J.-H. & Weinbach, G. C. (1994).** "Regime Switching with Time-Varying Transition Probabilities." In C. P. Hargreaves, ed., *Nonstationary Time Series Analysis and Cointegration*, Oxford University Press, 283–302. [publisher — paywalled] — Transition probabilities that depend on observables, with an EM algorithm. The origin of §6.6.
- **Filardo, A. J. (1994).** "Business-Cycle Phases and Their Transitional Dynamics." *Journal of Business & Economic Statistics* 12(3), 299–308. [publisher — paywalled] — Time-varying transitions driven by leading indicators; the standard applied template.
- **Durland, J. M. & McCurdy, T. H. (1994).** "Duration-Dependent Transitions in a Markov Model of U.S. GNP Growth." *Journal of Business & Economic Statistics* 12(3), 279–288. [publisher — paywalled] — Transition probabilities that depend on how long the state has lasted (§6.5).
- **Hamilton, J. D. (2016).** "Macroeconomic Regimes and Regime Shifts." In J. B. Taylor & H. Uhlig, eds., *Handbook of Macroeconomics* vol. 2A, Elsevier, 163–201. — The founder's own retrospective: what the models have and have not delivered in macroeconomics.

4.2 Inference machinery, from the other community

The speech-recognition and machine-learning literature developed the same algorithms independently, and its expositions are often clearer.

- **Baum, L. E., Petrie, T., Soules, G. & Weiss, N. (1970).** "A Maximization Technique Occurring in the Statistical Analysis of Probabilistic Functions of Markov Chains." *Annals of Mathematical Statistics* 41(1), 164–171. — The EM algorithm for HMMs, nineteen years before Hamilton and in a different field.
- **Rabiner, L. R. (1989).** "A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition." *Proceedings of the IEEE* 77(2), 257–286. — Still the best single exposition of forward-backward, Viterbi, and Baum-Welch. If Hamilton's notation is fighting you, read this instead.
- **Zucchini, W., MacDonald, I. L. & Langrock, R. (2016).** *Hidden Markov Models for Time Series: An Introduction Using R*, 2nd ed. Chapman & Hall/CRC. — The practical book. Numerical stability, scaling, label switching, model checking.
- **Frühwirth-Schnatter, S. (2006).** *Finite Mixture and Markov Switching Models.* Springer. — The Bayesian reference, and the best treatment of the label-switching and identification problems.
- **Cappé, O., Moulines, E. & Rydén, T. (2005).** *Inference in Hidden Markov Models.* Springer. — The theory: consistency, asymptotics, and what can and cannot be identified.
- **Fox, E. B., Sudderth, E. B., Jordan, M. I. & Willsky, A. S. (2011).** "A Sticky HDP-HMM with Application to Speaker Diarization." *Annals of Applied Statistics* 5(2A), 1020–1056. — Nonparametric K plus an explicit persistence prior. The right answer to "how many regimes" if you insist on a Bayesian one, and the origin of the "stickiness" idea that jump models later made explicit.
- **Albert, J. H. & Chib, S. (1993).** "Bayes Inference via Gibbs Sampling of Autoregressive Time Series Subject to Markov Mean and Variance Shifts." *Journal of Business & Economic Statistics* 11(1), 1–15. [publisher — paywalled] — The first Gibbs sampler for Markov-switching models, drawing one state at a time.
- **Carter, C. K. & Kohn, R. (1994).** "On Gibbs Sampling for State Space Models." *Biometrika* 81(3), 541–553. [publisher — paywalled] — Forward filtering, backward sampling: draw the whole latent path at once.
- **Chib, S. (1996).** "Calculating Posterior Distributions and Modal Estimates in Markov Mixture Models." *Journal of Econometrics* 75(1), 79–97. [publisher — paywalled] — The joint-path sampler for HMMs, which mixes far better than single-site sampling (§6.7).
- **Stephens, M. (2000).** "Dealing with Label Switching in Mixture Models." *Journal of the Royal Statistical Society: Series B* 62(4), 795–809. [publisher — paywalled] — Relabelling MCMC draws after the fact.
- **Frühwirth-Schnatter, S. (2001).** "Markov Chain Monte Carlo Estimation of Classical and Dynamic Switching and Mixture Models." *Journal of the American Statistical Association* 96(453), 194–209. [publisher — paywalled] — The random permutation sampler, and identification by ordering constraints.
- **Scott, S. L. (2002).** "Bayesian Methods for Hidden Markov Models: Recursive Computing in the 21st Century." *Journal of the American Statistical Association* 97(457), 337–351. [publisher — paywalled] — The clearest review of the recursions behind Bayesian HMM computation.

- **Celeux, G. & Durand, J.-B. (2008).** "Selecting Hidden Markov Model State Number with Cross-Validated Likelihood." *Computational Statistics* 23(4), 541–564. [publisher — paywalled] — Choosing K by held-out likelihood rather than by an information criterion.
- **Langrock, R. & Zucchini, W. (2011).** "Hidden Markov Models with Arbitrary State Dwell-Time Distributions." *Computational Statistics & Data Analysis* 55(1), 715–724. [publisher — paywalled] — Semi-Markov durations approximated by an ordinary HMM with expanded states, so the standard machinery still applies (§6.5).
- **Cappé, O. (2011).** "Online EM Algorithm for Hidden Markov Models." *Journal of Computational and Graphical Statistics* 20(3), 728–749. — Recursive updating of HMM parameters as data arrive (§6.8).
- **Pohle, J., Langrock, R., van Beest, F. M. & Schmidt, N. M. (2017).** "Selecting the Number of States in Hidden Markov Models: Pragmatic Solutions Illustrated Using Animal Movement." *Journal of Agricultural, Biological and Environmental Statistics* 22(3), 270–293. — Why information criteria over-select states, and a practical procedure. The application is ecology; the argument is general.
- **Software. Visser, I. & Speekenbrink, M. (2010)**, "depmixS4: An R Package for Hidden Markov Models", *Journal of Statistical Software* 36(7); **O'Connell, J. & Højsgaard, S. (2011)**, "Hidden Semi Markov Models for Multiple Observation Sequences: The mhsmm Package for R", *Journal of Statistical Software* 39(4); and **Ardia, D., Bluteau, K., Boudt, K., Catania, L. & Trottier, D.-A. (2019)**, "Markov-Switching GARCH Models in R: The MSGARCH Package", *Journal of Statistical Software* 91(4). — The packages of §6.12.

4.3 Regime models in finance — the empirical case

- **Turner, C. M., Startz, R. & Nelson, C. R. (1989).** "A Markov Model of Heteroskedasticity, Risk, and Learning in the Stock Market." *Journal of Financial Economics* 25(1), 3–22. — The first serious equity application.
- **Hamilton, J. D. & Susmel, R. (1994).** "Autoregressive Conditional Heteroskedasticity and Changes in Regime." *Journal of Econometrics* 64(1–2), 307–333. — SWARCH: switching plus ARCH, and the demonstration that most of what looks like ARCH persistence is regime persistence.
- **Gray, S. F. (1996).** "Modeling the Conditional Distribution of Interest Rates as a Regime-Switching Process." *Journal of Financial Economics* 42(1), 27–62. [publisher — paywalled] — Solves the path-dependence problem that makes naive MS-GARCH impossible to estimate.
- **Haas, M., Mittnik, S. & Paolella, M. S. (2004).** "A New Approach to Markov-Switching GARCH Models." *Journal of Financial Econometrics* 2(4), 493–530. — The cleaner solution, and the one now generally used.
- **Ang, A. & Bekaert, G. (2002).** "International Asset Allocation With Regime Shifts." *Review of Financial Studies* 15(4), 1137–1187. — The canonical allocation application: correlations and volatilities rise together in the bad state, and what that does to optimal portfolios.
- **Longin, F. & Solnik, B. (2001).** "Extreme Correlation of International Equity Markets." *Journal of Finance* 56(2), 649–676. — Correlation is not constant in the tails, established with extreme-value theory rather than by splitting the sample.
- **Ang, A. & Chen, J. (2002).** "Asymmetric Correlations of Equity Portfolios." *Journal of Financial Economics* 63(3), 443–494. — The same asymmetry domestically, with a statistic designed to avoid the conditioning bias.
- **Guidolin, M. & Timmermann, A. (2007).** "Asset Allocation under Multivariate Regime Switching." *Journal of Economic Dynamics and Control* 31(11), 3503–3544. — Four states for stocks and bonds jointly, and the dynamic programme that uses them.
- **Ang, A. & Timmermann, A. (2012).** "Regime Changes and Financial Markets." *Annual Review of Financial Economics* 4, 313–337. — **The single best entry point.** Balanced, current enough, and honest about what the models do not deliver.
- **Sims, C. A. & Zha, T. (2006).** "Were There Regime Switches in U.S. Monetary Policy?" *American Economic Review* 96(1), 54–81. — Rigorous multivariate switching applied to policy, with the recurring punchline: variances switch, coefficients barely do. [publisher — paywalled]
- **Henkel, S. J., Martin, J. S. & Nardari, F. (2011).** "Time-Varying Short-Horizon Predictability." *Journal of Financial Economics* 99(3), 560–580. — Return predictability concentrated in recessions. The strongest published case for regime-conditional expected returns.
- **Schaller, H. & van Norden, S. (1997).** "Regime Switching in Stock Market Returns." *Applied Financial Economics* 7(2), 177–191. [publisher — paywalled] — Stock-market regimes whose transitions depend on deviations from fundamentals.
- **Rydén, T., Teräsvirta, T. & Åsbrink, S. (1998).** "Stylized Facts of Daily Return Series and the Hidden Markov Model." *Journal of Applied Econometrics* 13(3), 217–244. [publisher — paywalled] — What a Gaussian HMM reproduces in daily returns, and the one fact it misses: the slow decay of autocorrelation in absolute returns.
- **Maheu, J. M. & McCurdy, T. H. (2000).** "Identifying Bull and Bear Markets in Stock Returns." *Journal of Business & Economic Statistics* 18(1), 100–112. — Duration-dependent switching on 160 years of monthly data; bull markets have a declining hazard.
- **Perez-Quiros, G. & Timmermann, A. (2000).** "Firm Size and Cyclical Variations in Stock Returns." *Journal of Finance* 55(3), 1229–1262. [publisher — paywalled] — Time-varying transitions driven by monetary indicators, with small firms

most exposed to the recession state.

- **Hardy, M. R. (2001).** "A Regime-Switching Model of Long-Term Stock Returns." *North American Actuarial Journal* 5(2), 41–53. [publisher — paywalled] — The two-state regime-switching lognormal model, used for long-dated equity guarantees (§6.10).
- **Lunde, A. & Timmermann, A. (2004).** "Duration Dependence in Stock Prices: An Analysis of Bull and Bear Markets." *Journal of Business & Economic Statistics* 22(3), 253–273. [publisher — paywalled] — Duration dependence in bull and bear markets dated by price rules.
- **Bulla, J. & Bulla, I. (2006).** "Stylized Facts of Financial Time Series and Hidden Semi-Markov Models." *Computational Statistics & Data Analysis* 51(4), 2192–2209. — Negative binomial sojourns reproduce the autocorrelation of squared returns better than geometric ones.
- **Kim, C.-J., Piger, J. & Startz, R. (2008).** "Estimation of Markov Regime-Switching Regression Models with Endogenous Switching." *Journal of Econometrics* 143(2), 263–273. — What changes when the state is correlated with the shocks, and how to test for it.
- **Bulla, J. (2011).** "Hidden Markov Models with t Components. Increased Persistence and Other Aspects." *Quantitative Finance* 11(3), 459–475. [publisher — paywalled] — Student-t emissions make the fitted states more persistent. The most useful single refinement for daily data (§6.3).
- **Guidolin, M. (2011).** "Markov Switching Models in Empirical Finance." *Advances in Econometrics* 27B, 1–86. [publisher — paywalled] — A long survey of the applications, model by model.
- **Nystrup, P., Madsen, H. & Lindström, E. (2015).** "Stylised Facts of Financial Time Series and Hidden Markov Models in Continuous Time." *Quantitative Finance* 15(9), 1531–1541. — Documents how fitted HMM parameters drift through time.
- **Nystrup, P., Madsen, H. & Lindström, E. (2017).** "Long Memory of Financial Time Series and Hidden Markov Models with Time-Varying Parameters." *Journal of Forecasting* 36(8), 989–1002. — Adaptive estimation with exponential forgetting; time-varying parameters reproduce long memory and improve density forecasts (§6.8).
- **Kole, E. & van Dijk, D. (2017).** "How to Identify and Forecast Bull and Bear Markets?" *Journal of Applied Econometrics* 32(1), 120–139. [publisher — paywalled] — Price rules date regimes better in sample; regime-switching models forecast better.
- **Farmer, L. E., Schmidt, L. & Timmermann, A. (2023).** "Pockets of Predictability." *Journal of Finance* 78(3), 1279–1341. — Predictability arrives in short local episodes rather than as a stable relationship. Read with the replication in §4.4.

4.4 The critiques — read these next to §4.3

A bibliography that lists only a field's successes is marketing. These are the papers that constrain what you should believe.

- **Hansen, B. E. (1992).** "The Likelihood Ratio Test under Nonstandard Conditions: Testing the Markov Switching Model of GNP." *Journal of Applied Econometrics* 7(S1), S61–S82. — The number of regimes cannot be tested with a standard LR test, because the transition probabilities are unidentified under the null. Applying a valid bound, **Hamilton's own model does not reject a plain AR(4)**. This is the most important critique in the field and the least cited relative to its importance.
- **Garcia, R. (1998).** "Asymptotic Null Distribution of the Likelihood Ratio Test in Markov Switching Models." *International Economic Review* 39(3), 763–788. — Works out the correct asymptotics in specific cases.
- **Cho, J. S. & White, H. (2007).** "Testing for Regime Switching." *Econometrica* 75(6), 1671–1720. — The modern treatment of the same problem.
- **Dacco, R. & Satchell, S. (1999).** "Why Do Regime-Switching Models Forecast So Badly?" *Journal of Forecasting* 18(1), 1–16. — Excellent in-sample fit, worse than a random walk out of sample, with an analysis of *why*: state misclassification costs more than correct state parameters gain. [publisher — paywalled]
- **Diebold, F. X. & Inoue, A. (2001).** "Long Memory and Regime Switching." *Journal of Econometrics* 105(1), 131–159. — Regime switching and long memory are analytically confusable even asymptotically. The observational-equivalence problem stated precisely.
- **Granger, C. W. J. & Hyung, N. (2004).** "Occasional Structural Breaks and Long Memory with an Application to the S&P 500 Absolute Stock Returns." *Journal of Empirical Finance* 11(3), 399–421. — The same confusion, demonstrated on real data.
- **Perron, P. (1989).** "The Great Crash, the Oil Price Shock, and the Unit Root Hypothesis." *Econometrica* 57(6), 1361–1401. — The precedent from a different literature: breaks and persistence are nearly indistinguishable in finite samples. The lesson generalises.
- **Forbes, K. J. & Rigobon, R. (2002).** "No Contagion, Only Interdependence: Measuring Stock Market Comovements." *Journal of Finance* 57(5), 2223–2261. — Correlation measured on high-volatility subsamples is biased upward for mechanical reasons. Correct for it before claiming a correlation regime.
- **Psaradakis, Z. & Spagnolo, N. (2003).** "On the Determination of the Number of Regimes in Markov-Switching Autoregressive Models." *Journal of Time Series Analysis* 24(2), 237–252. — Information criteria for K, and how badly they

behave.

- **Cakici, N. et al. (2025).** "Pockets of Predictability: A Replication." *Journal of Finance* 80(6), 3771–3790. — Disputes the economic significance of the Farmer et al. result. Cite the original and this together or neither.

4.5 Structural breaks and change-point detection

Different object, adjacent literature, frequently the better tool (§5.6).

- **Chow, G. C. (1960).** "Tests of Equality Between Sets of Coefficients in Two Linear Regressions." *Econometrica* 28(3), 591–605. — The break test everyone learns first; requires a known break date, which you never have.
- **Andrews, D. W. K. (1993).** "Tests for Parameter Instability and Structural Change with Unknown Change Point." *Econometrica* 61(4), 821–856. — The sup-Wald test that fixes that.
- **Bai, J. & Perron, P. (1998).** "Estimating and Testing Linear Models with Multiple Structural Changes." *Econometrica* 66(1), 47–78. — Multiple unknown breaks, with a dynamic programme to locate them.
- **Adams, R. P. & MacKay, D. J. C. (2007).** "Bayesian Online Changepoint Detection." arXiv:0710.3742. — Online, exact, and the run-length posterior it produces is a genuinely useful real-time object. The recommended default when the question is "did something just change" rather than "which of K states are we in".
- **Killick, R., Fearnhead, P. & Eckley, I. A. (2012).** "Optimal Detection of Changepoints with a Linear Computational Cost." *Journal of the American Statistical Association* 107(500), 1590–1598. — PELT. Exact segmentation in linear time; the workhorse for offline analysis.
- **Truong, C., Oudre, L. & Vayatis, N. (2020).** "Selective Review of Offline Change Point Detection Methods." *Signal Processing* 167, 107299. — The survey, and the paper behind the `ruptures` package.

4.6 Practitioner and allocation work

Often the most directly usable material, and produced by people who sell the thing. Interests noted.

- **Chow, G., Jacquier, E., Kritzman, M. & Lowry, K. (1999).** "Optimal Portfolios in Good Times and Bad." *Financial Analysts Journal* 55(3), 65–73. — The original "split the sample by turbulence and optimise separately" idea.
- **Kritzman, M. & Li, Y. (2010).** "Skulls, Financial Turbulence, and Risk Management." *Financial Analysts Journal* 66(5), 30–41. — The Mahalanobis turbulence index: a single scalar regime indicator with no latent state and no estimation problem. Underrated because it is simple.
- **Kritzman, M., Page, S. & Turkington, D. (2012).** "Regime Shifts: Implications for Dynamic Strategies." *Financial Analysts Journal* 68(3), 22–39. — The practitioner framework. **[Contested]** — the authors' firm sells regime-based analytics, and the backtests are not adjusted for specification search.
- **Kritzman, M., Li, Y., Page, S. & Rigobon, R. (2011).** "Principal Components as a Measure of Systemic Risk." *Journal of Portfolio Management* 37(4), 112–126. — The absorption ratio. A regime indicator built from the eigenvalue concentration of the correlation matrix rather than from returns.
- **Bulla, J., Mergner, S., Bulla, I., Sesboüé, A. & Chesneau, C. (2011).** "Markov-Switching Asset Allocation: Do Profitable Strategies Exist?" *Journal of Asset Management* 12(5), 310–321. — An honest out-of-sample test with transaction costs across three markets and forty years. The answer is a qualified yes, and the mechanism is volatility avoidance, not return prediction.
- **Ang, A. & Bekaert, G. (2004).** "How Regimes Affect Asset Allocation." *Financial Analysts Journal* 60(2), 86–99. — The practitioner version of their 2002 paper: regime-switching allocation with a risk-free asset available.
- **Nystrup, P., Hansen, B. W., Madsen, H. & Lindström, E. (2015).** "Regime-Based Versus Static Asset Allocation: Letting the Data Speak." *Journal of Portfolio Management* 42(1), 103–109. [publisher — paywalled] — Two-state regime inference driving allocation, against a static benchmark.
- **Nystrup, P., Madsen, H. & Lindström, E. (2018).** "Dynamic Portfolio Optimization Across Hidden Market Regimes." *Quantitative Finance* 18(1), 83–95. — Careful work on the allocation problem with adaptive estimation. [publisher — paywalled]
- **Nystrup, P., Kolm, P. N. & Lindström, E. (2020).** "Greedy Online Classification of Persistent Market States Using Realized Intraday Volatility Features." *Journal of Financial Data Science* 2(3), 25–39. — The statistical jump model applied online, with the striking claim that it beats a correctly specified maximum-likelihood HMM on classification accuracy. **[Contested]**, and worth replicating yourself. The likely source of the advantage is that a jump penalty is a better-conditioned way to impose persistence than a transition matrix estimated from few transitions (§7.2).
- **Bemporad, A., Breschi, V., Piga, D. & Boyd, S. (2018).** "Fitting Jump Models." *Automatica* 96, 11–21. — The method itself, from the control literature. Read this before the finance applications.
- **Asness, C., Ilmanen, A. & Maloney, T. (2017).** "Market Timing: Sin a Little." *Journal of Investment Management* 15(3), 23–40. — The best statement of the sceptical practitioner position on conditioning: timing works a bit, and much less than it appears to. **[Contested]** — AQR sells the strategies that do not require timing.

4.7 Dataset shift, conditioning, and learned states

- **Quiñonero-Candela, J., Sugiyama, M., Schwaighofer, A. & Lawrence, N. D., eds. (2009).** *Dataset Shift in Machine Learning.* MIT Press. — The source of the covariate-shift / concept-drift vocabulary that §9 is built on.
- **Shimodaira, H. (2000).** "Improving Predictive Inference under Covariate Shift by Weighting the Log-Likelihood Function." *Journal of Statistical Planning and Inference* 90(2), 227–244. — Importance weighting, and the precise conditions under which it helps. It helps less often than people assume.
- **Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M. & Bouchachia, A. (2014).** "A Survey on Concept Drift Adaptation." *ACM Computing Surveys* 46(4), article 44. — The taxonomy of drift types and adaptation strategies. Almost everything finance calls "regime handling" is in here under another name, usually with more careful evaluation.
- **Jacobs, R. A., Jordan, M. I., Nowlan, S. J. & Hinton, G. E. (1991).** "Adaptive Mixtures of Local Experts." *Neural Computation* 3(1), 79–87. — The original mixture of experts. A gated regime model in everything but name, and thirty years older than the finance papers that reinvent it.
- **Peters, J., Bühlmann, P. & Meinshausen, N. (2016).** "Causal Inference by Using Invariant Prediction: Identification and Confidence Intervals." *Journal of the Royal Statistical Society: Series B* 78(5), 947–1012. — The inverse of regime modelling: instead of adapting to each regime, find the relationship that is *invariant* across them. §9.6 argues this is underused, and §10.6 rates it the best-value scheme for a model meant to run for years.
- **Arjovsky, M., Bottou, L., Gulrajani, I. & Lopez-Paz, D. (2019).** "Invariant Risk Minimization." arXiv:1907.02893. — The deep-learning version. Read with Rosenfeld, Ravikumar and Risteski, "The Risks of Invariant Risk Minimization" (ICLR 2021), which shows it can fail badly outside its assumptions.
- **Grinsztajn, L., Oyallon, E. & Varoquaux, G. (2022).** "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" NeurIPS Datasets and Benchmarks. — Why tree ensembles beat networks on tabular data, mechanistically: trees handle irregular target functions and tolerate uninformative features. Both properties matter for regime features (§10.8).
- **Gu, S., Kelly, B. & Xiu, D. (2020).** "Empirical Asset Pricing via Machine Learning." *Review of Financial Studies* 33(5), 2223–2273. — The benchmark ML asset-pricing study. Note what it does about regimes: interacts characteristics with eight observable macro predictors and lets trees and networks find the structure. No latent state model anywhere.
- **Chen, L., Pelger, M. & Zhu, J. (2024).** "Deep Learning in Asset Pricing." *Management Science* 70(2), 714–750. — Learns a latent macroeconomic state with an LSTM as part of an end-to-end no-arbitrage objective. The most sophisticated published example of the "let the network find the regime" approach.
- **Kelly, B., Malamud, S. & Zhou, K. (2024).** "The Virtue of Complexity in Return Prediction." *Journal of Finance* 79(1), 459–503. — Argues against the instinct that instability demands simple models: heavily overparameterised models with strong shrinkage do better than parsimonious ones, even in short samples. **[Contested]** and directly relevant to whether you should be subsetting your data by regime at all (§10.4).
- **Linderman, S. et al. (2017).** "Bayesian Learning and Inference in Recurrent Switching Linear Dynamical Systems." AISTATS. — Latent discrete states whose transitions depend on the continuous state. The most natural modern generalisation of Hamilton's model, from neuroscience.
- **Perez, E., Strub, F., de Vries, H., Dumoulin, V. & Courville, A. (2018).** "FiLM: Visual Reasoning with a General Conditioning Layer." *AAAI*. — Feature-wise linear modulation: conditioning information produces a per-channel scale and shift on hidden activations. The correct way to inject a regime into a network (§10.8), from a completely unrelated field.
- **Shazeer, N. et al. (2017).** "Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer." ICLR. — Mixture of experts at scale, with the load-balancing and expert-collapse problems that §10.5 warns about, solved for a much larger regime.
- **Gibbs, I. & Candès, E. (2021).** "Adaptive Conformal Inference Under Distribution Shift." NeurIPS. — Prediction intervals with coverage guarantees that survive distribution shift. A stronger guarantee than anything else in this document, and directly usable for sizing.

4.8 Evaluation and backtesting

- **Diebold, F. X. & Mariano, R. S. (1995).** "Comparing Predictive Accuracy." *Journal of Business & Economic Statistics* 13(3), 253–263. — The base test.
- **Diebold, F. X., Gunther, T. A. & Tay, A. S. (1998).** "Evaluating Density Forecasts with Applications to Financial Risk Management." *International Economic Review* 39(4), 863–883. — The probability integral transform as a check of a whole predictive density; the basis of HMM pseudo-residuals (§6.9).
- **Giacomini, R. & White, H. (2006).** "Tests of Conditional Predictive Ability." *Econometrica* 74(6), 1545–1578. — Tests whether model A beats model B *given the current state*, which is exactly the regime question. Should be standard in this literature and is not.
- **Giacomini, R. & Rossi, B. (2010).** "Forecast Comparisons in Unstable Environments." *Journal of Applied Econometrics* 25(4), 595–620. — The fluctuation test: does the relative performance of two models change over time? The single most useful diagnostic in this whole document (§11.6).

- **White, H. (2000).** "A Reality Check for Data Snooping." *Econometrica* 68(5), 1097–1126. — And **Hansen, P. R. (2005)**, "A Test for Superior Predictive Ability," *JBES* 23(4), 365–380, which fixes its power problem.
- **Bailey, D. H. & López de Prado, M. (2014).** "The Deflated Sharpe Ratio." *Journal of Portfolio Management* 40(5), 94–107. — Adjusts a Sharpe ratio for the number of trials and for non-normality. Essential once you start slicing by regime, which multiplies trials fast (§11.6).
- **Harvey, C. R., Liu, Y. & Zhu, H. (2016).** "…and the Cross-Section of Expected Returns." *Review of Financial Studies* 29(1), 5–68. — The multiple- testing problem in finance, with the recommended $t > 3$ threshold.
- **Pesaran, M. H. & Timmermann, A. (2007).** "Selection of Estimation Window in the Presence of Breaks." *Journal of Econometrics* 137(1), 134–161. — How much history to train on when the data-generating process changes. The formal version of the question every practitioner answers by guessing.
- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Purged and embargoed cross-validation, combinatorial purged CV, triple-barrier labelling. **[Contested]** in places — several claims are asserted rather than demonstrated — but the CV machinery in chapters 7 and 12 is correct, necessary, and not available elsewhere in one place.
- **Arnott, R., Harvey, C. R. & Markowitz, H. (2019).** "A Backtesting Protocol in the Era of Machine Learning." *Journal of Financial Data Science* 1(1), 64–74. — A short checklist worth rereading annually.

4.9 If you only read eight things

In this order:

1. **Ang & Timmermann (2012)** — the survey. Two hours, and you will know the shape of the field.
2. **Hamilton (1989), §2–3** — the filter, from the source.
3. **Zucchini, MacDonald & Langrock (2016)** — the practical HMM book: estimation, model checking, and the extensions of §6, worked in code.
4. **Hansen (1992)** — why you cannot test what you want to test.
5. **Dacco & Satchell (1999)** — why in-sample fit does not transfer.
6. **Diebold & Inoue (2001)** — why you might not have regimes at all.
7. **Gama et al. (2014)** — the machine-learning vocabulary for the same problem, with better evaluation discipline.
8. **Bulla et al. (2011)** — what an honest out-of-sample regime backtest looks like, and how modest the answer is.

Items 4, 5, and 6 are the ones that will change what you build. Read them before you write code, not after your backtest looks good.

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5. The formal models

This is the model zoo. §7 collapses it into a four-slot design space, so read this for the mechanics and §7 for the map. The hidden Markov model of §5.2 and §5.3 is the zoo's central member, and §6 treats it in depth: how to estimate it, which refinements pay, and how to use its output.

Before any of the models, one piece of discipline that governs all of them.

5.1 The filtration ladder

Every regime model produces a belief about the state. **Which information set that belief conditions on is the single largest determinant of measured performance, and it is not a modelling choice — it is a research-protocol choice.** More regime research is invalidated by getting this wrong than by every modelling error combined.

There are five rungs, in decreasing order of how good they look and increasing order of how honest they are:

```
Rung                                        Uses                          Tradable?
──────────────────────────────────────────────────────────────────────────────────────
1  Oracle     s_t                           the truth                     never
   └─ upper bound only; the number you can never reach

2  Smoothed   xi(t|T)   with theta-hat(all)   all data, both directions    never
   └─ what every regime plot in every paper shows

3  Filtered   xi(t|t)   with theta-hat(all)   all data, one direction      never
   └─ the leak that survives careful-looking pipelines
```

```
4  Filtered   xi(t|t)   with theta-hat(1..t)  data through t             partly
   └─ honest, but r_t is in the conditioning set

5  Predicted  xi(t|t-1) with theta-hat(..t-1) data through t-1           yes
   └─ the only object you can actually trade at the open of bar t
```

The four distinctions worth stating precisely:

Smoothed versus filtered. $\xi_{t|T} = \Pr(s_t \mid \mathcal{F}_T)$ uses the *entire sample*, including everything that happened after t. It is the right object for describing history and the wrong object for every other purpose. Because the smoother is what produces the clean, decisive-looking regime plots that appear in papers and pitch decks, it is easy to absorb an intuition about how sharply regimes can be identified that is simply false in real time.

Filtered versus predicted. $\xi_{t|t}$ conditions on r_t. If your position for bar t must be set before r_t is observed — which it must — then $\xi_{t|t}$ is unavailable and you need $\xi_{t|t-1} = \mathbf{P}^\top \xi_{t-1|t-1}$. The gap is one application of the transition matrix, which sounds trivial and is not: §8.3 measures it at roughly a quarter of the Sharpe ratio in a well-specified simulation where the states are easy to tell apart.

Parameter leakage. This is the one that survives. Suppose you carefully use only $\xi_{t|t-1}$, but you estimated θ and $\mathbf{P}$ by maximum likelihood on the whole sample. Then $\xi_{t|t-1}$ depends on the future through $\widehat{\theta}$, and the dependence is not small: the fitted state means and variances *define* what "the turbulent state" means, and they were chosen with knowledge of every turbulent episode in the sample. A model that "discovers" a crash state has been told where the crashes are.

Label leakage. Related and worse. State k has no intrinsic identity — the likelihood is invariant to permuting the labels. Researchers routinely resolve this by sorting states by their fitted mean and calling the low-mean one "bear". Doing that on the full sample and then using the labels in a backtest is a look-ahead of exactly the kind that produces beautiful equity curves. Sort by a quantity that is monotone and stable — realised volatility is the usual choice — and do the sorting inside the expanding window.

> **The protocol.** Every number you report should be labelled with its rung. Report rung 5 as the headline. Report rung 2 alongside it, deliberately, as a *diagnostic*: the ratio between them is a direct measure of how much of your result is look-ahead. If rung 2 looks wonderful and rung 5 looks like noise, you have learned something real — that the regime is only identifiable in hindsight — and that is a finding, not a failure. **[Practice]**

5.2 The Hamilton filter

The recursion is two lines, and writing them out demystifies the whole apparatus.

Let $\xi_{t|t}$ be the K-vector of filtered state probabilities and let η_t be the K-vector of conditional observation densities, $\eta_t^{(k)} = f(y_t \mid s_t = k, \mathcal{F}_{t-1}; \theta_k)$. Then:

$$\text{Predict:} \quad \xi_{t|t-1} = \mathbf{P}^\top \xi_{t-1|t-1}$$

$$\text{Update:} \quad \xi_{t|t} = \frac{\xi_{t|t-1} \odot \eta_t}{\mathbf{1}^\top (\xi_{t|t-1} \odot \eta_t)}$$

where $\odot$ is elementwise multiplication, and the recursion is started at $\xi_{0|0} = \pi$ (the stationary distribution) or at a diffuse $1/K$ each. That is the entire filter: propagate the belief forward through the chain, then reweight it by how well each state explains the new observation. It is the Kalman filter with a discrete state space, and it is the forward pass of the HMM forward-backward algorithm.

The denominator is not a nuisance. Expand it:

$$\mathbf{1}^\top (\xi_{t|t-1} \odot \eta_t) = \sum_{k=1}^{K} \Pr(s_t = k \mid \mathcal{F}_{t-1})\, f(y_t \mid s_t = k) = p(y_t \mid \mathcal{F}_{t-1}),$$

which is exactly the predictive density of §1.2. So the log-likelihood is

$$\ell(\theta, \mathbf{P}) = \sum_{t=1}^{T} \ln p(y_t \mid \mathcal{F}_{t-1}),$$

the sum of one-step-ahead predictive log-densities. **Maximum likelihood for a regime model is therefore optimising one-step-ahead density forecasting, and nothing else.** If your decision problem is h-step-ahead, or asymmetric in the cost of state errors, or cares about a functional of the distribution rather than the whole density, maximum likelihood is optimising the wrong objective — which is precisely the mechanism Dacco and Satchell (1999) identified for why these models forecast badly. §8.5 returns to this.

Smoothing. The Kim (1994) backward recursion starts from $\xi_{T|T}$ (at the last date, filtered and smoothed coincide) and runs from $t = T - 1$ down:

$$\xi_{t|T}^{(j)} = \xi_{t|t}^{(j)} \sum_{k=1}^{K} \frac{p_{jk}\, \xi_{t+1|T}^{(k)}}{\xi_{t+1|t}^{(k)}}.$$

Read it as: the filtered belief at t, reweighted by how much the future supported each of the states t could have led to.

Numerical practice. Three things bite. (i) The elementwise products underflow for long series — either rescale ξ to sum to one at every step (as written above, which is why it is written that way) or work in log space with `logsumexp`. Rescaling loses nothing, because the log of each denominator is the term $\ln p(y_t \mid \mathcal{F}_{t-1})$ of ℓ, so the log-likelihood is the running sum of those logs. (ii) The likelihood surface is multimodal and EM converges to a local optimum; restart from many initialisations and keep the best. (iii) Any state can collapse onto a single observation with $\sigma_k \to 0$ and likelihood $\to \infty$ — the classic degeneracy of Gaussian mixtures. Bound σ_k from below or put a prior on it. This is not a rare edge case; on daily financial returns with $K \geq 3$ it happens routinely.

5.3 Markov-switching regressions

Intuition. The relationship you are estimating is not one relationship but K of them, and which one is active follows a hidden Markov chain. This is the hidden Markov model with regressions as its emissions; §6 covers estimation and the practical refinements.

Definition. For a regression target y_t and predictors x_t,

$$y_t = x_t^{\top} \beta_{s_t} + \sigma_{s_t} \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, 1), \qquad \Pr(s_t = k \mid s_{t-1} = j) = p_{jk}.$$

Special cases: $x_t = 1$ gives switching means; $x_t = (1, y_{t-1}, \dots)$ gives the Markov-switching autoregression of Hamilton (1989); a vector y_t gives MS-VAR.

Assumptions. The chain is first-order and homogeneous (transition probabilities constant through time); the state is independent of the innovation given the past; K is known; the emission family is correctly specified. The first and third are the ones that fail. Transition probabilities that are themselves time-varying are the rule rather than the exception. §6.6 solves the problem within this framework, by letting the transition probabilities depend on lagged observables; §5.5 sidesteps it, by dropping the latent Markov chain altogether in favour of a state that is an explicit function of an observable z.

Strengths. Interpretable, likelihood-based, gives a full predictive density, and the filter is exact. Handles the "different relationship in different conditions" idea directly rather than through interactions.

Weaknesses. Parameter count grows as K times (the number of regression coefficients, plus one for σ_k) plus K^2 transition parameters — strictly $K(K-1)$ of those are free, since every row of $\mathbf{P}$ sums to one, and "K^2" is the shorthand used throughout. Identification is weak (§8.1); the number of states is untestable by standard means (§8.2).

Cost. $O(TK^2)$ per likelihood evaluation, $O(TK^2 I)$ for EM with I iterations (§6.1). Trivial for $K \leq 5$ and T in the tens of thousands.

Failure modes. (i) A state collapses onto outliers and becomes a "crash detector" that fires once and never generalises. (ii) With $K \geq 3$ and Gaussian emissions on financial data, the extra states almost always split the volatility dimension further rather than finding new mean behaviour — check this before interpreting them. (iii) Estimated transition probabilities near 1 produce near-absorbing states, which means the model has decided a break occurred rather than a recurrent regime, and your out-of-sample state belief will be stuck.

When preferred. When you have a real economic reason to expect a small number of discrete relationship regimes, when you want a density forecast rather than a point forecast, and when $K \leq 3$. For daily returns, the variance-only version with Student-t emissions (§6.3) is the specification to start from.

5.4 Switching volatility: SWARCH and MS-GARCH

Intuition. Volatility clusters *and* jumps between levels. GARCH captures the clustering; a regime captures the level shifts. Doing both is natural and technically awkward.

Definition. The naive combination is

$$\sigma_t^2 = \omega_{s_t} + \alpha_{s_t} a_{t-1}^2 + \beta_{s_t} \sigma_{t-1}^2,$$

where $a_t = \sigma_t \varepsilon_t$ is the unstandardised innovation and ω, α, β are GARCH coefficients (here β is not the regression coefficient of §5.3). This is **inestimable**: σ_{t-1}^2 depends on s_{t-1} and on σ_{t-2}^2, which depends on s_{t-2} and on σ_{t-3}^2, and so on back to the start of the sample. Evaluating the likelihood at time t therefore requires summing over all K^t state paths. This path-dependence problem is the reason MS-GARCH was hard for a decade. Gray (1996) solves it by collapsing the conditional variance across states at each step; Haas, Mittnik and Paolella (2004) solve it more cleanly by running K *parallel*, non-interacting GARCH recursions and letting the state select which one is observed.

Assumptions. As §5.3, plus a GARCH functional form within each state.

Strengths. Hamilton and Susmel's (1994) central finding is worth internalising: **a large part of what a single-regime GARCH reads as extreme volatility persistence is actually regime persistence.** Fit GARCH to a series with two volatility levels and you get $\alpha + \beta \approx 0.99$, an integrated process, and forecasts that revert far too slowly. Allowing the level to switch returns the within-regime persistence to plausible values.

Weaknesses. Many parameters (K GARCH triples plus K^2 transitions), badly conditioned. In head-to-head out-of-sample volatility forecasting, simple alternatives — HAR on realised volatility, or an EWMA — are hard to beat. **[Contested]**: the published comparisons are sensitive to the loss function and the sample.

Cost. $O(TK^2)$ for the Haas et al. formulation; the Gray formulation is similar but with a collapsing approximation.

Failure modes. (i) The high-volatility state absorbs a handful of crisis days and its GARCH parameters are then estimated from almost nothing. (ii) With $K = 3$ the middle state is frequently uninterpretable and unstable. (iii) The model's volatility forecast at a transition jumps discontinuously, which produces large discrete position changes in a vol-targeted strategy — a costly artefact of the discretisation, not a feature.

When preferred. When you specifically need a *density* forecast whose shape changes with the state, not just a variance forecast. If you need only the variance, use HAR or EWMA and spend the saved complexity elsewhere. **[Practice]**

5.5 Observable-threshold models: TAR, SETAR, STAR

Intuition. Stop pretending the state is hidden. Declare it a function of something you can see.

Definition. A threshold autoregression makes the regime a deterministic function of an observable $z_{t-\kappa}$:

$$y_t = \begin{cases} x_t^\top \beta_1 + \sigma_1 \varepsilon_t & \text{if } z_{t-\kappa} \le c \\ x_t^\top \beta_2 + \sigma_2 \varepsilon_t & \text{if } z_{t-\kappa} > c \end{cases}$$

with threshold c and delay $\kappa \ge 1$ (so that the regime is known before bar t opens), both estimated by grid search over the likelihood. When z is a lag of y itself this is a *self-exciting* TAR (SETAR; Tong, 1983). The *smooth transition* version (STAR; Teräsvirta, 1994) replaces the indicator with a logistic or exponential function of $(z_{t-\kappa} - c)$, giving a continuous blend between the two regimes:

$$y_t = \big(1 - G(z_{t-\kappa}; \gamma, c)\big)\, x_t^\top \beta_1 + G(z_{t-\kappa}; \gamma, c)\, x_t^\top \beta_2 + \sigma \varepsilon_t, \qquad G(z; \gamma, c) = \frac{1}{1 + e^{-\gamma(z-c)}}.$$

The weight G rises from 0 to 1 as z crosses c, and $\gamma > 0$ sets how fast. As $\gamma \to \infty$ it becomes the indicator $\mathbb{1}[z > c]$ and the model reduces to the threshold model above; as $\gamma \to 0$ it flattens to $\frac{1}{2}$ and the model to a single regression.

Assumptions. That the correct conditioning variable is observable and known. Everything hangs on that.

Strengths. No filtering, no latent state, no look-ahead in the state estimate — the regime at time t is a function of data you had before t. Estimation is a grid search over (c, κ) with OLS inside, which is fast and has no local-optimum problem worth worrying about. **This is a much larger advantage than the literature's relative neglect of these models suggests**, and it is the

reason practitioner "regime" definitions (VIX above 25, price below the 200-day average, yield curve inverted) are threshold models whether or not anyone says so.

Weaknesses. You must choose z. If the true state driver is not in your candidate set, the model cannot find it, whereas an HMM can in principle. The threshold c is estimated by profiling the likelihood, which is a multiple- testing exercise: searching 100 thresholds and reporting the best fit without adjustment is a specification search.

Cost. $O(T \cdot |C| \cdot |L|)$ for a grid C of thresholds and L of delays. Cheap.

Failure modes. (i) Threshold instability — refit on a new sample and c moves, so the historical labelling changes retroactively. Fix c at a round number chosen a priori and you lose fit but gain everything else. (ii) A threshold on a *non-stationary* z (a price level, an unnormalised spread) gives a regime definition that drifts out of range; always threshold a normalised quantity. (iii) The hard-indicator version produces one-day round trips when z sits at the threshold — see hysteresis in §12.2.

When preferred. This should be your default. If you can name the observable that defines your regime, a threshold or smooth-transition model gives you 80% of what an HMM gives with a tenth of the fragility, and the smooth version fixes the choppiness for free. Reach for a latent-state model only when you have tried this and the observable genuinely is not available. **[Practice]**

5.6 Structural breaks and change-point detection

Intuition. A different question: not "which of K recurring states are we in" but "did the process change, and when".

Definition. Offline, the multiple-break model of Bai and Perron (1998) partitions $1..T$ into $m+1$ segments with constant parameters within each, choosing break dates by dynamic programming to minimise the sum of segment residual sums of squares plus a penalty; PELT (Killick et al., 2012) does the same in $O(T)$ by pruning candidate break dates that can never be optimal, under a mild condition on the segment cost. Online, Bayesian online changepoint detection (Adams & MacKay, 2007) maintains a posterior over the *run length* ρ_t — how long since the last change — via a message-passing recursion, and gives $p(y_{t+1} \mid y_{1:t})$ as a run-length-weighted mixture.

Assumptions. Breaks are permanent, not recurrent. Segments are independent given the break structure.

Strengths. BOCPD in particular produces something a regime model does not: an explicit, calibrated answer to "how confident am I that the world just changed", available in real time and with no requirement to specify K. Its run- length posterior is a genuinely useful input to a sizing rule (§12.1).

Weaknesses. Non-recurrence is the fundamental limitation. If you break, you have no data on the new regime — that is the point of a break — so any model that needs θ_{new} is starting from scratch. Pesaran and Timmermann (2007) show the surprising and important consequence: **it can be optimal to include pre-break data in estimation**, because the bias from the old regime is sometimes smaller than the variance from a short post-break sample.

Cost. Bai-Perron $O(T^2)$, PELT $O(T)$ amortised, BOCPD $O(1)$ per step ($O(T)$ total) with run-length truncation, $O(t)$ per step ($O(T^2)$ total) without.

Failure modes. (i) Every change-point method fires on volatility changes and is nearly blind to mean changes — the same asymmetry as §8.1, for the same reason. (ii) The penalty parameter directly controls how many breaks you find and is nearly always tuned by eye. (iii) End-of-sample breaks are detected with a long delay by construction, which is exactly when you need them.

When preferred. When the question is genuinely "has something changed" — model monitoring, deciding when to retrain (§12.6), detecting the death of a strategy — rather than "which state are we in". For that job it beats an HMM comfortably.

5.7 Statistical jump models

Intuition. Keep the discrete state and the persistence, throw away the probability model. Instead of a transition matrix, penalise switching directly.

Definition. Given features $u_t \in \mathbb{R}^d$ (typically volatility, downside deviation, and trend measures computed over several windows), fit K centroids $\{c_k\}$ and a state path $\{s_t\}$ minimising

$$\min_{\{c_k\},\{s_t\}} \sum_{t=1}^{T} \|u_t - c_{s_t}\|^2 \;+\; \lambda \sum_{t=2}^{T} \mathbb{1}[s_t \neq s_{t-1}].$$

The first term is k-means; the second is a **jump penalty** with a single tunable λ controlling persistence. The optimisation alternates between assigning states by dynamic programming given centroids (exact, $O(TK^2)$, a Viterbi-style recursion) and updating centroids given states (a mean). The recursion carries $V_t(k)$, the lowest cost of any path that ends in state k at t: $V_t(k) = \|u_t - c_k\|^2 + \min\{V_{t-1}(k),\ \min_{j \neq k} V_{t-1}(j) + \lambda\}$. Staying is free and switching costs λ. Bemporad et al. (2018) develop the general form; Nystrup, Kolm and Lindström (2020) apply it online to market states.

Assumptions. That the regime is characterised by the *location* of a feature vector, and that persistence can be imposed by a single scalar rather than by a $K \times K$ matrix.

Strengths. Three that matter. (i) **One persistence parameter instead of K^2**, estimated from a validation objective rather than from the handful of observed transitions — this is the specific fix for the small-sample problem of §6.1. (ii) No explicit distributional assumption on returns (§7.2 shows the implicit one), so no Gaussian-mixture degeneracy and no sensitivity to fat tails. (iii) You choose the feature space, so you can put realised volatility, drawdown, and dispersion in directly rather than hoping a Gaussian mixture on returns recovers them. Nystrup et al. report that it beats a *correctly specified* maximum-likelihood HMM on classification accuracy in simulation. **[Contested]** — that is a strong claim from an interested source, but the mechanism is plausible and §7.2 shows why it should hold.

Weaknesses. No predictive density, no likelihood, no principled way to compare K, and no probabilistic state — you get a hard label unless you build a soft variant. Feature scaling matters a great deal and is a hidden researcher degree of freedom.

Cost. $O(TK(K + d))$ per sweep — $O(TKd)$ to score every observation against every centroid, $O(TK^2)$ for the dynamic programme — and a handful of sweeps. Fast enough to refit daily on decades of data.

Failure modes. (i) Features on different scales silently determine the clustering; standardise inside the training window. (ii) λ tuned on the same data used to evaluate is a leak, and it is a powerful one because λ controls turnover and therefore costs. (iii) The state path is the solution to a global optimisation, so the *offline* fitted path is a smoothed object — using it directly is rung 2 of §5.1. The online variant exists precisely to avoid this; use it.

When preferred. As a default latent-state method for trading applications that need a hard label. It gives up the density forecast in exchange for protection against the two things that break freely estimated HMMs in practice: transition estimation from few events, and emission misspecification. A regularised HMM — Student-t emissions and a sticky prior (§6.3, §6.7) — buys much of the same protection and keeps the probabilities. **[Practice]**

5.8 Clustering-based regimes

Intuition. Regimes are groups of similar days. Cluster the days.

Definition. Build a feature vector u_t per period, then apply an unsupervised method: Gaussian mixture (which is an HMM with no transition structure), k-means, agglomerative clustering, HDBSCAN, or — for correlation-structure regimes — clustering the sequence of estimated correlation matrices under a suitable metric.

Assumptions. That temporal order does not matter for defining the states (though it obviously matters for using them).

Strengths. Trivially simple, no likelihood, arbitrary features, and it sidesteps the emission-specification problem entirely.

Weaknesses. No persistence. A plain clustering of days produces a state sequence that flickers, because nothing in the objective rewards staying put. Practitioners then smooth the labels post hoc — which, done over a centred window, is a look-ahead, and done over a trailing window, is just a slow-moving feature you could have built directly. The statistical jump model is exactly this method *with* a persistence term, which is why it dominates.

Cost. Trivial.

Failure modes. (i) Post-hoc label smoothing that peeks forward. (ii) Clustering on full-sample-standardised features (a leak). (iii) Cluster identity not stable across refits, so "regime 2" means something different each month.

When preferred. As an exploratory tool to find out whether your feature space has any group structure at all, before committing to a model. Rarely as the production state estimator — add the persistence term and you have §5.7.

5.9 Scalar regime indices

Intuition. Skip the state entirely. Build one number that goes up in bad times, and condition on it.

Definition. Three that are worth knowing:

- **Financial turbulence** (Kritzman & Li, 2010): the squared Mahalanobis distance of today's cross-asset return vector $\mathbf{r}_t$ (a vector here, not the scalar r_t of the notation block) from its historical mean, $d_t = (\mathbf{r}_t - \mu)^\top \Sigma^{-1} (\mathbf{r}_t - \mu)$, where

μ and Σ are estimated on a trailing window. Large when returns are individually extreme *or* when the cross-sectional pattern is unusual, which is the informative part — a day where everything moves a normal amount but in the wrong relative directions scores high.

- **Absorption ratio** (Kritzman, Li, Page & Rigobon, 2011): the fraction of total cross-sectional variance explained by the first n principal components. High absorption means the market is being driven by few factors — a fragility indicator.
- **Implied volatility level and term structure**: VIX, and the sign of the VIX futures basis. Crude, observable, and hard to beat.

Assumptions. That a scalar suffices, and that its trailing normalisation is adequate.

Strengths. No latent state, no K, no filtering, no label switching, and every value is $\mathcal{F}_t$-measurable by construction if you compute the moments on a trailing window. As a *feature* for a forecasting model (Part II) this class is far more useful than a fitted state posterior, because it is continuous, monotone, and has no identification problem.

Weaknesses. No model, so no density forecast, no probability, and no way to say what the state implies for returns beyond what you fit downstream.

Failure modes. (i) Computing μ, Σ on the full sample — the standard mistake, and it matters most exactly at the crisis dates you care about. (ii) Σ^{-1} is unstable when the number of assets approaches the window length; shrink it. (iii) Turbulence and realised volatility are highly correlated, so the incremental information over "vol is high" is smaller than the framing suggests — test the increment explicitly.

When preferred. As the first thing you try, and often the last. If a scalar turbulence index gets you 90% of the benefit of a four-state MS-VAR, and it usually does, the MS-VAR is a research project rather than a trading edge. **[Practice]**

5.10 Deep latent-state models

Intuition. Keep the graphical structure of the HMM — a latent state and observations that depend on it — but let neural networks parameterise the pieces.

Definition. Four variants travel under this heading:

- **HMM with neural emissions.** $f(y_t \mid s_t = k)$ becomes a network; the filter is unchanged. Rarely worth it, because emission misspecification is usually fixed more cheaply by a Student-t (§6.3).
- **Recurrent switching linear dynamical systems** (Linderman et al., 2017). A discrete state indexes a linear dynamical system for a continuous latent state h_t (a hidden state, not the forecast horizon h), and the discrete transitions depend on it: $\Pr(s_t \mid s_{t-1}, h_{t-1})$. This generalises Hamilton's model in the direction of §6.6, with the covariate itself latent.
- **Deep state-space models** (deep Kalman filters, structured inference networks). A continuous latent state with amortised variational inference. Not a regime model in the discrete sense, but it occupies the same slot.
- **A learned state inside a task network** (Chen, Pelger & Zhu, 2024). A recurrent network compresses macro series into a low-dimensional state that feeds an asset-pricing objective, trained end to end. No separate regime model exists; the state is whatever serves the task (§10.7).

Strengths. Expressive, with state-dependent transitions, and in the end-to-end case the state is optimised for the decision rather than for one-step density forecasting (§5.2).

Weaknesses. All of the identification problems of §8, plus variational approximation error, plus many more hyperparameters, on samples that support none of it. The financial applications rarely report the stability diagnostics of §2.7.

Failure modes. (i) Posterior collapse: the model ignores the latent state and becomes a plain autoregression, with no symptom in the loss. (ii) The learned state is a noisy proxy for realised volatility; regress it on volatility before claiming anything else. (iii) Large, unadjusted specification search.

When preferred. In the end-to-end form, with a large panel and a clear downstream objective. As a standalone state estimator for a single series, essentially never: the data do not support it.

5.11 Supervised regime labels

Intuition. Define the regime by what happened next, then train a classifier to predict it.

Definition. Construct a label from forward information — "the next 20 days had realised volatility above the 80th percentile", or a trend-scanning label that fits a line over a forward window and takes its t-statistic's sign, or a triple-barrier label (López de Prado, 2018) recording which of a profit-taking, stop-loss, or time barrier was hit first. Then train a supervised model to predict that label from $\mathcal{F}_t$ features.

Strengths. It makes the target explicit and decision-relevant, which is a real advantage over unsupervised states whose meaning is whatever the likelihood found. It also converts the problem into standard supervised learning, with all the evaluation machinery that brings.

Weaknesses and the central trap. The label spans $(t, t+h]$ while the features stop at t, so **adjacent training examples share outcome data and are massively dependent**. Standard k-fold cross-validation on such a dataset is not slightly optimistic — it is meaningless, because a fold boundary at t puts overlapping labels on both sides. This is the problem purging and embargo exist to solve (§11.3), and it is *the* reason financial ML papers with 0.9 AUC disappear on live data.

Failure modes. (i) Unpurged CV, as above. (ii) Labels defined with full-sample percentile thresholds ("the top quintile of forward volatility") — the threshold itself is a look-ahead; use an expanding-window quantile. (iii) Class balance shifting across time so the classifier learns the base rate of the training era rather than the signal.

When preferred. When you want a regime for a specific decision and can state that decision as a forward-looking event. Which, notably, is most of the time in trading. **Recommendation:** when the regime feeds one decision that can be stated as a forward event, prefer a supervised, purged formulation to an unsupervised state posterior. **[Practice]**

5.12 Comparison

Ratings are this chapter's assessment, not measured quantities. "Real-time" means the state estimate at t uses only $\mathcal{F}_t$ without additional machinery. The three HMM refinements of §6 are listed with the zoo because they change the attributes that discriminate.

Method	State type	Persistence	Real-time	Params	Density forecast	Main failure
MS regression (§5.3)	latent, soft	Markov	filter needed	$K(d+1)+K^2$	yes	weak identification
HMM, Student-t, sticky prior (§6.3, §6.7)	latent, soft	Markov, regularised	filter needed	$3K+K^2$	yes	constant hazard
Hidden semi-Markov (§6.5)	latent, soft	explicit durations	filter needed	$4K+K^2$	yes	few spells to fit durations
HMM with TVTP (§6.6)	latent, soft	Markov, driven by z	filter needed	$2K+K^2\dim(z)$	yes	overfits transitions
MS-GARCH (§5.4)	latent, soft	Markov	filter needed	$3K+K^2$	yes	over-parameterised
Threshold / STAR (§5.5)	observed	via z	**yes**	$2d+4$	yes	must choose z
Change-point (§5.6)	run length	by construction	yes (BOCPD)	few	yes (BOCPD)	non-recurrent
Jump model (§5.7)	latent, hard	jump penalty	yes (online)	$Kd+1$	no	no likelihood
Clustering (§5.8)	latent, hard	**none**	yes	Kd	no	flickers
Scalar index (§5.9)	observed	via window	**yes**	0–2	no	not a model
Deep latent (§5.10)	latent, soft	learned	filter needed	many	yes	data-hungry
Supervised label (§5.11)	defined	by label span	yes	model-dep.	no	CV leakage

Parameter counts assume Gaussian emissions unless the row says Student-t, and count K^2 for the transition matrix, the shorthand of §5.3.

§5 Key takeaways

1. The filtration ladder — oracle, smoothed, filtered-with-leaked-parameters, filtered, predicted — determines your measured performance more than your model choice does. Label every reported number with its rung.
2. The Hamilton filter is two lines: propagate the belief through the chain, then reweight by the observation likelihood. Its normalising constant is the predictive density, so maximum likelihood optimises one-step density forecasting and nothing else.
3. The hidden Markov model, under its econometric name of Markov-switching regression, is the zoo's central member; §6 covers how to fit and use it.
4. Transition probabilities are estimated from the number of *transitions*, not the number of observations. Twenty years of daily data may contain forty regime changes, which is why every useful refinement buys persistence cheaply.
5. If you can name the observable that defines your regime, use a threshold or smooth-transition model. It gives most of the benefit with none of the latent-state fragility, and it is what practitioner regime definitions already are.
6. For a latent state, two defaults compete. A statistical jump model gives a hard label from one persistence parameter, with no distributional assumption. A regularised HMM — Student-t emissions, a sticky prior — gives calibrated probabilities and a predictive density. Both beat a freely estimated Gaussian HMM.

7. Change-point detection answers a different and often better question — "did something just change" — and BOCPD's run-length posterior is directly usable as a sizing input.
8. Supervised regime labels are the natural formulation when the regime serves one forward-looking decision, and they carry one severe trap: overlapping labels make standard cross-validation meaningless.

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6. Hidden Markov models in practice

§5.2 and §5.3 defined the hidden Markov model and its filter. This section covers what it takes to fit one and use it: estimation, the choice among the four state beliefs, the emission model, the number of states, how long states last, transitions that respond to observables, Bayesian and adaptive estimation, model checking, direct use for forecasting and allocation, a worked example on a century of US equity returns, and software.

Two terms need fixing first. A **hidden Markov model** (HMM) is any model with a latent Markov chain s_t and an observation density that depends on the current state. A **Markov-switching model** is the econometric name for the same object, usually with regressions as the state-conditional densities (§5.3). §7.2 shows the identity is exact. This section says HMM throughout.

Three ideas organise the section, and every refinement below serves one of them:

1. **Transitions are scarce.** A twenty-year daily sample holds a few dozen visits to the rarer regime, and so a few dozen exits from each regime. Every parameter that governs the state process is estimated from those few dozen events, so the refinements that pay are the ones that buy persistence and stability cheaply: heavy-tailed emissions, priors, and restrictions.
2. **State means are not identified; state variances are.** §8.1 makes this precise. Restrict what you cannot estimate, and let the model spend its parameters on volatility.
3. **Only the predicted belief, computed with parameters estimated on past data, is tradable** (§5.1). Several standard software functions return a different object by default.

6.1 Estimation by EM, and what goes wrong

The algorithm. Maximum likelihood for an HMM is almost always computed by the expectation–maximisation algorithm in its HMM form, **Baum–Welch** (Baum et al., 1970; Hamilton, 1990, for the econometric version). Each iteration has two steps.

The **E-step** runs the forward filter of §5.2 and the backward smoother to obtain two sets of posteriors under the current parameters: the smoothed state probabilities $\xi^{(k)}_{t|T} = \Pr(s_t = k \mid \mathcal{F}_T)$, and the pairwise probabilities of each transition,

$$\xi^{(jk)}_{t-1,t|T} = \Pr(s_{t-1} = j,\ s_t = k \mid \mathcal{F}_T) = \frac{\xi^{(j)}_{t-1|t-1}\, p_{jk}\, f(y_t \mid \theta_k)\, b^{(k)}_t}{\sum_{j',k'} \xi^{(j')}_{t-1|t-1}\, p_{j'k'}\, f(y_t \mid \theta_{k'})\, b^{(k')}_t},$$

where $b^{(k)}_t$ is the backward variable, proportional to $p(y_{t+1:T} \mid s_t = k)$. In words: the probability that the chain moved from j to k at t is the belief in j just before, times the chance of the move, times how well state k explains y_t, times how well it explains everything after. The four factors are the pieces of the joint density of $s_{t-1} = j$, $s_t = k$ and the data from t on, given the past; the denominator rescales them to sum to one, which is why b_t only needs to be proportional. Summing the pairwise probabilities over k returns the smoothed probability, $\sum_k \xi^{(jk)}_{t-1,t|T} = \xi^{(j)}_{t-1|T}$.

The **M-step** treats those posteriors as if they were observed fractional counts. For Gaussian emissions the updates are weighted moments:

$$\hat{p}_{jk} = \frac{\sum_{t=2}^{T} \xi^{(jk)}_{t-1,t|T}}{\sum_{t=2}^{T} \xi^{(j)}_{t-1|T}}, \qquad \hat{\mu}_k = \frac{\sum_t \xi^{(k)}_{t|T}\, y_t}{\sum_t \xi^{(k)}_{t|T}}, \qquad \hat{\sigma}^2_k = \frac{\sum_t \xi^{(k)}_{t|T}\, (y_t - \hat{\mu}_k)^2}{\sum_t \xi^{(k)}_{t|T}}.$$

The transition estimate is the expected number of $j \to k$ moves divided by the expected number of days spent in j that have a following day (days 1 to $T-1$). That ratio is the clearest statement of the small-sample problem below. No iteration lowers the likelihood, and the sequence converges, but only to a local optimum.

The four things that go wrong.

1. **Local optima.** The likelihood is multimodal in $(\theta, \mathbf{P})$. Fifty random restarts is not paranoid; it is the minimum.
2. **Label switching.** The likelihood is invariant under permutation of state labels, so the mapping from "state 1" to "the calm state" is arbitrary and can flip between refits. Impose an identifying restriction — sorting by σ_k is the standard and the most stable choice — and apply it *inside* every window (§5.1).
3. **Degeneracy.** The likelihood is unbounded as $\sigma_k \to 0$: one state can collapse onto a single observation. Constrain σ_k from below or put a prior on it (§6.7).
4. **Small effective sample.** This is the one that matters. The information about θ_k comes only from observations assigned to state k, and the information about p_{jk} comes only from observed *transitions*. A 20-year daily sample with a 25-day turbulent state entered twice a year contains about 40 turbulent spells, so 40 entries and 40 exits. **You are estimating a transition probability from forty events, not from five thousand observations**, and the standard error scales accordingly.

The fourth item deserves a number. A turbulent state occupied for 1,000 days (the 40 spells of 25 days) with $p_{22} = 0.96$ gives 1,000 stay-or-leave trials, so $\hat{p}_{22}$ has a standard error of roughly $\sqrt{0.96 \times 0.04/1{,}000} = 0.0062$. The expected spell length $1/(1-p_{22}) = 25$ days is a smooth function of p_{22}, and the standard error of a smooth function is its derivative times the standard error of its argument (the delta method). The derivative is $1/(1-p_{22})^2 = 625$, so the standard error is about $625 \times 0.0062 \approx 3.9$ days, or 16% of its value. The spells give the same answer directly: a geometric spell has a standard deviation close to its mean (24.5 against 25 days), so the average of 40 spells has a standard error of about $24.5/\sqrt{40} \approx 3.9$ days. That is the *best* case, with the state path known. With the path inferred, the error is larger, and the half-life of §1.3 inherits all of it.

Initialisation. [Practice] Start the volatilities at quantiles of $|y_t|$ (spread across the range), the means near the sample mean, and the transition matrix sticky (p_{kk} between 0.95 and 0.98). Run many random perturbations of that start, and add one start from a cheap clustering of a trailing-volatility feature. Keep the best likelihood, then look at the spread of the others: several distinct optima within a few log-likelihood points of each other mean the data do not pin the model down, and no single fit should be trusted.

Direct maximisation as the alternative. The log-likelihood is the sum of log predictive densities that the filter already computes (§5.2), so it can be handed straight to a quasi-Newton optimiser after reparameterising: logits for each transition row, logs for the scales. EM is robust far from the optimum and slow near it; quasi-Newton is the reverse. The usual combination is EM to get close, then a few Newton steps to finish (Zucchini, MacDonald & Langrock, 2016). Direct maximisation is also the easy route for the extensions of §6.5 and §6.6, whose M-steps have no closed form. **[Practice]**

Standard errors. Two routes. The inverse of the negative of the numerically differentiated Hessian of the log-likelihood at the optimum (the inverse of the observed information) is cheap. It is unreliable when $\hat{p}_{kk}$ sits near 1, because the parameter is near the boundary of its space; report standard errors for the logit of p_{kk}, not for p_{kk} itself, or use the parametric bootstrap (simulate from the fitted model, refit, read off the spread). The bootstrap is the safer default, and it is the same machinery §8.2 recommends for testing K.

Cost. $O(TK^2)$ per E-step, so $O(TK^2IR)$ for I iterations and R restarts. For $K \leq 3$ and decades of daily data this is seconds in compiled code. A walk-forward study refits the model once per period, which multiplies the cost by the number of refits; warm-starting each refit from the previous parameters cuts the iterations to a handful.

6.2 Which state belief for which job

An HMM can report four different things about the state, and they answer different questions. The table puts them side by side. The first three are the ξ objects of the notation block.

Object	Definition	Conditions on	Use	Tradable at the open of bar t?
Predicted	$\xi_{t\|t-1} = \mathbf{P}^\top \xi_{t-1\|t-1}$	$\mathcal{F}_{t-1}$	positions, forecasts	yes
Filtered	$\xi_{t\|t}$ (§5.2)	$\mathcal{F}_t$	end-of-bar reporting	no; it uses y_t
Smoothed	$\xi_{t\|T}$ (§5.2)	$\mathcal{F}_T$	describing history, EM	never
Viterbi path	$\arg\max_{s_{1:T}} \Pr(s_{1:T} \mid \mathcal{F}_T)$	$\mathcal{F}_T$	labelling episodes	never

The **Viterbi path** is the single most probable *sequence* of states. It is computed by the forward recursion with max in place of $\sum$:

$$\delta_t(k) = f(y_t \mid \theta_k) \max_j \delta_{t-1}(j)\, p_{jk},$$

where $\delta_t(k)$ is the probability of the best path that ends in state k at t, together with the data through t, started at $\delta_1(k) = \pi_k f(y_1 \mid \theta_k)$. The path is then read off by tracing back the maximising j from the best final state. It is not the same as taking the most probable state at each date from $\xi_{t|T}$, and the two can differ for long stretches: the path must be coherent as a whole. It can omit, for example, a one-day visit to a state that the marginal probabilities favour for that day alone, because the visit costs two transitions and no single path containing it need be the most probable one. Like the smoother, the Viterbi path is revised as data arrive. Its value at t computed with data through $T > t$ is a look-ahead. Its *terminal* value, recomputed each day with data through that day, is $\mathcal{F}_t$-measurable, but it is a hard label with the defects §1.2 describes.

The software trap. [Fact] Common libraries return the smoothed or Viterbi object by default, because those are what speech recognition and bioinformatics need. In `hmmlearn`, `predict` returns the Viterbi path and `predict_proba` returns smoothed posteriors over whatever array you pass. Passing the full history and reading off the value at t is rung 2 of the filtration ladder. In `statsmodels`, a fitted Markov-switching model exposes `predicted_marginal_probabilities`, `filtered_marginal_probabilities` and `smoothed_marginal_probabilities` as separate attributes; only the first is tradable, and only with parameters estimated on data before t. Writing the forward filter yourself is twenty lines (§5.2) and removes the ambiguity. §6.12 lists the packages.

6.3 Choosing the emission model

The emission density $f(y_t \mid \theta_k)$ decides what a state *is*. Three choices matter: the distribution family, which observables enter y_t, and which parameters are allowed to switch.

Gaussian emissions. The default, and the source of two practical problems. First, a Gaussian calm state assigns a tiny density to a single large return, so one outlier can force the filter into the turbulent state for a day and out again. The model then reports a switch where nothing persistent happened. Second, with $K \geq 3$ the likelihood will often devote a state to a handful of outliers (the degeneracy of §6.1 in a milder form).

Student-t emissions. The fix for both. Within state k,

$$f(y_t \mid \theta_k) = \frac{\Gamma\left(\frac{\nu_k+1}{2}\right)}{\Gamma\left(\frac{\nu_k}{2}\right)\sqrt{\nu_k \pi}\,\sigma_k}\left(1 + \frac{(y_t-\mu_k)^2}{\nu_k \sigma_k^2}\right)^{-\frac{\nu_k+1}{2}},$$

with location μ_k, scale σ_k and degrees of freedom ν_k. Note that σ_k is now a *scale*, not a standard deviation: the state's volatility is $\sigma_k\sqrt{\nu_k/(\nu_k-2)}$ for $\nu_k > 2$. The density has polynomial rather than exponential tails, so a single large return is plausible in either state and no longer forces a switch.

Estimation stays within EM because the Student-t is a scale mixture of normals (the Student-t distribution): $y_t = \mu_k + \sigma_k \varepsilon_t / \sqrt{\omega_t}$, with $\varepsilon_t \sim \mathcal{N}(0,1)$ and a latent precision ω_t drawn from a gamma distribution with shape and rate both $\nu_k/2$ (mean 1). Given ω_t the observation is Gaussian, so EM treats ω_t as missing data. The E-step adds one quantity, the expected precision of each observation in each state,

$$\omega_{tk} = \frac{\nu_k + 1}{\nu_k + (y_t - \mu_k)^2/\sigma_k^2},$$

which is below 1 for any return more than one scale σ_k from μ_k and falls towards 0 for outliers. The M-step uses $\xi_{t|T}^{(k)}\omega_{tk}$ as the weight in the mean and in the numerator of the scale update (the scale update still divides by $\sum_t \xi_{t|T}^{(k)}$), and maximises over ν_k numerically. In effect, each state's moments are computed with outliers automatically downweighted.

Bulla (2011) documents the consequence on daily index returns: Student-t components make the fitted states markedly more persistent, because the transitions that Gaussian components spent on outliers disappear. **[Fact]** The worked example of §6.11 reproduces it on US data from 1926 to 2025: moving from Gaussian to Student-t emissions roughly doubles both expected spell lengths (71 to 144 days calm, 23 to 56 days turbulent) and halves the number of regime switches in the smoothed path, from 4.6 to 2.1 a year. Persistence is what §1.3 says carries all the forecasting content, and turnover is what §12.3 says decides whether a strategy pays. **Recommendation: use Student-t emissions for daily returns.** The cost is one parameter per state.

Which observables enter y_t. A univariate HMM on returns can only define states by the distribution of returns. A multivariate emission on a feature vector defines them by joint behaviour, and the choice of features *is* the choice of what a regime means. Four rules apply. First, put in what you want the state to track: log realised volatility from intraday data sharpens the identification of volatility states far beyond what daily returns allow, which is part of why the jump-model applications of

§5.7 use realised-volatility features. Second, transform features toward symmetry (log volatility, changes in spreads rather than levels) so that a Gaussian or Student-t emission is not grossly wrong. Third, standardise inside the training window, never on the full sample (§11.2, leak 4). Fourth, watch the parameter count: a full covariance per state costs $K\,d(d+1)/2$ parameters for d features, so beyond three or four features use a diagonal or factor covariance.

Which parameters switch. Every parameter allowed to vary by state is estimated from that state's observations alone. §8.1 shows that state means are estimated with an error larger than the differences you hope to find, while state variances are pinned down precisely. Two consequences follow. A model with a common mean, $\mu_k = \mu$, loses little likelihood and removes $K-1$ noisy parameters. And the in-sample mean spread a free-mean model reports is not a forecast: the worked example finds a 42-point annualised gap between the Student-t states' fitted means, and a realised gap of -2.5 points (standard error 5.3) between the days the walk-forward model *predicted* to be calm and turbulent (§6.11). **Recommendation: start with variance-only switching.** Free the means only if the out-of-sample log score improves.

Autoregressive emissions. Hamilton's original model lets the autoregressive coefficients switch. For daily returns the autoregressive terms are negligible and add parameters for nothing; for macroeconomic growth rates and realised volatility they matter, and §5.3 covers them.

6.4 How many states

The likelihood-ratio test for K versus $K+1$ does not have a χ^2 distribution, because under the null the extra state's parameters are unidentified and the transition probabilities involving it lie on the boundary (Hansen, 1992; Garcia, 1998; Cho & White, 2007; §8.2). Information criteria are routinely used instead and are known to over-select in this setting (Psaradakis & Spagnolo, 2003). Pohle et al. (2017) explain why in general terms: any misspecification of the emission or the dynamics shows up as extra structure, and an additional state is the cheapest way for the likelihood to absorb it. **[Fact]**

The worked example shows the pattern. On a century of daily US returns, BIC improves from the two-state Gaussian model to the two-state Student-t model, to the three-state Gaussian, to the three-state Student-t, and there is no reason to expect it to stop. The three-state Gaussian model does not discover a new kind of market. It splits the volatility axis into three levels, 8.1%, 16.5% and 41.5% annualised (§6.11).

The recommended procedure, in order of preference:

1. **Fix K a priori from the decision problem.** If the portfolio has two settings, $K=2$. This is not cheating; it is admitting that K is a modelling choice (§1.1) and choosing it for a reason.
2. **Out-of-sample predictive log-likelihood** on a walk-forward split, or cross-validated likelihood over blocks of the sample (Celeux and Durand, 2008). Slow, honest, and directly relevant.
3. **Stability across subsamples.** If $K=3$ is right, fitting it on halves should give recognisably the same three states. If it does not, K is absorbing misspecification.
4. **Inspect the states, then choose pragmatically.** Pohle et al. (2017) recommend fitting a range of K, checking each model's states and residuals (§6.9), and reporting more than one model when no choice is clearly better.
5. **Nonparametric** K via a sticky HDP-HMM (Fox et al., 2011), which puts a prior over K and a separate prior on persistence. Principled and expensive, and it tends to return more states than a trading decision can use.

6.5 How long states last: durations and semi-Markov models

The problem. A first-order Markov chain implies *geometric* sojourn times: the probability of remaining in state k for exactly d periods is $p_{kk}^{d-1}(1-p_{kk})$, so the modal duration is always 1 and the hazard of leaving is constant (sojourn times). A crisis that has lasted six months is, under the model, exactly as likely to end tomorrow as one that started yesterday.

The evidence. Rydén, Teräsvirta and Åsbrink (1998) found that Gaussian HMMs reproduce most stylised facts of daily returns but not the slow decay of the autocorrelation of absolute returns, which is a statement about durations: a geometric sojourn forgets too fast. Bulla and Bulla (2006) show that hidden semi-Markov models with negative binomial sojourns reproduce that autocorrelation better. Direct tests of duration dependence point the same way: Durland and McCurdy (1994) in GNP growth, Maheu and McCurdy (2000), who find that bull markets have a declining hazard in over 160 years of monthly US returns, and Lunde and Timmermann (2004) for bull and bear markets defined by price rules. **[Fact]** that the constant hazard is rejected; **[Contested]** what shape the hazard has, since each study measures it on a few dozen spells.

The semi-Markov model. A hidden semi-Markov model (HSMM) separates the two jobs a transition matrix does. A duration distribution $\Pr(D_k = d)$ says how long a visit to state k lasts; an embedded chain with zero diagonal says where the process goes when the visit ends. The workhorse duration family is the shifted negative binomial,

$$\Pr(D_k = d) = \binom{d + m_k - 2}{d - 1} q_k^{m_k} (1 - q_k)^{d-1}, \qquad d = 1, 2, \ldots,$$

with shape $m_k > 0$ and success probability $q_k \in (0, 1)$. At $m_k = 1$ it is the geometric distribution with $q_k = 1 - p_{kk}$, so the HMM is nested. For $m_k > 1$ the hazard rises with duration (the longer the spell, the likelier it ends); for $m_k < 1$ it falls (Maheu and McCurdy's bull markets). For integer m_k, $D_k - 1$ is the sum of m_k independent geometric waiting times, so the spell must work through m_k stages in sequence and rarely ends early. With $m_k = 2$ and $q_k = 0.5$ the hazard at $d = 1, 2, 3, 4$ is $0.25, 0.33, 0.375, 0.40$, climbing towards q_k from below. The mean duration is $1 + m_k(1 - q_k)/q_k$, which is $1/q_k$ at $m_k = 1$.

Estimation. Two routes. The explicit-duration forward recursion tracks the pair (state, time in state) and costs $O(TKD_{\max})$ for a maximum duration $D_{\max}$, which must be set long enough to cover the longest plausible spell. The cheaper route is due to Langrock and Zucchini (2011): represent state k by a block of M_k sub-states that the chain moves through in order. The block's internal transitions reproduce any duration distribution exactly up to M_k periods, with a geometric tail beyond. The result is an ordinary HMM with $\sum_k M_k$ states, so the filter, the predicted probabilities and direct likelihood maximisation all work unchanged. Summing the sub-state probabilities recovers the probability of each regime.

Duration-dependent switching (Durland & McCurdy, 1994; Maheu & McCurdy, 2000) is the close cousin: keep the Markov chain but let p_{kk} depend on the current spell length through a logistic function, which the filter handles by augmenting the state with a duration counter up to a cap.

Failure modes. (i) The duration shape is estimated from the number of *spells*, a few dozen at best, so m_k is weakly identified and its confidence interval routinely spans 1. (ii) The current spell is right-censored in real time: you know how long it has lasted, not how long it will last, which is exactly the quantity the duration model is meant to supply. (iii) Truncating $D_{\max}$ too short forces long spells to end.

Cheaper ways to buy persistence. Student-t emissions (§6.3) remove the outlier-driven switches that make spells look short. A stickiness prior on the diagonal of $\mathbf{P}$ (§6.7) or a jump penalty (§5.7) inflates self-transitions without modelling duration. And a downstream model can be given the time since the last transition as a feature (§10.1), which hands it the duration information directly.

Recommendation. If the fitted state path flickers, the geometric-duration assumption is the first suspect: **naive HMMs produce implausibly choppy state sequences, and that choppiness is what destroys a trading strategy through turnover (§12.2)**. Try Student-t emissions and a stickiness prior first. Fit an HSMM only if those leave a measurable gap in out-of-sample log score, and judge it on that score rather than on the likelihood. **[Practice]**

6.6 Transitions that respond to observables

§5.3 lists constant transition probabilities among the assumptions that fail in practice. The direct fix is to let them depend on lagged observables.

Definition. With a vector of $\mathcal{F}_{t-1}$-measurable covariates z_{t-1} (including a constant), a **time-varying transition probability** (TVTP) model sets each row of the transition matrix by a multinomial logit,

$$p_{jk,t} = \Pr(s_t = k \mid s_{t-1} = j,\ z_{t-1}) = \frac{\exp\big(z_{t-1}^\top \psi_{jk}\big)}{\sum_{l=1}^{K} \exp\big(z_{t-1}^\top \psi_{jl}\big)}, \qquad \psi_{jj} = 0,$$

where the coefficient vectors ψ_{jk} are the state-process parameters ψ of the master form (§1.4), now driven by a covariate. Setting $\psi_{jj} = 0$ makes staying the baseline, since adding the same number to every score in a row would leave the probabilities unchanged. For $K = 2$ each row has a single free coefficient vector, $\psi_j \equiv \psi_{jk}$ for the one $k \neq j$, and reduces to a logistic function. The probability of leaving state j is $1/(1 + \exp(-z_{t-1}^\top \psi_j))$, so the probability of staying is $p_{jj,t} = 1/(1 + \exp(z_{t-1}^\top \psi_j))$. A positive coefficient raises the chance of leaving; flip its sign to parameterise persistence directly. With z_{t-1} equal to a constant alone, this is the constant-$\mathbf{P}$ model. The filter is unchanged except that $\mathbf{P}$ becomes $\mathbf{P}_t$: $\xi_{t|t-1} = \mathbf{P}_t^\top \xi_{t-1|t-1}$.

Diebold, Lee and Weinbach (1994) introduced the model with an EM algorithm, and Filardo (1994) applied it to business-cycle phases driven by leading indicators. In finance, Schaller and van Norden (1997) let stock-market transitions depend on deviations of prices from fundamentals, and Perez-Quiros and Timmermann (2000) used interest rates and money growth to drive recession probabilities in small- and large-firm returns.

Estimation. The E-step is unchanged. The M-step for ψ becomes a weighted multinomial logistic regression of the pairwise posteriors $\xi^{(jk)}_{t-1,t|T}$ on z_{t-1}, with no closed form; direct maximisation of the likelihood (§6.1) is usually simpler.

What it buys. TVTP is the bridge between the latent and observable branches of §7.1. The state is still filtered from the data, so the model can find a regime the covariate does not name, but the regime's dynamics now respond to something you can see. In one limit the bridge closes completely: if the transition probabilities do not depend on the current state, $p_{jk,t} = g_k(z_{t-1})$ for every j, then $\Pr(s_t = k \mid \mathcal{F}_{t-1}) = g_k(z_{t-1})$ and the model is a mixture of experts with an observable gate (§7.2).

Choosing z. Lagged, normalised, and few: a trailing volatility rank, the slope of the VIX futures curve, a credit spread change, the term spread for macro regimes. Every covariate must be $\mathcal{F}_{t-1}$-measurable, built from real-time vintages if macroeconomic (§11.2, leak 8), and stationary, since a logit of a trending level drifts toward 0 or 1.

Failure modes.

1. **Parameter count against forty transitions.** TVTP has $K(K-1)\dim(z)$ transition parameters, against $K(K-1)$ for a constant **P**, all estimated from the few transitions in the sample. Overfitting is the default outcome. Penalise ψ (a ridge term in the likelihood) or put a prior on it.
2. **A covariate that measures the state rather than predicting it.** A lagged VIX level predicts that the turbulent state will persist, because high implied volatility *is* turbulence. It does not predict entries before they start. Check whether the covariate improves the forecast of transitions *into* the turbulent state, not just of staying there.
3. **Degeneration into a threshold rule.** If the fitted logit is steep, the latent state becomes a near-deterministic function of z. Then the threshold or smooth-transition model of §5.5 is the honest description, with less machinery.
4. **Endogenous switching.** If the covariate responds to the same shocks that drive the state, the standard likelihood is misspecified. Kim, Piger and Startz (2008) develop the estimator that allows for it.

The evidence. In-sample improvements from TVTP are common and large; out-of-sample improvements are rarer. Kole and van Dijk (2017) compare rule-based and regime-switching identification of bull and bear markets and find rules better for dating states in sample and regime-switching models, including versions with predictors, better for forecasting. **[Contested]**

Recommendation. One or two normalised covariates, a ridge penalty on ψ, and judgment by out-of-sample log score against the constant-**P** model. §8.6 scores "predict regime transitions ahead of time" as not working. TVTP is the principled attempt, and that verdict stands unless your own out-of-sample evidence overturns it.

6.7 Bayesian estimation

Why. A Bayesian treatment addresses the failures of §6.1 in three ways. A prior on each σ^2_k removes the degeneracy. A prior on each row of **P** regularises transition probabilities estimated from forty events and encodes the belief that regimes persist. And the posterior over parameters lets the predictive density integrate over parameter uncertainty instead of plugging in point estimates, which matters most for exactly the poorly determined parameters: the state means and the transition probabilities.

The priors. Conjugate choices keep the algebra closed. Each transition row gets a Dirichlet prior, $(p_{j1}, \dots, p_{jK}) \sim \text{Dirichlet}(a_{j1}, \dots, a_{jK})$, whose parameters act as **pseudo-counts**: the prior behaves as if a_{jk} extra $j \to k$ transitions had been observed. Given a state path with n_{jk} observed transitions, the posterior is $\text{Dirichlet}(a_{j1} + n_{j1}, \dots, a_{jK} + n_{jK})$. A sticky prior puts large pseudo-counts on the diagonal: $a_{kk} = 49$ and $a_{kj} = 1$ for $K = 2$ says "prior-mean persistence of 0.98, held with the weight of fifty days of data". Means and variances get normal and inverse-gamma priors.

The Gibbs sampler. Bayesian HMMs are estimated by Markov chain Monte Carlo, alternating three draws:

1. **The whole state path at once, by forward filtering, backward sampling** (FFBS). Run the filter of §5.2 forward. Draw s_T from $\xi_{T|T}$. Then, for $t = T-1$ down to 1, draw s_t from $\Pr(s_t = j \mid s_{t+1}, \mathcal{F}_t) \propto \xi^{(j)}_{t|t}\, p_{j s_{t+1}}$. Carter and Kohn (1994) introduced the idea for state-space models and Chib (1996) for Markov mixtures.
2. **Each transition row** from its Dirichlet posterior, using the transition counts of the sampled path.
3. **Each state's emission parameters** from their conjugate posteriors, using the observations the sampled path assigns to that state.

Drawing the path jointly matters. The earlier single-site sampler of Albert and Chib (1993) draws each s_t given its neighbours, and with persistent states it mixes very slowly, because with its neighbours held fixed a day inside a spell almost always keeps its state, so a spell can change length only at its edges, one day at a time. Scott (2002) reviews the recursions and recommends the forward–backward sampler. **[Fact]**

Label switching, again. Under a symmetric prior the posterior is invariant to relabelling the states, so the sampler can swap labels between sweeps and posterior means of "state 1" average two different states. Three remedies: impose the ordering

$\sigma_1 < \cdots < \sigma_K$ inside the sampler; use the random permutation sampler of Frühwirth-Schnatter (2001), which permutes labels deliberately and then post-processes; or relabel the draws afterwards (Stephens, 2000).

The cheap version. The posterior *mode* under these priors is computed by EM with one change to the M-step: add the prior pseudo-counts to the expected counts. The mode of a Dirichlet distribution with parameters $b_1, \ldots, b_K$ is $(b_k - 1)/(\sum_l b_l - K)$, so the pseudo-counts that enter are $a_{jk} - 1$, one fewer than the prior's parameters (the formula needs $a_{jk} \geq 1$). The transition update becomes

$$\hat{p}_{jk} = \frac{\sum_t \xi^{(jk)}_{t-1,t|T} + a_{jk} - 1}{\sum_t \xi^{(j)}_{t-1|T} + \sum_l (a_{jl} - 1)},$$

so the sticky prior above adds 48 stays and no exits to the expected counts. The variance update gains a floor from the inverse-gamma prior. **Recommendation: use this penalised EM, with a weak sticky prior and a variance prior, as the default fitting routine.** It costs nothing and fixes degeneracy and erratic transitions. Reserve full MCMC for when you need the predictive distribution to carry parameter uncertainty, as in allocation (§6.10). **[Practice]**

Failure modes. (i) The prior dominates any rarely visited state, which is often the state you care about; report how much the posterior moved from the prior. (ii) MCMC convergence is not guaranteed and must be checked across chains. (iii) A walk-forward study needs a full MCMC run per refit, which is expensive enough that it tempts people into fitting once on the full sample, which is leak 2 of §11.2.

6.8 Adaptive and online estimation

The problem. HMM parameters fitted to financial returns drift. Rydén, Teräsvirta and Åsbrink (1998) found different parameters across subperiods of the same index, and an expanding-window refit gives the 1930s the same weight as last year. Three responses, in increasing order of continuity:

- **Rolling windows.** Refit on the last W observations. Simple, but the window edge creates parameter jumps when a large observation drops out.
- **Exponential forgetting.** Maximise a weighted log-likelihood in which the contribution of observation t is discounted by λ^{T-t} for a forgetting factor $\lambda < 1$. The weights sum to $1/(1-\lambda)$, which is the effective memory in observations ($\lambda = 0.998$ gives 500). Nystrup, Madsen and Lindström (2017) estimate a two-state Gaussian HMM this way and find that the time-varying parameters reproduce the long memory of squared daily returns, the stylised fact that defeats constant-parameter HMMs (§6.5), and improve one-step density forecasts. Nystrup, Madsen and Lindström (2015) document the parameter drift that motivates it.
- **Online EM.** Update the expected sufficient statistics recursively as each observation arrives, with a step size that shrinks over time or stays constant (Cappé, 2011). A constant step size is exponential forgetting in recursive form.

The trade-off is the identification asymmetry again. Forgetting reduces the effective sample, and §8.1 says what survives a small sample: volatilities, not means or transition probabilities. A memory of 500 days with a turbulent state occupying 20% of the time contains about 100 turbulent days and perhaps four spells, hence four entries and four exits. The state volatilities are estimable from that; the transition probabilities are not. **Recommendation: forget fast for the emission scales and slowly, or through a prior, for the transition matrix and the means. [Practice]**

Failure modes. (i) The forgetting factor is a hyperparameter, and tuning it on the evaluation period is leak 7 of §11.2. (ii) Short memories make label switching between refits more likely; track the state identity across refits (§12.6). (iii) Parameters that jump between refits make positions jump; §7.2 records that exponential weighting is itself approximately a break model with the break date integrated out, so the same smoothing logic applies.

6.9 Model checking

A good likelihood does not show that a model is adequate, and §2.6 shows how a model with no regimes can fit well. Four checks do the work.

1. Pseudo-residuals. If the model is correct, the probability integral transform of each observation under its one-step predictive distribution,

$$\mathrm{PIT}_t = \Pr(y \leq y_t \mid \mathcal{F}_{t-1}) = \sum_{k=1}^{K} \xi^{(k)}_{t|t-1} \, F(y_t \mid \theta_k),$$

is independent and uniform on $[0, 1]$ (Diebold, Gunther & Tay, 1998), where $F(\cdot \mid \theta_k)$ is the state-k distribution function. The **pseudo-residual** $e_t = \Phi^{-1}(\text{PIT}_t)$, with Φ the standard normal distribution function, should then be independent standard normal (Zucchini, MacDonald & Langrock, 2016). Read three things off them: a quantile–quantile plot of e_t against the normal (tails wrong means the emission family is wrong); the autocorrelation of e_t^2 or $|e_t|$ (remaining volatility clustering means the state process is too simple: too few states, the wrong durations, or constant transitions); and the same statistics by subperiod (parameter drift, §6.8). Compute them from predicted probabilities and walk-forward parameters, so that the check is itself out of sample.

2. The horse race. Compare the average one-step log predictive density out of sample against continuous benchmarks: an EWMA volatility with normal and Student-t innovations, and a GARCH. Test the difference with a Diebold–Mariano statistic and a HAC standard error (§11.6). This is the experiment §2.7 recommends, and the worked example runs it.

3. Stability across refits. Track each parameter across walk-forward refits against its bootstrap interval, and the correlation of overlapping state paths between consecutive refits (§12.6).

4. The decoded history. Count switches per year and look at the spell lengths. A state that is entered and left within days, repeatedly, is flicker, not a regime; §6.3 and §6.5 say what to change.

6.10 Using the model directly: forecasts and allocation

An HMM is a complete predictive model, and it can drive decisions without any downstream learner.

Forecasting h steps ahead. The state forecast is $\xi_{t+h|t} = (\mathbf{P}^\top)^h \xi_{t|t}$, which decays toward π at the rate $|\lambda_2|$ of §1.3. The predictive density of y_{t+h} is the mixture $\sum_k \xi^{(k)}_{t+h|t} f(\cdot \mid \theta_k)$. For the cumulative return over the next h bars, $R_{t,h} = \sum_{i=1}^{h} y_{t+i}$, suppose the observations are independent given the state path (no autoregressive terms). Conditioning on the path and applying the law of total variance, the variance decomposes as

$$\text{Var}(R_{t,h} \mid \mathcal{F}_t) = \sum_{i=1}^{h} \mathbb{E}\left[\sigma^2_{s_{t+i}} \mid \mathcal{F}_t\right] + \text{Var}\left(\sum_{i=1}^{h} \mu_{s_{t+i}} \,\middle|\, \mathcal{F}_t\right).$$

The first term is the expected within-state variance along the path: given the path the returns are independent, so their variances add. The second is the uncertainty about which means the path will visit, and it includes covariances between dates, because a persistent chain that is in the bad state on one day is likely to be there the next. §12.1 shows that for single-asset daily data the second term is negligible until the horizon reaches years. The variance term is what an HMM forecasts well.

Sizing a single asset. §12.1 derives the mean-variance position from the mixture's first two moments and shows that it de-risks smoothly as the turbulent probability rises. That is the HMM used directly, and on the worked example it is the best of the four strategies tested, though not by a margin the sample can distinguish (§6.11).

Multi-asset allocation. With a vector of returns, each state has its own mean vector and covariance matrix, and the allocation problem conditions on the predicted state probabilities. Ang and Bekaert (2002) fit this to international equity returns and find a bear state with higher volatilities and higher correlations. Their allocation result is the useful one: the cost of ignoring regimes is small for an all-equity portfolio, because diversification still pays in both states, and large when a risk-free asset can be held, because the model then moves into cash in the bad state. Ang and Bekaert (2004) give the practitioner version. Guidolin and Timmermann (2007) solve the dynamic problem with four states for stocks and bonds, and find optimal weights that depend strongly on the current state and on the investment horizon, because the state forecast reverts to π as the horizon lengthens. Nystrup, Hansen, Madsen and Lindström (2015) report that regime-based allocation adds value over a static benchmark out of sample. **[Contested]** — the allocation literature's backtests share the specification-search problem of §3.6.

Long-horizon tail risk. Actuaries use the HMM for a different job. The two-state regime-switching lognormal model of Hardy (2001) prices long-dated equity guarantees, where what matters is the left tail of cumulative returns over ten years or more. A persistent high-volatility state fattens that tail relative to a single lognormal, and the model became standard in actuarial work on such guarantees. **[Practice]** This is an application that uses exactly what the HMM estimates well, the volatility states and their persistence, and none of what it estimates badly.

Parameter uncertainty. Plug-in allocations treat $\widehat{\mu}_k$ as known, and §8.1 shows they are noise. Two remedies: integrate over the posterior (§6.7), or apply the pooling rule of §9.5 and set the state means equal while letting the covariance matrices switch. **Recommendation: in allocation, switch the covariance and pool the mean** unless out-of-sample evidence supports switching means.

6.11 A worked example: a century of US equity returns

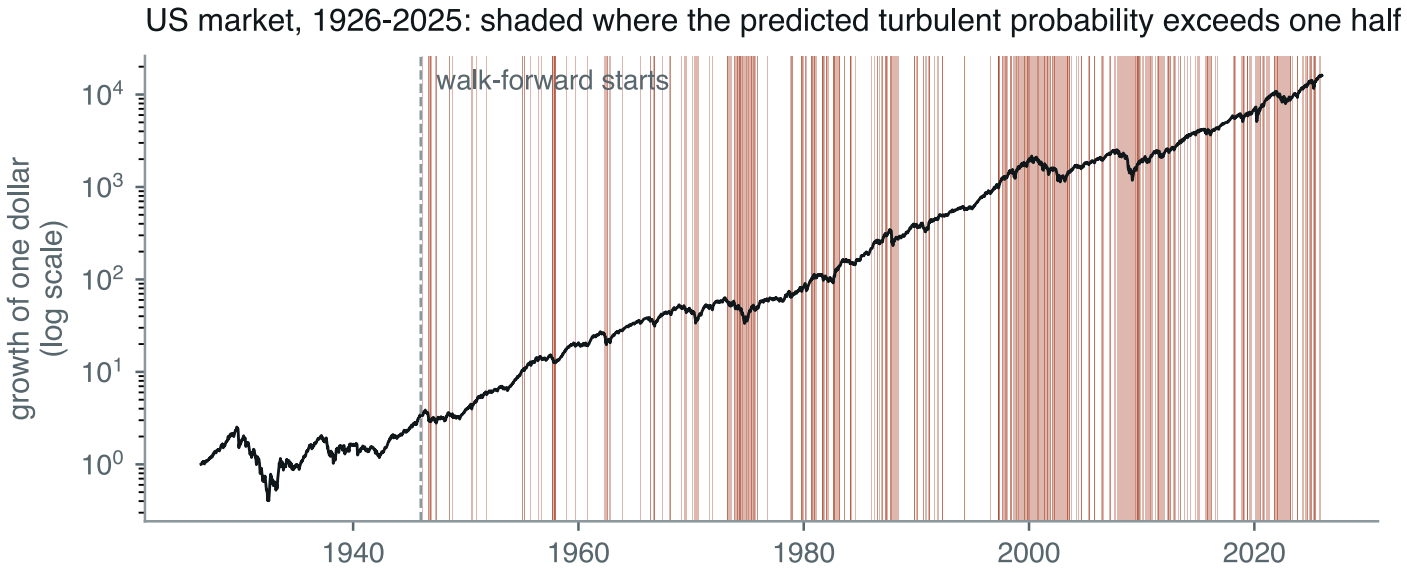
The example fits the models of this section to daily returns on the US stock market, the way a researcher would have fitted them in real time, and reports every number the earlier sections cite.

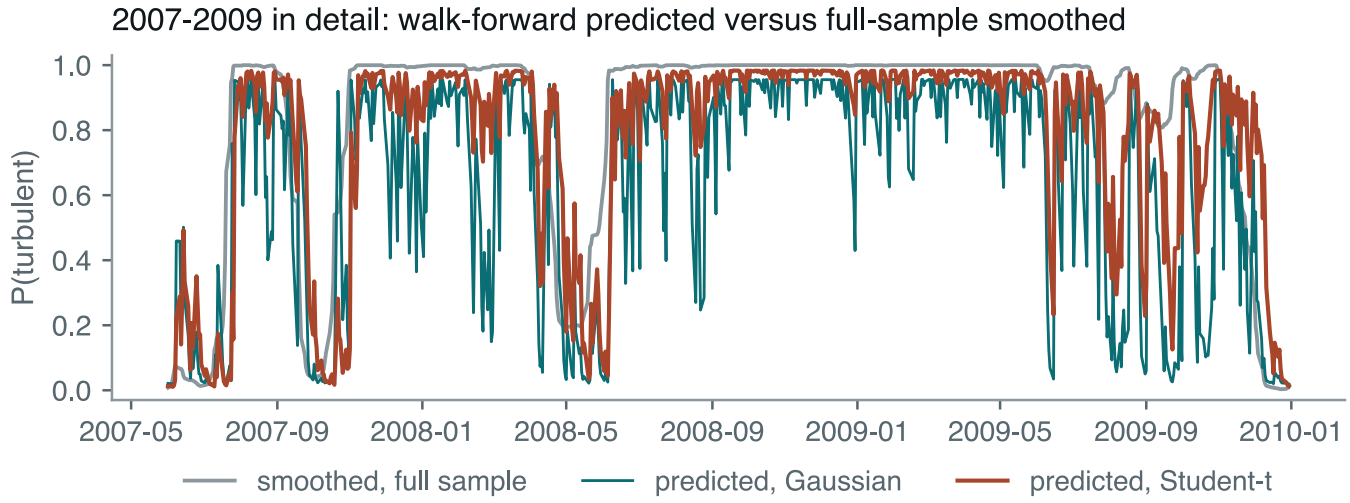
Setup. Daily value-weighted US market returns from the Kenneth French data library, July 1926 to December 2025: 26,151 days of log returns with an annualised volatility of 17.1%. The models are fitted to total returns; the out-of-sample return tables use returns in excess of the risk-free rate. Four models are fitted to the full sample for description: Gaussian and Student-t emissions, with two and three states, all with freely switching means and variances. For the out-of-sample work, the two-state Gaussian and Student-t models are refitted at the start of every year from 1946 on, on an expanding window of all earlier data, each refit warm-started from the previous one. States are sorted by volatility inside every fit. Only the predicted probability $\xi_{t|t-1}$ computed with that year's parameters is used for any out-of-sample number. That is rung 5 of §5.1, over 20,337 days. Everything is printed by `figures/regime_hmm_worked.py`, which implements the EM of §6.1 with the Student-t extension of §6.3 in numpy. It uses plain EM with a variance floor and no priors (§6.7), and keeps the best of eight random starts for each full-sample fit, fewer than §6.1 advises, to keep the run to a few minutes.

The full-sample fits. [Fact] for this sample; these are rung-2 numbers, and describe history only.

Model	Volatility by state	Mean by state	Expected spell (days)	Half-life of λ_2	Smoothed switches per year	BIC
Gaussian, $K = 2$	9.7%, 30.0%	+21.8%, −29.1%	71, 23	11.7 days	4.63	−175,095
Student-t, $K = 2$	10.1%, 28.1%	+23.8%, −18.1%	144, 56	27.6 days	2.06	−176,562
Gaussian, $K = 3$	8.1%, 16.5%, 41.5%	+26.1%, −3.0%, −39.9%	41, 23, 27	19.8 days	1.35	−177,008
Student-t, $K = 3$	8.2%, 14.2%, 35.0%	+26.0%, +12.4%, −37.6%	71, 46, 51	37.7 days	1.19	−177,531

Volatilities and means are annualised, from total log returns. The expected spell is $1/(1-p_{kk})$, and a smoothed switch is a crossing of one half by the smoothed probability of the most volatile state. BIC is $-2 \ln L + n_{\text{par}} \ln T$ for n_{par} free parameters (counting $K(K-1)$ for the transition matrix), so lower is better. The Student-t degrees of freedom are 5.5 (calm) and 4.1 (turbulent) in the two-state model, so returns are heavy-tailed *within* each state, not only across them.





Out of sample, 1946 to 2025. First, the density forecast: the average one-step log predictive density, measured against an EWMA volatility ($\lambda = 0.94$) with normal innovations, with Newey–West standard errors.

Model	Average log score per day	Difference from EWMA normal (se)
EWMA, normal	3.4131	—
EWMA, Student-t ($\nu = 5.1$)	3.4477	+0.0346 (0.0081)
HMM, Gaussian	3.4059	−0.0073 (0.0093)
HMM, Student-t	3.4318	+0.0187 (0.0090)

Second, what the predicted state separates. The table conditions realised *excess* returns on whether the walk-forward Student-t model put the turbulent probability above one half before the day began.

Predicted state	Share of days	Annualised mean (se)	Annualised volatility
Calm	76.2%	+7.4% (1.6)	10.8%
Turbulent	23.8%	+9.9% (5.0)	24.5%

Third, four ways to hold the market, all set from information through the previous close: buy and hold; a gate that is out of the market whenever the predicted turbulent probability exceeds one half; and two volatility targets aiming at 15% a year with leverage capped at 2, one using the EWMA volatility and one using the HMM's predictive variance (§12.1).

Strategy	Return	Volatility	Sharpe	Max drawdown	Turnover per year
Buy and hold	8.0%	15.3%	0.52	59.2%	0.0
HMM gate (Student-t)	5.6%	9.5%	0.60	37.7%	9.5
EWMA volatility target	8.8%	15.5%	0.57	62.5%	8.7
HMM volatility target (Student-t)	8.4%	13.5%	0.62	44.3%	14.2

Returns and volatilities are annualised excess returns. Average exposure is 1.00, 0.76, 1.33 and 1.12 respectively. The standard error of a Sharpe ratio near 0.5 over 81 years is about 0.12.

[Fact] for this sample, read in order of importance:

1. **The volatility levels are large and stable.** Calm volatility near 10% and turbulent volatility near 28–30% appear in every two-state fit, and the last walk-forward refit (data through 2024) gives 10.0% and 28.1%, matching the full-sample Student-t fit. The predicted state separates realised volatility by a factor of 2.3 out of sample. That shows the volatility levels are real; it does not show that they are discrete, which §2.6 says return data alone cannot settle.
2. **The mean spread is an in-sample artefact.** The Student-t states' fitted means differ by 42 percentage points a year. Out of sample, the days predicted turbulent earned *more* on average than the days predicted calm, and the difference, −2.5 points with a standard error of 5.3, is noise. The in-sample spread arises because the smoother and the filter use y_t itself to classify day t: large negative returns arrive with rising volatility (the leverage effect), so they are assigned to the turbulent state *after the fact*. A prediction made before the day cannot use that association. This is §8.1 and §8.4 on real data.
3. **The emission family changes the dynamics, not just the tails.** Student-t emissions double the expected spells and halve the smoothed switching rate (§6.3).
4. **BIC does not stop.** Each larger model wins, and the third state subdivides the volatility axis (§6.4).
5. **The continuous model wins the density forecast.** An EWMA volatility with Student-t innovations beats the Student-t HMM by 0.016 log points a day, with a standard error of 0.003. Heavy tails matter more than regimes here: Student-t innovations add 0.035 to the EWMA and 0.026 to the HMM, while the regime structure adds nothing, since the Gaussian HMM trails the Gaussian EWMA by 0.007 and the Student-t HMM trails the Student-t EWMA by 0.016. This is the result §2.7 and §8.7 predict.
6. **Every overlay helps a little, and the sample cannot rank them.** All three overlays raise the Sharpe ratio from 0.52 to between 0.57 and 0.62, which is at most one standard error. The difference shows in the drawdown: the gate and the HMM volatility target cut the worst loss from 59% to 38% and 44%. The EWMA target's drawdown is the worst of all, and its timing is instructive: buy and hold's worst loss ran from 2000 to 2009, the EWMA target's from 1968 to 1974,

a slow decline in which trailing volatility stayed moderate and the strategy stayed levered all the way down. Volatility targeting protects against fast crashes, not slow ones.

7. **The predicted probability flickers; the smoothed one does not.** The walk-forward Student-t probability crosses one half 9.5 times a year, against 2.1 for the smoothed path and 15.5 for the Gaussian predicted path. The bottom panel of the figure shows why: the smoother sees the whole crisis and draws a block; the filter re-decides every day. That flicker is the gate's turnover of 9.5 a year, one full position change per crossing, and it carries into the HMM volatility target's turnover of 14.2, against 8.7 for the EWMA target. §12.2 and §12.3 say what to do about it. In February 2020 the predicted probability crossed one half on the 25th, four trading days after the market's peak on the 19th.

The caveats are the ones §11.5 insists on. This is one market and one history, with perhaps a few dozen independent turbulent episodes in the out-of-sample period; the target volatility, the leverage cap and the refit schedule were set once, not tuned, but they were set by someone who knew the history.

6.12 Software and an implementation checklist

Package	Language	Models	State beliefs exposed	Watch for
`statsmodels` (`MarkovRegression`, `MarkovAutoregression`)	Python	Gaussian switching regressions and autoregressions, switching variance, TVTP through `exog_tvtp`	predicted, filtered, smoothed, as separate attributes	Gaussian only; fits once on what you pass, so the walk-forward loop is yours
`hmmlearn`	Python	Gaussian, Gaussian-mixture and discrete emissions	smoothed (`predict_proba`), Viterbi (`predict`)	no filtered or predicted output; both defaults look ahead
`depmixS4` (Visser & Speekenbrink, 2010)	R	GLM emissions, covariates on transitions and emissions	filtered and smoothed	covariate-driven transitions overfit easily
`MSGARCH` (Ardia et al., 2019)	R	Markov-switching GARCH (§5.4), Student-t and skewed emissions, ML and MCMC	filtered, smoothed, predictive	many parameters per state
`mhsmm` (O'Connell & Højsgaard, 2011)	R	hidden semi-Markov models (§6.5)	smoothed	duration distributions need many spells

None of them runs a walk-forward study, sorts states inside each window, or refuses to hand you a smoothed probability. The forward filter is short enough to write and test against a library on the same parameters, and the script of §6.11 is a numpy reference implementation.

The checklist.

1. Write down the decision the state will serve and fix K from it (§6.4).
2. Compute the half-life of the fitted $|\lambda_2|$ and compare it to the holding period (§1.3).
3. Use Student-t emissions on daily returns; start with variance-only switching (§6.3).
4. Fit by penalised EM with a weak sticky prior and a variance floor, from many starts; keep the best and inspect the spread of the rest (§6.1, §6.7).
5. Sort states by volatility inside every estimation window (§6.1).
6. Refit on an expanding or exponentially weighted window; never use full-sample parameters out of sample (§6.8, §11.2).
7. Emit $\xi_{t|t-1}$ only, and label every reported number with its rung (§5.1, §6.2).
8. Check pseudo-residuals from the walk-forward fit (§6.9).
9. Race the model against an EWMA with Student-t innovations on out-of-sample log score (§6.9).
10. Count switches per year and turnover before costing anything (§6.11, §12.3).
11. Add durations (§6.5) or covariates (§6.6) only when the simple model fails a specific check, and judge them out of sample.

§6 Key takeaways

1. An HMM's transition probabilities are estimated from the number of transitions, a few dozen in twenty years of daily data. The expected spell length of a turbulent state carries a standard error of about 16% even with the state path known.
2. Four state beliefs exist — predicted, filtered, smoothed, Viterbi — and only the first is tradable. Popular libraries return the smoothed or Viterbi object by default.
3. Student-t emissions are the highest-value refinement for daily returns. On a century of US data they double the expected spell lengths and halve the switching rate, because single outliers stop forcing switches.
4. Let variances switch and pool the means. On US data the fitted states' means differ by 42 points a year in sample, and the predicted states' realised means differ by −2.5 points with a standard error of 5.3.
5. Information criteria keep adding states, and the extra states subdivide volatility. Fix K from the decision.
6. Constant hazards are rejected in the data. Semi-Markov models fix them at a cost; heavy tails and sticky priors buy most of the persistence more cheaply.
7. Covariate-driven transitions bridge latent and observable regimes, and overfit forty transitions easily. Regularise them and judge them out of sample.
8. Penalised EM with Dirichlet and inverse-gamma priors is the default fitter: it removes degeneracy and steadies the transition estimates at no cost.
9. Adaptive estimation should forget fast for volatilities and slowly for transitions and means, because only the volatilities survive a short memory.
10. Check the model with pseudo-residuals and an out-of-sample race against an EWMA with Student-t innovations. On US equities the EWMA wins the density forecast; the HMM earns its place through sizing and drawdown, not through forecasting returns.

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7. Taxonomy and equivalences

§5 listed nine families. This section argues they are one model with four slots (§1.4), that several of them are provably identical, and that one slot, inference and use, matters far more than the other three. §7.4 splits the four slots into seven finer knobs and ranks them.

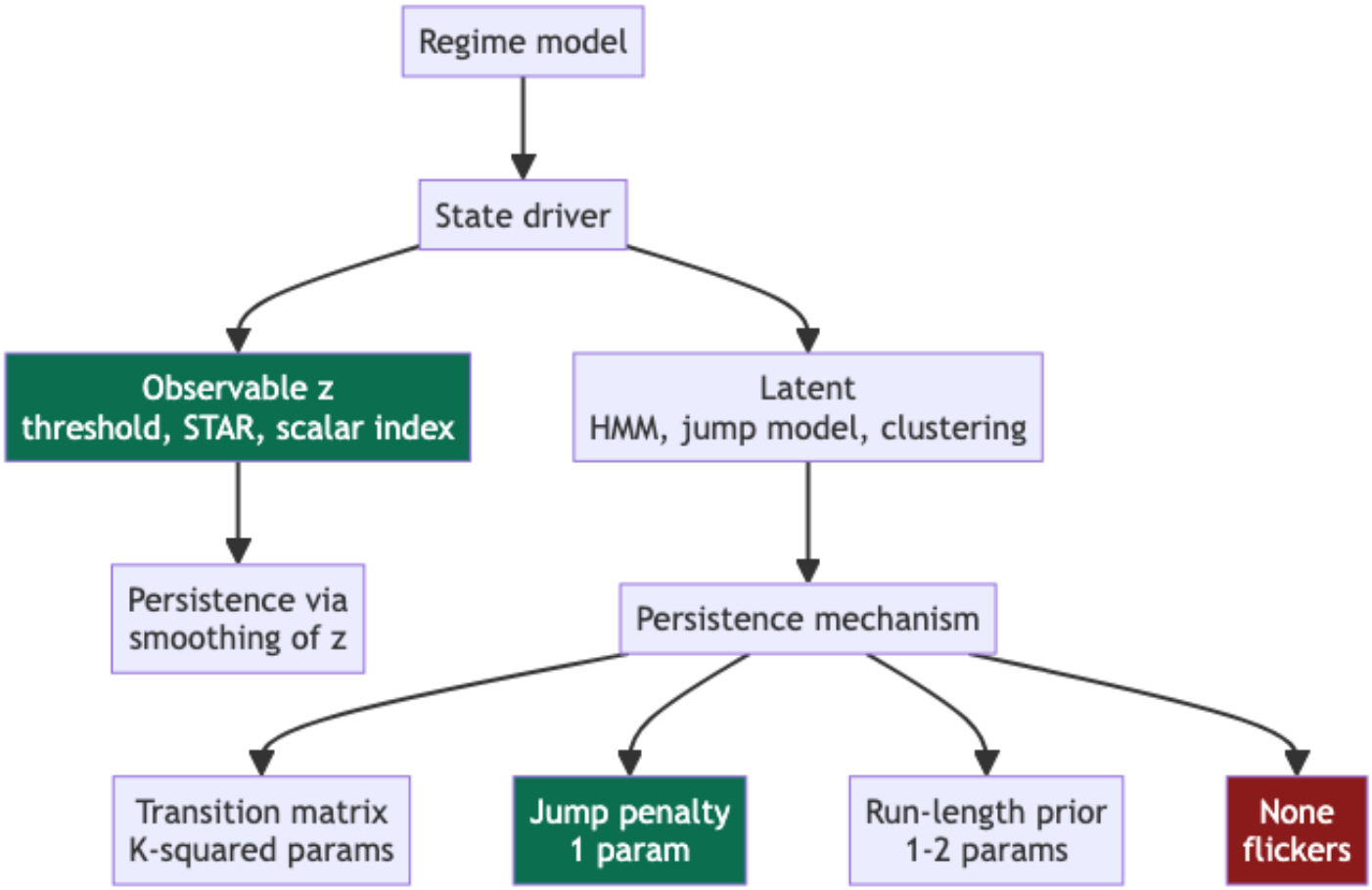
7.1 The design space is a product, not a tree

Recall the master form of §1.4. Filling in the first three slots for each family gives a coordinate table rather than a taxonomy — the fourth, inference and use, is orthogonal to family choice and is treated separately at the end of this section — and the empty cells are the interesting part, because they are combinations nobody has tried.

Family	What varies	State dynamics	Driver	Typical use
MS regression / HMM	β_k, σ_k	Markov	latent	density forecast
HMM with TVTP	β_k, σ_k	Markov, covariate-driven	**latent and observable**	density forecast
Hidden semi-Markov	β_k, σ_k	**explicit durations**	latent	density forecast
MS-GARCH	ω, α, β per state	Markov	latent	vol forecast
Threshold / STAR	β_k, σ_k	indicator or logistic	**observable**	conditional regression
Change point	everything	single break / run length	latent	monitoring
Jump model	centroid c_k	**jump penalty**	latent	classification
Clustering	centroid c_k	**none**	latent	exploration
Scalar index	nothing (no states)	trailing window	**observable**	feature
Deep latent	network weights	learned, state-dependent	latent or both	end-to-end
Supervised label	label definition	label span	**defined**	classification

Two structural observations. First, the "driver" column is close to binary in practice — the state is either a function of something you can see, or it is inferred from the series itself — and the observable branch is systematically underused relative to how well it performs (§5.5). Second, the persistence mechanism varies more than the literature acknowledges: a Markov chain,

a jump penalty, a run-length prior, and a trailing window are four different ways to say “the state does not change every day”, with very different statistical costs.



The green nodes are the recommended choices; the red one is what plain clustering gives you and is the reason it needs a persistence term bolted on.

The two genuinely *orthogonal* modifiers, which are not branches of this tree and apply to every leaf, are: **(a)** what information set the state estimate conditions on (§5.1), and **(b)** whether the downstream consumer receives a hard label or a full posterior. Both are protocol choices, both cut across every model, and both matter more than the choice of leaf.

7.2 Equivalences

The highest-value table in this document. Things that look different and are the same, marked **exact** where the identity holds without approximation.

Claim	Status	Why
Hamilton filter = HMM forward algorithm = discrete-state Bayes filter	**exact**	Identical recursion; three literatures (econometrics, speech, control) named it separately
Kim smoother = HMM forward-backward posterior	**exact** (discrete state only)	Both compute $\Pr(s_t \mid \mathcal{F}_T)$; different factorisation, same number. With a continuous state as well, Kim's smoother collapses mixtures and is approximate
Gaussian mixture model = HMM with $\mathbf{P} = \mathbf{1}\pi^\top$	**exact**	A mixture is an HMM with no memory; hence forecasts nothing (§1.3)
Jump model = MAP state path of a restricted HMM	**exact** for the path given the centroids	See derivation below — this is the important one
HSMM with geometric durations = HMM	**exact**	The geometric is the semi-Markov duration with constant hazard (§6.5)

Claim	Status	Why
HSMM $\approx$ HMM on an expanded state space	approximate	Exact up to M_k periods of duration, geometric beyond (Langrock & Zucchini, 2011)
TVTP HMM whose transitions ignore the current state $=$ mixture of experts gated on z	**exact**	If $p_{jk,t} = g_k(z_{t-1})$ for every j, then $\Pr(s_t = k \mid \mathcal{F}_{t-1}) = g_k(z_{t-1})$ (§6.6)
Student-t emission $=$ Gaussian emission with a gamma-distributed latent precision	**exact**	The scale-mixture representation that makes EM work (§6.3)
Penalised EM with Dirichlet transition priors $=$ posterior-mode Bayesian HMM	**exact**	The prior pseudo-counts are added to the expected counts (§6.7)
STAR with logistic transition $=$ two-expert mixture of experts with a logistic gate on one input	**exact** for the conditional mean	Same blended point forecast, invented twice (§10.5). The densities differ: STAR is one Gaussian with a blended mean, a density mixture of experts is a two-component mixture
Regime dummy interacted with every feature $=$ separate models per regime	**exact** for least squares; approximate for a deep tree	Least squares with the full set of interactions returns the per-regime coefficients. A deep tree can represent the same pair of fits and does so if it splits on the dummy first (§9.3, §10.4)
Vol-targeted position sizing $=$ regime conditioning with $K = \infty$ and a $1/\hat{\sigma}$ response	interpretation	Every vol-targeted strategy is already regime-conditional
$K = 2$ HMM differing only in variance $\Rightarrow$ unconditional return distribution is a scale mixture of normals	**exact**	Hence symmetric, leptokurtic, and indistinguishable from a fat-tailed i.i.d. law on unconditional moments alone (§2.6)
Single GARCH with near-unit persistence $\approx$ MS-GARCH with lower within-regime persistence plus level shifts	approximate	Hamilton & Susmel (1994): apparent IGARCH is partly unmodelled regime switching
Markov-switching volatility with large $K \approx$ discretised stochastic volatility	approximate	The chain converges to a discretisation of the continuous latent scale
Long memory $\approx$ occasional regime switching	approximate	Diebold & Inoue (2001); mutually confusable even asymptotically
Exponentially decayed sample weights $\approx$ a break model with an unknown break date integrated out	approximate	Both downweight the past geometrically: with a constant per-bar break hazard, the chance that no break occurred in the last s bars is λ^s with $\lambda = 1 -$ hazard
BOCPD predictive density $\approx$ automatic estimation-window selection	approximate	Its forecast is a run-length-weighted mixture of fixed-window estimators

The jump-model identity, derived. Take an HMM whose emissions are isotropic Gaussians, of common variance σ^2 (the spread of the feature vector around its centroid — not a return volatility), around centroids c_k,

$$f(u_t \mid s_t = k) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|u_t - c_k\|^2}{2\sigma^2}\right),$$

and whose transition matrix is symmetric, $p_{kk} = p$ and $p_{jk} = (1-p)/(K-1)$ for $j \neq k$. The MAP (maximum a posteriori) state path maximises the joint log-density of path and data, $\ln \pi_{s_1} + \sum_{t \geq 2} \ln p_{s_{t-1} s_t} + \sum_t \ln f(u_t \mid s_t)$. The first two of those terms nearly collapse. A symmetric chain has a uniform stationary distribution, so $\ln \pi_{s_1} = -\ln K$ whichever state the path starts in. And the transition term takes only two values, so it can be written as a constant plus an indicator:

$$\ln p_{s_{t-1}s_t} = \ln p \;-\; \mathbb{1}[s_t \neq s_{t-1}]\,\ln\frac{p\,(K-1)}{1-p}.$$

Check both branches: with $s_t = s_{t-1}$ the indicator vanishes and this is $\ln p$; otherwise it is $\ln p - \ln p - \ln(K-1) + \ln(1-p) = \ln\frac{1-p}{K-1}$, as required. Sweeping every path-independent piece — the $-\ln K$, the $(T-1)\ln p$, and the per-observation Gaussian normaliser $-\frac{d}{2}\ln(2\pi\sigma^2)$ — into a constant leaves

$$\sum_t \ln f(u_t \mid s_t) + \sum_{t\geq 2} \ln p_{s_{t-1}s_t} = -\sum_t \frac{\|u_t - c_{s_t}\|^2}{2\sigma^2} - \ln\frac{p(K-1)}{1-p}\sum_{t\geq 2}\mathbb{1}[s_t \neq s_{t-1}] + \text{const.}$$

Multiplying by $2\sigma^2$ and flipping the sign turns this maximisation into the minimisation of §5.7, with

$$\boxed{\lambda = 2\sigma^2 \ln\frac{p\,(K-1)}{1-p}}$$

The two limits are the sanity check. $\lambda > 0$ exactly when $p > 1/K$ — that is, exactly when the chain is *sticky*, more likely to stay than an independent uniform draw would be. At $p = 1/K$ the states are i.i.d., $\lambda = 0$, and the objective is plain k-means: §5.8, flickering and all. As $p \to 1$, $\lambda \to \infty$ and no switch is ever worth paying for, leaving one state for the whole path.

So the statistical jump model is not an *alternative* to the HMM — it is the Viterbi decode of an HMM whose emissions are constrained to be isotropic with equal variance and whose transitions are constrained to be symmetric. The identity is exact for the path given the centroids and λ. The jump model also chooses the centroids by minimising the same objective over path and centroids together, whereas maximum likelihood for the HMM sums over paths, so the two fitted centroids can differ. That reframes its empirical advantage (§5.7): **the jump model wins not by escaping the probabilistic framework but by imposing restrictions on it**, replacing K^2 transition parameters estimated from a few dozen observed transitions with one penalty tuned on a validation objective. It is a bias-variance trade made in the right direction for the sample sizes available in finance. That is a much better reason to use it than "clustering is simpler than likelihood".

7.3 Same name, different thing

The inverse table, and the source of a great deal of talking past each other.

Term	Sense A	Sense B	Why it matters
Regime	macro state (inflationary, tightening); horizon: years	volatility state (calm, turbulent); horizon: weeks	Differ by two orders of magnitude in persistence. A model fitted for one cannot serve the other (§1.3)
Regime detection	"which of K states are we in now"	"did the process just change"	Different questions; §5.6 answers the second much better than §5.3 does
Clustering into regimes	partition of *time*	partition of *assets*	Both are k-means; only the first is a regime
Bull / bear market	a latent state	a $\pm 20\%$ drawdown rule	The drawdown rule is assigned retroactively and is a look-ahead by construction
Regime probability	$\xi_{t\|t-1}$, tradable	$\xi_{t\|T}$, historical	The two differ by the entire value of the research (§5.1)
Turbulence	squared Mahalanobis distance of the return vector	realised volatility	Correlated at 0.7–0.9 in most samples; test the increment before claiming novelty

Term	Sense A	Sense B	Why it matters
Regime-switching model	states switch parameters	states switch *the model itself* (gating)	Statistically similar, computationally and organisationally very different (§10.5)

7.4 Which knob actually matters

The payoff of the taxonomy. In descending order of effect on realised out-of-sample performance, judged by how much measured results move when each knob is changed:

1. **The information set (§5.1).** Dominates. The difference between rung 2 and rung 5 is routinely larger than the difference between the best and worst model in §5.12. **[Practice]**, but held with high confidence — it is arithmetic, not opinion, once you measure both.
2. **Whether variance switches.** Yes, always, in every published study and in the worked example of §6.11. This is the one thing regime models reliably find. **[Fact]**
3. **Whether the driver is observable.** An observable driver removes the filtering problem, the label-switching problem, and most of the estimation error. When one exists, use it.
4. **The persistence mechanism.** A jump penalty, a sticky prior or a run-length prior beats a freely estimated transition matrix at realistic sample sizes, for the parameter-counting reason in §7.2.
5. **Whether the mean switches.** Assume not (§8.1). If you must, demand out-of-sample evidence at the standard of §11, not a likelihood improvement.
6. **The number of states.** Matters less than every knob above it, which surprises people. Beyond $K = 2$ or 3, additional states almost always subdivide the volatility axis, and the marginal decision value is near zero. **[Practice]**
7. **The emission family.** Matters least for *where* the states sit — Gaussian and Student-t fits agree on the state volatilities to within two points on US data (§6.11). It matters a great deal for *persistence*, which places it inside knob 4: heavy tails stop single outliers from forcing switches, and halve the switching rate. Use Student-t emissions for that reason, not for the density.

§7 Key takeaways

1. The model zoo is a four-slot product space, not a taxonomy. The empty cells are combinations nobody has tried, and some of them are good.
2. The statistical jump model is *exactly* the MAP path of an HMM with isotropic equal-variance emissions and symmetric transitions, with $\lambda = 2\sigma^2 \ln[p(K-1)/(1-p)]$. Its advantage is parameter restriction, not escape from probability.
3. A Gaussian mixture is an HMM with no memory and therefore forecasts nothing beyond the unconditional distribution.
4. Every volatility-targeted strategy is already a regime-conditional strategy with a continuum of states. Many people who say they do not use regime models are using the most reliable one.
5. Long memory, stochastic volatility, and regime switching are mutually confusable. Do not treat a likelihood improvement as evidence for one over the others.
6. Ranked by effect on out-of-sample performance: information set $\gg$ variance switching $>$ observable driver $>$ persistence mechanism $>$ mean switching $>$ number of states $>$ emission family — except that a heavy-tailed emission is itself a persistence mechanism.
7. The HMM is the hub of the equivalence table. A mixture, a jump model, a semi-Markov model, a covariate-driven model and a Student-t emission are each an HMM with one restriction or one extension, so the HMM's diagnostics and failure modes carry over to all of them.

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8. What is known to work, and what is not

This is the section that should change what you build. Almost all of it reduces to one asymmetry, so that comes first.

8.1 The identification asymmetry

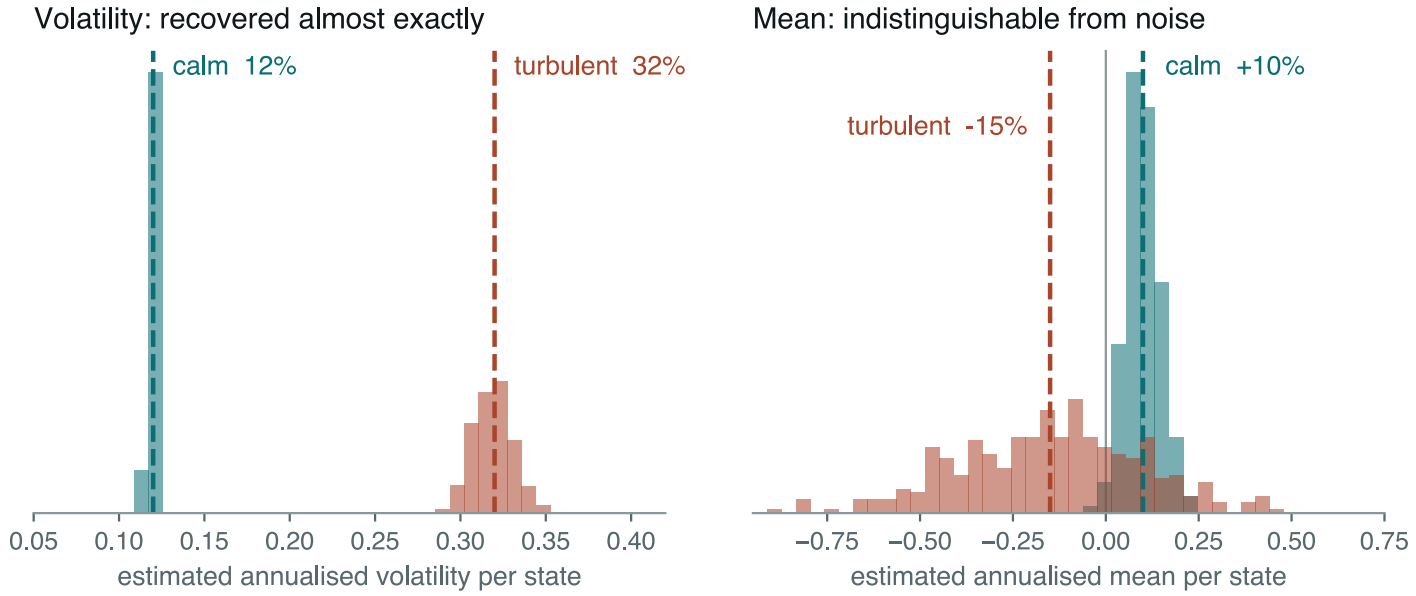
The claim. Regime models estimate state *variances* precisely and state *means* not at all, and this is a mathematical property of the estimation problem, not a defect of any particular implementation.

The mechanism, in three lines. Suppose you observe n i.i.d. draws from $\mathcal{N}(\mu, \sigma^2)$ over a fixed calendar span τ, at spacing Δ, so that $n = \tau/\Delta$. Here μ and σ are *per bar*, so the annualised quantities are $\mu_{\text{ann}} = \mu/\Delta$ and $\sigma_{\text{ann}} = \sigma/\sqrt{\Delta}$. Then

$$\text{sd}(\hat{\mu}_{\text{ann}}) = \underbrace{\frac{\sigma}{\sqrt{n}}}_{\text{per bar}} \cdot \underbrace{\frac{1}{\Delta}}_{\text{annualise}} = \frac{\sigma}{\Delta\sqrt{n}} \overset{\sigma = \sigma_{\text{ann}}\sqrt{\Delta}}{=} \frac{\sigma_{\text{ann}}}{\sqrt{n\Delta}} = \frac{\sigma_{\text{ann}}}{\sqrt{\tau}}, \qquad \frac{\text{sd}(\hat{\sigma}^2)}{\sigma^2} \approx \sqrt{\frac{2}{n}}.$$

Sampling faster shrinks Δ and raises n while leaving $\tau = n\Delta$ alone, which is the whole point: the standard error of the mean depends only on the *calendar span* τ — going from daily to five-minute data does not help it at all — while the *relative* error of the variance shrinks with the *number of observations*, so higher frequency helps it directly. This is Merton's (1980) observation, and it is the single most important fact in quantitative finance that is not taught first. Applied to regimes: a state occupied for two years' worth of days gives you thousands of observations to pin down σ_k and two years of calendar time to pin down μ_k.

The measurement. A simulation draws from a two-state Gaussian HMM with parameters chosen to be *generous* — a calm state at $+10\%$ drift and 12% volatility, a turbulent state at -15% drift and 32% volatility, expected durations of 100 and 25 days — and fitted a correctly specified $K = 2$ model by EM to 400 independent ten-year daily samples. The model knows the right number of states, the right emission family, and the right state dynamics. Nothing is misspecified.



400 simulations, 10 years of daily data, correctly specified $K = 2$ Gaussian HMM, true parameters known to exist.
Volatility ratio recovered with 3.7% relative error; the estimated mean spread has the WRONG SIGN in 16% of samples.

[Simulated] The numbers, with a 30-year arm added for scale. Both arms are printed by `figures/regime_identification.py`.

Quantity	True	10 years: mean (sd)	30 years: mean (sd)
Calm volatility	12.0%	12.0% (0.20 pp)	12.0% (0.11 pp)
Turbulent volatility	32.0%	31.9% (1.17 pp)	32.0% (0.65 pp)
Calm mean	+10.0%	+10.0% (4.6 pp)	+10.1% (2.5 pp)
Turbulent mean	−15.0%	−15.1% (**24.7 pp**)	−13.5% (**13.2 pp**)
Mean spread, calm − turbulent	+25.0 pp	+25.1 (sd 25.1), **wrong sign in 16.3% of samples**, s/n **1.00**	+23.6 (sd 13.4), wrong sign in 4.0%, s/n 1.76

Quantity	True	10 years: mean (sd)	30 years: mean (sd)
Volatility ratio, turbulent / calm	2.67	2.66 (sd 0.11), s/n **15.8**	2.67 (sd 0.06), s/n 28.5

Read the fifth row twice. **With ten years of daily data, a perfectly specified model, and a true mean spread of 25 percentage points a year, the estimated spread has the wrong sign one time in six.** Its signal-to-noise ratio is 1.00 — you have, in effect, a single noisy observation of it. The volatility ratio in the very same fit has a signal-to-noise ratio of 15.8.

That factor of sixteen is the whole story of this field. And tripling the sample to thirty years — more daily history than most instruments have — raises the mean spread's signal-to-noise ratio only to 1.76, because it improves as $\sqrt{\tau}$ and $\sqrt{3} = 1.73$. There is no amount of data you can realistically obtain that fixes this.

Why the two states differ so much in mean precision. The turbulent state holds 20% of the sample and has 2.67 times the volatility, so its mean standard error is larger by $2.67 \times \sqrt{0.8/0.2} = 5.3$; measured, it is 5.4. The arithmetic is boring and the consequence is not: **the state whose mean you most want to know — the bad one — is the state whose mean you can least estimate**, because it is rare and violent. This is structural and no amount of data engineering fixes it.

8.2 You cannot test how many regimes there are

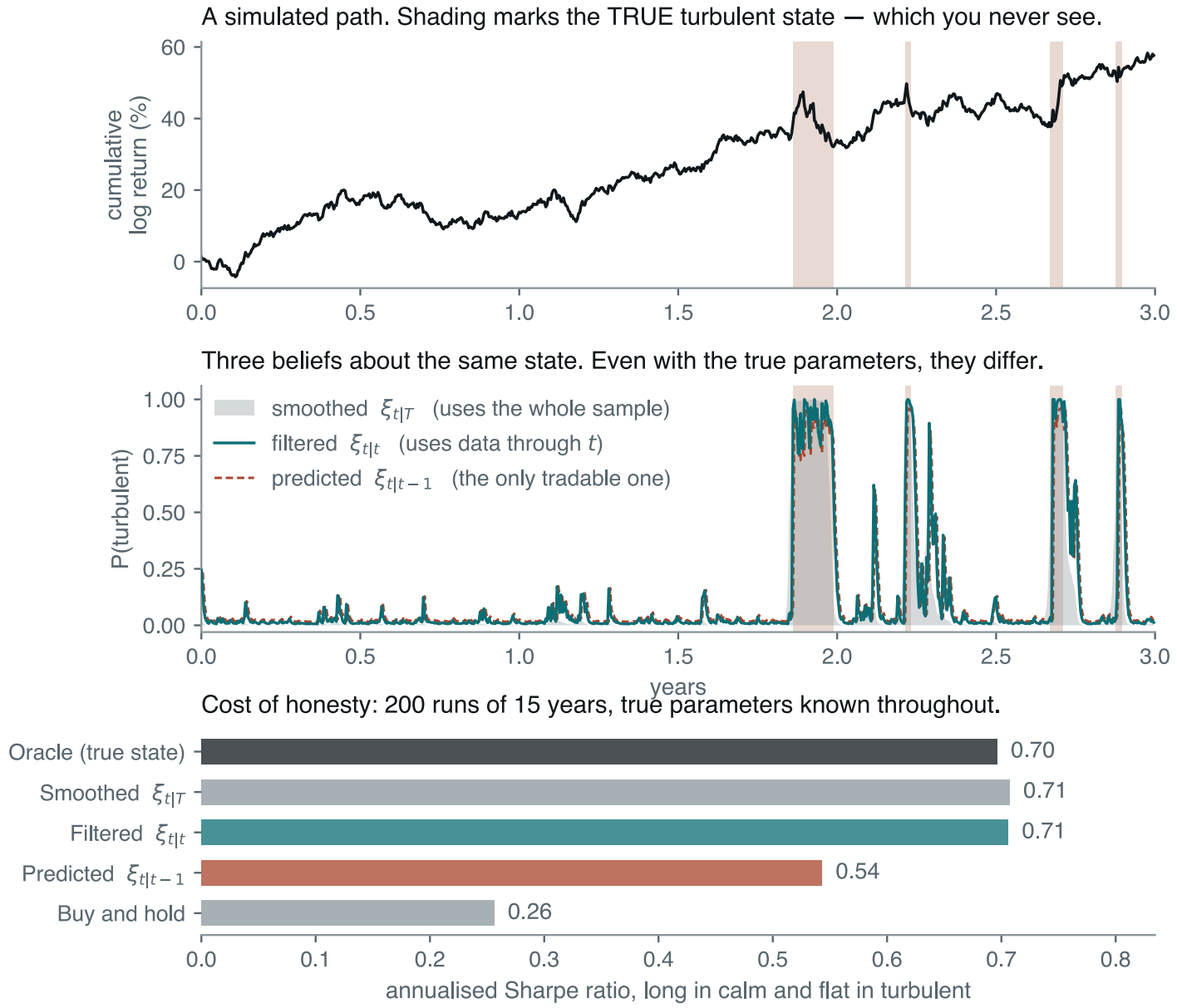
The likelihood-ratio statistic for K versus $K + 1$ regimes does not have a χ^2 null distribution. Two things go wrong simultaneously: under the null that state $K + 1$ does not exist, its parameters θ_{K+1} are completely unidentified (any value gives the same likelihood), and the transition probabilities into it are on the boundary of the parameter space. Standard asymptotics require neither condition. Hansen (1992) developed a valid bound using empirical-process theory and applied it to Hamilton's own GNP model; **the switching specification did not reject a plain AR(4). [Fact]**

The practical fallout:

- **A likelihood improvement means nothing on its own.** Adding a state always raises the likelihood; the question is by how much relative to a correct null, and you generally cannot compute that.
- **AIC and BIC over-select.** They assume the regularity conditions that fail here. Psaradakis and Spagnolo (2003) document the behaviour.
- **Parametric bootstrap is the practical route** — simulate from the fitted K-state model, refit K and $K + 1$, build the null distribution of the LR statistic empirically. It is expensive, it is correct, and almost nobody does it. If you are going to publish a claim about K, do this.
- **Or make K a decision, not an inference** (§6.4). This is the recommended route, and it is honest rather than a dodge: K is a modelling choice (§1.1), so choose it for the decision it serves and validate the whole pipeline out of sample.

8.3 What the filtration ladder costs

Using the same simulation with the **true parameters known** — so that this measures only the information-set effect and not estimation error — here is what each rung of §5.1 is worth. The strategy is deliberately crude: long when the predicted state is calm, flat when it is turbulent.



[Simulated] Three findings, one of them unexpected:

1. **Smoothing costs less than expected — when the states are far apart.** Smoothed and filtered give the same Sharpe to two decimals (0.71 versus 0.71). The reason is instructive: this regime is defined by a 2.67× volatility difference, and a single large return is nearly conclusive evidence, so the filter identifies the state within a day or two of a transition — **median detection lag 1.7 days**, against an expected turbulent spell of 25 days. Hindsight adds little when the present is already informative.
2. **Predicting costs a lot.** Going from $\xi_{t|t}$ to $\xi_{t|t-1}$ — one application of the transition matrix, the difference between "I know today was turbulent" and "I must decide before seeing today" — drops the Sharpe from 0.71 to 0.54, about **a quarter of the edge**. This is the honest cost of tradability, it cannot be engineered away, and it is routinely omitted.
3. **The oracle is not above the filter.** 0.70 for perfect knowledge of the true state versus 0.71 filtered — within Monte Carlo error of each other. When separation is strong, inference is nearly free; the binding constraint is the one-step-ahead prediction, not the filtering.

A fourth finding, on parameter leakage. Repeat the exercise with the parameters *estimated* rather than handed over, comparing a full-sample EM fit against an honest expanding-window refit, both using predicted probabilities:

[Simulated] 30 runs of 20 years, states identified by fitted volatility inside each fit.

Calibration	Buy & hold	Full sample, smoothed	Full sample, predicted	Expanding window, predicted
Volatility and mean	0.20	0.66	0.54	**0.53**
Mean only	0.50	0.02	−0.02	**−0.00**

Two readings. First, in the top row **parameter leakage costs essentially nothing**: 0.54 with the whole sample in hand against 0.53 without, standard errors around 0.03. The reason is §8.1 again — the parameters that drive this strategy are the state volatilities, and those are pinned down within a couple of years, so knowing the rest of the sample adds nothing. Parameter leakage matters where the fitted parameters are poorly determined, which means it matters most in exactly the specifications that were never going to work.

Second, the bottom row is the whole cautionary tale in one line. **A pipeline applied to a mean-only regime turns a 0.50 Sharpe buy-and-hold into zero**, and the full-sample smoothed version — the one every paper plots — does no better. The mechanism is label switching (§6.1). Identifying states by fitted volatility is the correct convention, but when the states do not differ in volatility the sort order is arbitrary, so “state 1” means something different in each refit and the position series is close to a coin flip. A regime pipeline can be much worse than no pipeline, and this is how.

The generalisation, which matters more than any of the four: **the smoothed- versus-filtered gap grows as state separation shrinks.** Strong separation makes smoothing unnecessary; weak separation makes it enormous and makes the whole exercise futile anyway. The central table follows in §8.4.

8.4 When does knowing the regime help at all?

Same simulation machinery, true parameters known throughout, 200 runs of 15 years, four calibrations that differ in *what separates the states*.

[Simulated]

States separated by	Buy & hold	Oracle	Smoothed	Filtered	**Predicted**	Accuracy †	Lag ‡
Volatility **and** mean (12/32% vol, +10/−15% mean)	0.26	0.70	0.71	0.71	**0.54**	0.932	1.7
Volatility only (12/32% vol, +5/+5% mean)	0.26	0.33	0.33	0.33	**0.29**	0.932	1.7
Mild volatility (15/22% vol, +12/−10% mean)	0.42	0.67	0.69	0.70	**0.50**	0.852	7.7
Mean only (20/20% vol, +20/−20% mean)	0.55	0.84	0.61	0.56	**0.55**	0.794	25.4

† Accuracy of the *predicted* state label. The turbulent state occupies 20% of the sample in every row, so a classifier that always says "calm" scores 0.800. ‡ Median days from a true transition into the turbulent state until the filtered probability first exceeds one half. The expected turbulent spell is 25 days.

Four readings, in order of importance:

(a) When the states differ only in volatility, knowing the state perfectly is worth almost nothing to a directional strategy. Row 2: the oracle earns 0.33 against a buy-and-hold 0.26, and the tradable version earns 0.29. You have perfect foresight of a 2.67$\times$ volatility regime and it buys you three hundredths of a Sharpe ratio. The entire gain is the mechanical one of avoiding high-variance periods that have the same mean — a *sizing* effect, not a timing effect, and §12.1 shows sizing captures it more cheaply.

(b) When the states differ only in the mean, you cannot find them. Row 4: the oracle earns 0.84, a large edge, and the tradable version earns 0.55 — exactly buy-and-hold. The predicted-state classifier scores 0.794 against a base rate of 0.800, so **it is worse than a constant that always says "calm."** Its median detection lag is 25.4 days against a 25-day expected spell: by the time the filter notices, the regime is over. Even the smoothed version, with the entire sample in hand, only reaches 0.61. A 40-percentage-point annual mean difference is invisible in daily returns because it amounts to 0.16% per day against a 1.26% daily standard deviation.

(c) Rows (a) and (b) together are the trap. The states you can detect are the ones that do not pay, and the states that would pay are the ones you cannot detect. Every real regime strategy lives in row 1 or row 3, where the mean difference *rides along with* the volatility difference and you detect the state through volatility while being paid through the mean. That is a real effect — Daniel and Moskowitz's (2016) momentum crashes are exactly it — but note that your edge is then entirely dependent on the mean-volatility association continuing to hold, which is the thing §8.1 says you cannot measure.

(d) The realistic case is row 3, and it is worth 0.08 Sharpe. A 15% versus 22% volatility split with a mild mean difference is roughly what equity index regimes actually look like. Perfect foresight of the state is worth 0.25 Sharpe over buy-and-hold; the tradable version is worth 0.08 — before costs, with the true parameters handed to you, in a correctly specified simulation, with the right K. Note also the detection lag of 7.7 days against a 25-day spell: you spend the first third of every regime not knowing you are in it. **Whatever you build will do worse than 0.08.** That is the number to hold in your head when someone shows you a regime backtest adding 0.5 Sharpe.

8.5 The out-of-sample forecasting record

[Fact] Regime-switching models fit in-sample beautifully and forecast badly. This has been the finding since Dacco and Satchell (1999) and has not been overturned. Their diagnosis is the part worth carrying: the loss from misclassifying the state exceeds the gain from having the right parameters conditional on classifying correctly, so a model can have every parameter right and still lose to a constant forecast.

Formally, if the model predicts $\hat{\mu}_{s_t}$ and misclassifies with probability q, the mean squared error picks up a term of order $q(1-q)(\mu_1 - \mu_2)^2$. Since $(\mu_1 - \mu_2)^2$ is exactly what you were hoping to exploit, the misclassification penalty scales with the size of the opportunity: **the bigger the regime effect you are trying to capture, the more a classification error costs you.** A model that gets the state right 80% of the time when the base rate is 80% (row 4 above) has $q \approx 0.2$ and captures none of the mean spread while paying the full variance penalty.

What *has* replicated:

- **Volatility forecasting improves.** Regime models beat single-regime GARCH on volatility, mostly by fixing the spurious near-integration (§5.4). **[Fact]** Though HAR and realised-volatility models beat both. **[Contested]**
- **Density and tail forecasts improve** over a constant-parameter model. If you need the shape of the conditional distribution — for options, for VaR, for a utility-based allocation — the mixture genuinely helps. **[Fact]** Against a continuous volatility model with heavy-tailed innovations it does not: on US equities the EWMA with Student-t innovations wins the density forecast (§6.11).
- **Correlation and co-movement structure improves.** Ang and Bekaert (2002), Longin and Solnik (2001). The bad state has higher correlations, and modelling that changes optimal portfolios materially. **[Fact]**, with the Forbes-Rigobon caveat that part of the effect is a volatility artefact.
- **Drawdown reduction is real; return improvement is not.** Bulla et al. (2011) find volatility down 41% on average with modest excess returns, across three markets and forty years, after costs. That is the honest summary of the entire applied literature. **[Fact]**

What has not replicated: regime-conditional expected-return prediction as a standalone source of alpha. **[Fact]**

8.6 The scorecard

This chapter's assessment of each claimed use, with the evidence status.

Use	Verdict	Status
Forecast volatility level shifts	**Works.** Large, persistent, estimable	[Fact]
Forecast correlation regime shifts	**Works**, partly artefactual	[Fact] / [Contested]
Improve density and tail forecasts	**Works**	[Fact]
Reduce drawdown and realised volatility	**Works** — this is the main benefit	[Fact]
Decide *when to retrain* a model	**Works**, and is underused	[Practice]
Size positions (risk targeting)	**Works** — and a continuous vol model does about as well (§6.11)	[Fact]
Time direction from a latent state	**Does not work** standalone	[Fact]
Predict regime *transitions* ahead of time	**Does not work**	[Fact]
Select among strategies by regime	**Contested** — plausible, and the evidence is mostly in-sample	[Contested]
Improve a forecasting model's conditional mean	**Contested** — depends entirely on §9	[Contested]
Explain history	Works, and is worth doing for its own sake	—

8.7 Should you use a discrete model at all?

The comparison nobody runs. Before building a regime model, fit the two obvious continuous alternatives and see whether the discrete one adds anything:

1. **EWMA or GARCH volatility**, with the position scaled by $1/\hat{\sigma}_t$.
2. **A continuous conditioning variable** — realised volatility, the VIX, a turbulence index — entered directly as a regressor or feature.

The evidence, including the worked example of §6.11, reads: **for volatility, the continuous model wins; for decisions that are genuinely discrete, the regime model wins.** The second half is the case for regime models and it is narrower than usually presented. Genuinely discrete decisions exist — turn a strategy on or off, switch a hedge, change a rebalancing frequency, escalate to a human — and for those a calibrated state probability is the right object because the decision cannot consume a continuous number anyway.

The corollary, which is the most useful sentence in this section: **if your downstream consumer can take a continuous input, give it one.** Discretising the world and then feeding the discretisation to a downstream model that would happily have taken the underlying continuous variable is a lossy transformation performed for no reason. §10.1 says what to feed instead.

§8 Key takeaways

1. The identification asymmetry is arithmetic, not a defect: the standard error of a mean depends on calendar span, the standard error of a variance on the number of observations. Regime models therefore find volatility states and not return states.
2. With ten years of daily data and a perfectly specified model, the estimated mean spread between two states with a true 25-point annual gap has the wrong sign in one sample out of six. The volatility ratio in the same fit has a signal-to-noise ratio of 16.
3. You cannot test the number of regimes with a standard likelihood-ratio test. Use a parametric bootstrap, or choose K from the decision and validate the pipeline out of sample.
4. Smoothing is a small cheat when states are well separated and a large one when they are not — but the step from filtered to predicted costs about a quarter of the Sharpe ratio even in the easy case, and that cost is real.
5. If the states differ only in volatility, perfect knowledge of the state is worth about 0.03 Sharpe to a directional strategy. If they differ only in the mean, the filter is worse than a constant guess and detects the regime only after it has ended. Real strategies live in the middle, where the mean difference rides along with the volatility difference — and there the tradable edge is roughly 0.08 Sharpe under ideal conditions.
6. Misclassification cost scales with the square of the mean spread you are trying to exploit, which is why bigger apparent opportunities are not easier.
7. The replicated benefits are volatility, correlation, density, and drawdown. The unreplicated one is directional timing. Design accordingly.
8. If the downstream consumer can take a continuous input, do not discretise.

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9. Which shift do you have?

Part I was about estimating a regime. Part II is about using one. A regime estimate has three possible consumers: a position rule, a risk model, and a forecasting model that predicts returns or some other target. §6.10 showed an HMM driving the first two on its own, and §12 returns to them. This section and the next are about the third, and they apply to any forecasting model: a linear regression, a regularised regression with interactions, a tree ensemble, or a network. §10.8 collects the few points that depend on the model family.

The section opens with the question that determines everything downstream and that almost nobody asks: **which distribution is actually changing?** The answer picks the integration scheme. Getting it wrong means doing work that cannot help — most commonly, adding a regime feature to address a problem that a feature cannot address.

9.1 The three shifts, and what each demands

Write the joint distribution of features and target as $p(x, y) = p(x)\,p(y \mid x)$. There are three ways it can change over time, and they are not interchangeable.

Covariate shift. $p(x)$ changes; $p(y \mid x)$ is unchanged. The market moves into a corner of feature space you have seen little of — volatility is at levels last visited in 2008, spreads are wider than your training data — but the *relationship* between features and returns is what it always was.

Conditional-scale drift. $\mathbb{E}[y \mid x]$ is unchanged; $\mathrm{Var}(y \mid x)$ changes. Your signal predicts the same thing; the noise around it is three times larger. This is technically a species of the next category, but it demands a completely different remedy and is by far the most common shift in financial data, so it gets its own name here.

Concept drift. $\mathbb{E}[y \mid x]$ changes. (In the machine-learning literature the term covers any change in $p(y \mid x)$; here it is reserved for the mean.) Momentum predicts positive returns in calm markets and negative ones after a crash. The function you are trying to learn is genuinely different in different regimes. This is the one everybody means when they say "regime", and it is the rarest of the three.

The remedies do not overlap:

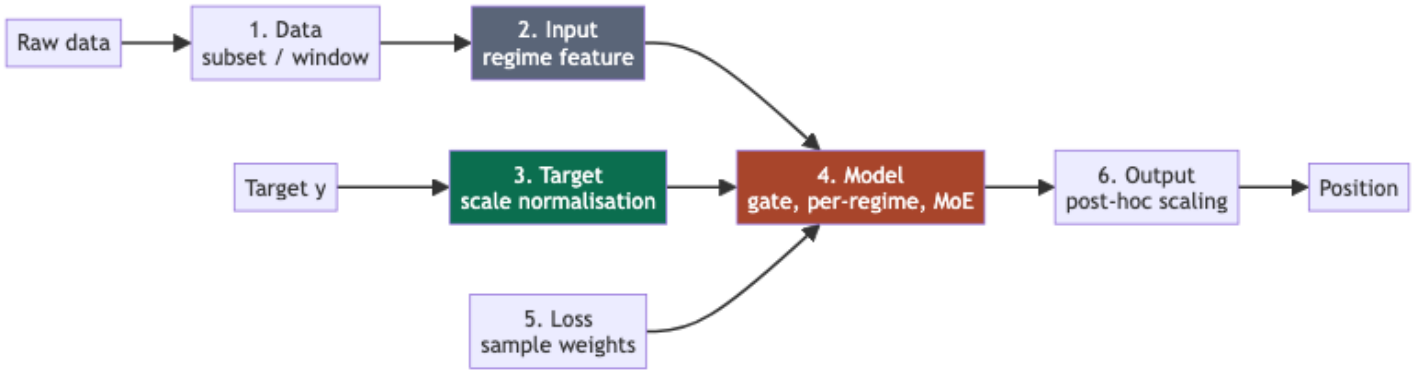
Shift	What changes	Right response	Common wrong response
Covariate	$p(x)$	Usually **nothing**. Ensure feature coverage; consider more capacity	Importance weighting — adds variance for no bias reduction
Conditional-scale	$\mathrm{Var}(y \mid x)$	Normalise the target by conditional scale, **or** weight by $1/\hat{\sigma}^2$ (they are different — §9.4)	Add a volatility feature and hope the model figures it out
Concept	$\mathbb{E}[y \mid x]$	Condition: interactions, gating, separate models, or a shorter window	Add a regime dummy (§9.3)

The covariate-shift row is the counterintuitive one. If $p(y \mid x)$ is genuinely unchanged and your model class contains the truth, covariate shift costs you nothing asymptotically: the model learns the same function whatever the input distribution. Shimodaira's (2000) result is precisely that importance weighting helps *only* under misspecification or in finite samples, and it always costs variance — the effective sample size falls to $\left(\sum_i w_i\right)^2 / \sum_i w_i^2$, which for realistic financial weights is often less than half the nominal sample. **[Fact]**

Since every model you fit is misspecified, the honest version is: weighting under covariate shift trades a small bias reduction for a large variance increase, and in the sample sizes available in finance the trade is usually bad. Diagnose the shift before reaching for the remedy.

9.2 Six places regime information can enter

Regime information can enter a forecasting model at exactly six points. They are not substitutes.



#	Channel	Addresses	Cost	Recommended order
3	**Target normalisation**	conditional-scale drift	near zero	**do this first, always**
6	**Output scaling** (position sizing)	conditional-scale drift	near zero	do this second
2	**Input feature**	concept drift, weakly	near zero	cheap, usually ineffective (§9.3)
5	**Sample weights**	covariate shift, scale drift	variance	situational
1	**Data subsetting / window**	concept drift, strongly	large variance cost	only with strong evidence

#	Channel	Addresses	Cost	Recommended order
4	**Model structure** (gating, MoE)	concept drift, strongly	complexity, overfitting	last, and only if subsetting (channel 1) works

The ordering is the practical content of this whole document's second half. The two channels that reliably pay are the two that address *scale*, which is the thing you can actually measure (§8.1). The channels that address the conditional *mean* are ranked last because the evidence that the conditional mean changes is weak and the cost of acting on a false positive is high.

9.3 Why adding a regime dummy usually does nothing

This deserves a mechanism, not just an observation, because the observation is universal and everyone re-derives the surprise independently.

Suppose you fit a flexible forecasting model — a regression with interactions, a tree ensemble, a network — with features x, and add a regime indicator $\widehat{s}_t$. Four things work against it:

(a) It is a coarsening of a feature you already have. Your regime label was derived from volatility, or from something highly correlated with volatility. If realised volatility is already in x — and it should be — the label carries almost no incremental information. Worse, a flexible learner actively prefers the continuous parent. The binary label is one particular threshold on that same variable, and the learner can choose a better function of it: a tree places its own threshold, a spline or a network fits a smooth curve. **The predictable consequence is that your regime feature shows near-zero importance while volatility shows high importance, and this is the correct behaviour, not a bug.**

(b) A mean-function feature cannot express a variance change. If the regime changes only $\mathrm{Var}(y \mid x)$, then by construction $\mathbb{E}[y \mid x, s] = \mathbb{E}[y \mid x]$ and the optimal predictor ignores s entirely. Nothing you do at the input can fix a problem that lives in the loss. This is the single most common category error in applied regime work.

(c) The interaction may be unlearnable. For the model to use the regime it must build an interaction — a different slope on a signal in each state — and whatever the learner, that interaction is estimated from the rare-regime observations alone. In a regression it is one coefficient per signal fitted on the turbulent subsample; in a tree it needs a split on the state followed by splits on the signal inside each branch, with enough samples in the rare-regime leaves. With 20% of observations in the turbulent state, an interaction of any depth inside that state has an effective sample of at most a few hundred observations per leaf or coefficient — which are, additionally, serially correlated, so the effective number of independent observations is far smaller still.

(d) The label is one bit. A hard $K = 2$ label carries at most one bit per observation, and a posterior probability not much more. Against a target with a signal-to-noise ratio around 0.05, one bit is not much to work with.

What actually helps, in descending order:

1. **Give the model the continuous variable the label was made from.** Realised volatility, the turbulence index, the term-structure slope. Continuous, monotone, no identification problem, no look-ahead if computed on a trailing window.
2. **Give it the posterior, not the label**, if you insist on the state ($\xi^{(k)}_{t|t-1}$ as a real number in $[0, 1]$).
3. **Build the interaction explicitly** as a feature: $x_j \cdot \xi^{(k)}$, rather than hoping the model discovers it. This is what Gu, Kelly and Xiu (2020) do with their macro predictors, and it is why their approach works without any latent-state machinery.

9.4 Normalisation is not weighting

Two ways to handle conditional-scale drift look interchangeable and encode different beliefs about the world. Getting this wrong silently changes what you are estimating.

Model A — constant expected return. The edge is a fixed number of basis points regardless of the volatility environment:

$$y_t = m(x_t) + \sigma_t \varepsilon_t.$$

Here $\mathbb{E}[y \mid x]$ does not depend on σ_t. The efficient estimator is **weighted least squares with** $w_t = 1/\sigma_t^2$ — downweight noisy periods. Normalising the target would be *wrong*: it would make the target function $m(x)/\sigma_t$, which depends on σ_t, so you would have created concept drift where there was none.

Model B — constant Sharpe. The edge scales with volatility, so the information ratio is what is stable:

$$y_t = \sigma_t\, m(x_t) + \sigma_t \varepsilon_t \qquad \Longleftrightarrow \qquad \frac{y_t}{\sigma_t} = m(x_t) + \varepsilon_t.$$

Here **normalising the target is exactly right** — it recovers a homoskedastic problem with a stable target function — and weighting is wrong, because after normalisation the noise is already homoskedastic and weighting would distort it.

These prescribe opposite actions, so you have to decide. **The diagnostic:** fit one model $\widehat{m}$ on all the data, bucket the held-out observations by $\widehat{\sigma}_t$, and regress realised y on $\widehat{m}(x)$ within each bucket. Under Model A the slope is the same in every bucket; under Model B it rises in proportion to σ. (Fitting a separate model per bucket would give a slope of one in both cases, because each fit absorbs its own bucket's scale.) Run it on your own data — the answer differs by signal and by asset class, and the published evidence is genuinely divided (Moreira and Muir, 2017, argue risk and return are *not* proportional, which favours Model A for the market factor; Cederburg et al., 2020, dispute the out-of-sample implementability). **[Contested]**

Regardless of which model holds, normalise anyway, for a reason independent of both. An unweighted squared loss on raw returns implicitly weights each observation by its conditional variance. Take the calibration from §8: 80% of days at 12% volatility, 20% at 32%. The share of total squared error contributed by the turbulent days is

$$\frac{0.2 \times 0.32^2}{0.8 \times 0.12^2 + 0.2 \times 0.32^2} = \frac{0.0205}{0.0320} = 64\%.$$

Twenty percent of your sample supplies sixty-four percent of the signal your loss function responds to. Your model is fitted predominantly to crises whether you intended that or not, and no hyperparameter search will reveal this to you because every configuration has the same problem. Normalising the target by a trailing volatility estimate — $\tilde{y}_t = y_t / \widehat{\sigma}_t$ — removes the implicit reweighting and is, in practice, the single highest-value change available in a financial forecasting pipeline. **[Practice]**, held strongly.

9.5 Pooling: the identification asymmetry, restated as a decision rule

Here is where §8.1 pays off in a form you can act on.

Suppose the true parameter in regime k is $\beta_k = \beta + \delta_k$ with $\mathrm{Var}(\delta_k) = \tau^2$ — τ^2 measures how much regimes genuinely differ — and your per-regime estimate has sampling variance $v = \sigma^2 / n_k$. The optimal shrinkage of the per-regime estimate toward the pooled one is the standard hierarchical weight

$$\widehat{\beta}_k^{\text{opt}} = w\, \widehat{\beta}_k^{\text{(regime)}} + (1 - w)\, \widehat{\beta}^{\text{(pooled)}}, \qquad w = \frac{\tau^2}{\tau^2 + v}.$$

Fitting separate models per regime is $w = 1$. Ignoring regimes entirely is $w = 0$. Adding a regime feature to a regularised model is somewhere in between, with w set implicitly by the regularisation. **The right value of w is determined by the ratio of genuine between-regime variation to estimation noise** — and §8.1 tells you that ratio directly:

- **For means**, τ^2 is no larger than v for the rare, volatile state whose mean matters most (even in the generous simulation of §8.1 the mean spread has a signal-to-noise ratio of 1 in a decade of data, and real spreads are smaller). So w is small: **pool. Do not fit separate mean models per regime.**
- **For variances and covariances**, τ^2 is large relative to v (the volatility ratio has a signal-to-noise ratio near 16). So $w \approx 1$: **subset. Estimate the risk model per regime.**

That is a clean, actionable, and slightly surprising conclusion, and it explains a pattern that otherwise looks like inconsistency in practitioner behaviour: the firms that use regimes successfully use them in the risk model and not in the alpha model. They are not being timid. They are doing the arithmetic.

9.6 The inverse framing: invariance instead of adaptation

Every scheme so far adapts the model to the regime. There is an opposite strategy, and it is underused.

Treat each regime as an **environment** and look for the predictor whose conditional distribution $p(y \mid x_S)$ is the *same* in every environment. Peters, Bühlmann and Meinshausen's (2016) invariant causal prediction formalises this: under assumptions, the features that appear in every invariant set are, with a stated confidence, direct causes (a conservative subset of them,

since a cause can go undetected), and a model built on them generalises to environments you have never seen — including future regimes that do not resemble any past one.

The practical version needs none of the causal machinery: fit the model separately in each regime, keep the features whose effect has a consistent sign and comparable magnitude in all of them, and refit on pooled data with only those. §10.6 gives the procedure and its failure modes.

This uses the regime decomposition as a *robustness filter* rather than as a conditioning variable, which is a far better match to what regime estimates can actually support. It is also the honest response to §8's scorecard: you cannot reliably predict which regime you will be in, so prefer a model that does not need to know. The caveat is the usual one for induction: a feature that is stable across the regimes *in your sample* may not be stable across the next one. **[Practice]**

§9 Key takeaways

1. Diagnose the shift before choosing the remedy. Covariate shift, conditional- scale drift, and concept drift demand different and non-overlapping actions.
2. Pure covariate shift usually needs no response at all. Importance weighting reduces effective sample size for a bias correction you often do not need.
3. A regime feature cannot fix a variance problem, because the variance does not live in the mean function. This is the most common category error in the field.
4. Your regime feature will show near-zero importance next to the continuous variable it was derived from. That is correct behaviour: a flexible learner can pick a better threshold, or a smooth curve, than you did.
5. Without target normalisation, twenty percent of your days supply sixty-four percent of the squared error your loss responds to. You are fitting crises whether you meant to or not.
6. Normalising the target and weighting by inverse variance are different models — constant Sharpe versus constant expected return. Test which one your data supports; the answer varies by signal.
7. The identification asymmetry becomes a pooling rule: pool the mean model across regimes, subset the risk model by regime. Firms that use regimes successfully do exactly this.
8. Consider inverting the problem: use regimes as environments to find the features that are *invariant*, and build a model that does not need to know the regime at all.

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10. Conditioning a forecasting model on the regime

Seven schemes, each with the same fields: what it does, how you build it, which shift it addresses, what it costs, how it fails, and a verdict. The first six apply to any forecasting model. The seventh collects the variants that only a network can implement. §10.8 lists what changes with the model family, and §10.9 ranks everything.

Throughout, $\widehat{\xi}_t$ is the regime posterior available when the forecast is made at the end of bar t for a target over $(t, t+h]$: the filtered $\xi_{t|t}$, or equivalently the predicted belief $\xi_{t+1|t}$ for the first bar of the target, computed with expanding-window parameters (§5.1). Anything that uses data after t, in the state belief or in the parameters, is a leak, and this section does not repeat the warning.

10.1 Scheme 1 — regime as a feature

What it does. Adds regime information to x_t and lets the learner use it.

How. Four encodings, in increasing order of usefulness:

1. **Hard label.** $\widehat{s}_t = \arg\max_k \widehat{\xi}_t^{(k)}$, integer-coded (use native categorical support where the library has it) or one-hot. Throws away the posterior.
2. **Posterior.** $\widehat{\xi}_t^{(1)}, \ldots, \widehat{\xi}_t^{(K-1)}$ as floats. Strictly more informative; costs nothing.
3. **The continuous parent.** The variable your regime was derived from — trailing realised volatility, the turbulence index, the VIX term-structure slope. Usually dominates both of the above (§9.3).
4. **Explicit interactions.** $x_j \cdot \widehat{\xi}_t^{(k)}$ for the signals x_j you actually believe are regime-dependent. This is what Gu, Kelly and Xiu (2020) do with macro predictors, and it is why their models capture conditional structure without any latent-state machinery.

Two features that are more useful than the state itself and are almost never included:

- **Time since the last transition**, $t - \max\{u \le t : \hat{s}_u \neq \hat{s}_{u-1}\}$. This carries the duration information that the geometric-sojourn assumption (§6.5) throws away, it is continuous, and it lets the model learn that early and late in a regime behave differently. **[Practice]**
- **Posterior entropy**, $-\sum_k \hat{\xi}_t^{(k)} \ln \hat{\xi}_t^{(k)}$. A direct measure of "the model does not know what is going on", which is exactly the condition under which you want to reduce risk. Feeds naturally into §12.1.

Addresses. Concept drift, weakly.

Cost. Negligible. One to three columns.

Failure modes. (i) Near-zero feature importance next to the continuous parent — expected, not a bug (§9.3). (ii) Look-ahead through the state estimate, which is the default unless you built rung 5. (iii) The label's *meaning* changes between refits (label switching), so the same column encodes different things at different times — this silently poisons the model and is invisible in every standard diagnostic.

Verdict. Cheap, do it, expect little. Include the posterior and the time-since-transition; skip the hard label.

10.2 Scheme 2 — regime-conditional target normalisation

What it does. Divides the target by a conditional scale estimate, so the loss weights every period equally.

How.

$$\tilde{y}_t = \frac{y_t}{\hat{\sigma}_t}, \qquad \hat{\sigma}_t^2 = \text{EWMA of } r^2 \text{ through } t, \quad \text{or} \quad \hat{\sigma}_t^2 = \sum_k \hat{\xi}_t^{(k)} \sigma_k^2,$$

the second being the regime model's own predictive variance. Train on $\tilde{y}$. The model now forecasts a **Sharpe ratio**, not a return; to size a position from it, divide by $\hat{\sigma}_t$ again — the normalisation appears twice, once in training and once in sizing, and both are correct.

Addresses. Conditional-scale drift. Directly and completely.

Cost. One column and one division. This is the cheapest high-value change in the whole document.

Failure modes. Exactly one, and it is fatal: $\hat{\sigma}_t$ **must be** $\mathcal{F}_t$**-measurable.** Normalising by realised volatility computed over the label window $[t, t+h]$ — which is the natural thing to write and looks identical in code — leaks the answer, because the denominator knows how volatile the period you are predicting turned out to be. This produces spectacular backtests. Use a trailing estimator and lag it explicitly.

Verdict. Do this first, before anything else in this section. If you take one thing from Part II, take this.

10.3 Scheme 3 — sample weighting and window selection

What it does. Changes how much each historical observation counts, so the fit reflects conditions resembling now.

How. Three variants:

- **Exponential decay.** $w_t = \lambda^{T-t}$. Choose the half-life, not λ; a half-life of 3–5 years is typical for daily equity data. **[Practice]**
- **Regime-similarity weighting.** $w_t = \hat{\xi}_t^\top \hat{\xi}_T$ — weight each historical observation by how much its regime posterior resembles today's. This is a *soft* version of subsetting (§10.4) and is strictly better behaved, because it degrades continuously instead of discarding 80% of the sample at a threshold. **[Practice]**
- **Window selection.** Pesaran and Timmermann (2007) give the formal treatment: the optimal window trades the bias from including pre-break data against the variance from a short post-break sample, and — the useful surprise — **including some pre-break data is often optimal.** The instinct to throw away everything before the last regime change is usually wrong.

Addresses. Covariate shift (weakly — see §9.1), conditional-scale drift, and concept drift under a slow-drift model.

Cost. Variance. Always report the effective sample size

$$n_{\text{eff}} = \frac{\left(\sum_t w_t\right)^2}{\sum_t w_t^2},$$

and treat n_{eff}, not T, as your sample size in every subsequent calculation. A 5-year half-life on 20 years of daily data gives $n_{\text{eff}} \approx 3,200$ out of 5,040 — you have thrown away over a third of your data's information, which may be the right call but should be a decision rather than a side effect.

Failure modes. (i) n_{eff} collapse, unnoticed. (ii) Tuning the half-life on the evaluation period. (iii) In tree models, weights change the *split criterion* as well as the leaf values, so the effect is larger and less predictable than in a linear model.

Verdict. Regime-similarity weighting is underused and worth trying. Exponential decay is a reasonable default. Hard window truncation is usually worse than both.

10.4 Scheme 4 — subsetting and per-regime models

What it does. Fits a separate model in each regime.

How. Partition the training data by $\hat{s}_t$ and fit K models. Or — much better — fit it as **explicit partial pooling**:

1. Fit a pooled model f_0 on all data.
2. For each regime k, fit a *correction* f_k on the residuals $y_t - f_0(x_t)$ restricted to regime k, with strong regularisation.
3. Predict $f_0(x) + \sum_k \hat{\xi}^{(k)} f_k(x)$.

Every model family has a cheap version. In a linear model, add regime-by-signal interaction terms and put a ridge penalty on them alone, so the per-regime slopes are shrunk toward the pooled slope. In boosting, train the pooled model, then continue boosting per regime from the pooled model's predictions as the starting offset (`init_score` in LightGBM, `base_margin` in XGBoost) with a small tree budget and strong regularisation. In a network, the same structure is a shared trunk with small per-regime heads and weight decay on the heads. Each gives hierarchical shrinkage by construction, and the shrinkage weight w of §9.5 is set by the correction's regularisation, where you can see it and tune it.

Addresses. Concept drift, strongly.

Cost. The variance cost is severe and is the whole story. With K regimes, per-regime sample sizes fall by roughly K (worse, since regimes are unequal), and the rare regime — the one you care about — gets the smallest sample. §9.5 is the calculation: this is only worth it when between-regime variation genuinely exceeds estimation noise.

Failure modes. (i) The rare-regime model is fitted on a few hundred autocorrelated observations and is noise. (ii) The regime assignment used for partitioning is itself a fitted object with error, so you are subsetting on a noisy label — errors compound. (iii) At inference the regime is uncertain and you must pick a model; hard selection at a boundary produces large discontinuous prediction changes.

Verdict. For risk models, yes — variances and correlations differ enough between regimes to justify it (§9.5). **For alpha models, no**, unless you have out-of-sample evidence at the standard of §11. If you do it, do it as partial pooling, never as hard subsetting.

10.5 Scheme 5 — gating and mixture of experts

What it does. Runs several models and blends their predictions by a regime-dependent gate. The generalisation of §10.4 to soft assignment.

How. $\hat{y} = \sum_k g_k(z) f_k(x)$ with $\sum_k g_k = 1$. The gate g can be the regime posterior itself ($g_k = \hat{\xi}^{(k)}$, no extra parameters), a learned logistic function of observables, or — in a network — a learned softmax trained end to end with the experts (Jacobs et al., 1991; sparsely-gated at scale in Shazeer et al., 2017).

Soft gating strictly dominates hard selection, for a reason worth stating: at a transition the posterior moves continuously from one expert to another, so the prediction path is continuous and the turnover is bounded. Hard selection jumps, and the jump is largest exactly when you are least sure. This is the same argument as smooth-transition versus threshold models (§5.5), and it has the same answer.

Addresses. Concept drift, strongly.

Cost. K models to train, tune, and monitor. Real organisational cost, not just compute.

Failure modes. (i) **Experts specialise on noise.** With a learned gate and enough capacity, the model partitions the training data in whatever way minimises training loss, which need not correspond to anything. Constrain the gate to a small number of interpretable inputs. (ii) The gate memorises the training era's regime sequence — check by evaluating gate outputs on a held-out period and confirming they still respond to the inputs rather than to the calendar. (iii) Expert collapse: one expert takes everything and the rest are dead weight.

Verdict. Worth it when you have genuinely different *strategies* to switch between (a trend model and a mean-reversion model, say) rather than genuinely different parameters of one model. In that case the gate is doing strategy allocation, which is a real problem with real evidence behind it. As a way to add capacity to a single model, use §10.1 interactions instead — cheaper and less fragile.

10.6 Scheme 6 — invariance filtering

What it does. Inverts the objective. Instead of adapting the model to each regime, it keeps only the relationships that do not change across regimes (§9.6).

How.

1. Partition history into regimes by any reasonable method; a scalar index (§5.9) or a two-state HMM (§6) is fine.
2. Fit the model separately in each regime.
3. Keep the features whose effect — a regression coefficient, a partial-dependence slope, a SHAP contribution — has a **consistent sign and comparable magnitude** in every regime. Discard the rest.
4. Refit on pooled data using only the surviving features.

Two heavier versions exist for networks: an adversarial head that makes the regime unpredictable from the shared representation, and the invariant risk minimisation penalty (Arjovsky et al., 2019). Both need more environments than finance supplies; Rosenfeld, Ravikumar and Risteski (2021) show that invariant risk minimisation can select the wrong features when environments are few. **[Contested]**

Addresses. All three shifts, by refusing to depend on them.

Cost. An afternoon for the filtering version.

Failure modes. (i) It may discard real, exploitable regime-specific signal; invariance is a robustness constraint and costs in-sample fit by construction. (ii) Invariance across the regimes in your sample is not invariance across the next one. (iii) With noisy per-regime estimates, "consistent sign" fails by chance; compare effects relative to their standard errors.

Verdict. The best-value idea in this section for any model meant to run for years, because it asks of the regime estimate only what §8 says it can deliver: a rough partition of history into environments. **[Practice]**

10.7 Scheme 7 — representation-learning variants

Four schemes need a network's shared representation or its training loop. They are listed together because they share a verdict: correct machinery aimed at sample sizes that rarely support it.

- **Regime as an auxiliary task.** Add a second output head that predicts the regime, trained jointly with a weight α on its loss, and discard the head at inference. The auxiliary target may be the forward-looking regime label of §5.11, because it is never used at inference and so creates no look-ahead in deployment; it only shapes the representation. Cheap and low-risk; if α is too large, the trunk optimises for classifying regimes instead of predicting returns.
- **A learned latent state.** Feed the conditioning series to a recurrent or attention encoder and let it build whatever state serves the objective (Chen, Pelger & Zhu, 2024; §5.10). The state is optimised for the decision rather than for one-step density forecasting, which is the appeal. It needs a large cross-section, it can collapse into an unused latent with no symptom in the loss, and the learned state is usually a noisy proxy for realised volatility; ablate it and regress it on volatility before claiming novelty.
- **Adversarial invariance.** The opposite of the auxiliary task: train the trunk so that a head *cannot* predict the regime (§10.6). Which of the two you want depends on whether you believe the regime-specific component is signal or noise.
- **Meta-learning.** Train so that a few gradient steps on recent data adapt the model. The adaptation set is tiny and noisy, and fast adaptation to a genuine break happens at the worst moment. Periodic retraining with exponentially decayed weights (§10.3) captures most of the benefit at a fraction of the risk.

Verdict. Try the auxiliary task before gating if you are already building a network. Treat the rest as research projects. **[Practice]**

10.8 What changes with the model family

The schemes above are model-agnostic. A few details are not.

Linear and regularised regression. Interactions must be built explicitly, which is an advantage: $x_j \cdot \hat{\xi}_t^{(k)}$ is one column with one coefficient, visible and testable. Sample weights act exactly as weighted least squares. A linear model extrapolates, so a feature outside its training range still moves the prediction, sometimes too far; clipping or ranking features bounds the damage.

Tree ensembles.

- *A binary regime indicator derived from a feature already in the model is exactly redundant.* A tree can split at any threshold on a continuous feature v, so $\mathbb{1}[v > c]$ adds no representable function. It can only change which function the greedy search finds, and the search will generally prefer the continuous parent. A *latent* state posterior is not a monotone function of any single feature, so it is not redundant, which is the one solid argument for using a fitted regime model rather than a threshold rule as a tree input.
- *Trees cannot extrapolate, and a new regime is precisely an extrapolation.* Outside the training range of a feature, a tree returns the boundary leaf's value, constant forever. If volatility reaches a level never seen in training, the model behaves as though volatility were at the training maximum. **This is the most serious regime-related weakness of tree models, and it is usually invisible in backtests**, because the backtest contains the extremes that training lacks only at the very end. The fix is to make features **regime-stationary by construction**, so that the training range covers what the future can produce: divide by a trailing scale ($x_j / \hat{\sigma}_t$), use trailing percentile ranks (bounded in $[0, 1]$), or rank cross-sectionally within each date. This is normally filed under feature scaling, and it is the most effective regime adaptation available to a tree model. **[Practice]**, held strongly.
- *Sample weights change splits, not just leaf values*, so their effect is larger and less predictable than in a linear model; inspect the trees, not just the loss.
- *Monotone constraints are a cheap robustness device.* If a feature's effect should have a stable sign across regimes, impose it (LightGBM and XGBoost both support per-feature constraints). This encodes the invariance of §10.6 without any machinery.

Networks.

- *Concatenating the regime to the input is the weakest injection.* The stronger one is multiplicative: let the regime posterior produce a scale and shift for hidden activations, $a^{(\ell)} \leftarrow \gamma^{(\ell)}(\hat{\xi}) \odot a^{(\ell)} + \beta^{(\ell)}(\hat{\xi})$, which is feature-wise linear modulation (FiLM; Perez et al., 2018). The regime can then switch pathways rather than nudge a sum.
- *Batch normalisation is a hidden shift hazard.* Its running statistics are calibrated to the training regime. Layer normalisation uses no cross-sample statistics and is safer.
- *Recurrent models can learn the calendar.* A hidden state can encode "this is the 2013 low-volatility era". Evaluate on held-out episodes (§11.3) and ablate the recurrent state.
- *Cheap uncertainty is worth having.* Deep ensembles widen their spread in unfamiliar conditions, which is a regime-aware sizing input obtained without a regime model.

For every family, adaptive conformal prediction (Gibbs & Candès, 2021) wraps any point forecast in prediction intervals whose long-run coverage survives distribution shift. It is the only tool in this document whose guarantee holds *under* shift, and the interval's width is a direct sizing input (§12.1).

10.9 Comparison and effort ordering

Scheme	Shift addressed	Effort	Data cost	Fragility	Verdict
2. Target normalisation	scale	trivial	none	low	**do first**
1. Feature (posterior, time since transition)	concept (weak)	trivial	none	low	do, expect little
6. Invariance filtering	all	low	none	low	**high value**
3. Sample weighting	scale, covariate	low	moderate	medium	situational
4. Partial pooling	concept	medium	high	medium	risk models yes, alpha no
5. Gating / mixture of experts	concept	high	high	high	only for strategy switching
7. Representation-learning variants	concept	medium to very high	moderate to very high	high	auxiliary task worth a try; the rest rarely

The recommended ordering, with the gate at each step:

1. **Normalise the target by trailing volatility** (§10.2). Gate: does out-of-sample information coefficient improve? It almost always does.
2. **Make every feature regime-stationary** — trailing z-scores or ranks (§10.8). Gate: does performance in the most extreme out-of-sample period improve?
3. **Size positions by predicted volatility** (§12.1). Gate: does realised volatility of the strategy stabilise?
4. **Add the continuous conditioning variables as features** — volatility, turbulence, term structure (§10.1, encoding 3). Gate: incremental out-of-sample information coefficient over step 1.
5. **Run the invariance filter** (§10.6). Gate: does the reduced feature set hold up better in the worst out-of-sample regime?
6. **Only now**, add a fitted regime posterior and its interactions (§10.1, encodings 2 and 4). Gate: §11's ablation ladder, in full.
7. **Only if 6 clearly passes**, consider partial pooling or gating.

Steps 1–3 are not regime modelling and will deliver most of the benefit people attribute to regime modelling. That is the honest summary of this section.

§10 Key takeaways

1. Normalise the target by a trailing volatility estimate. It is the cheapest change available and it addresses the shift that actually dominates.
2. The denominator must be $\mathcal{F}_t$-measurable. Normalising by realised volatility over the label window is the most spectacular leak in this document.
3. Feed the posterior and the time since the last transition, not the hard label. Duration is information the Markov assumption discards.
4. Regime-similarity sample weighting is a strictly better-behaved version of subsetting. Always report effective sample size.
5. If you fit per-regime models, do it as partial pooling from a pooled base — penalised interactions, boosting from an offset, or shrunk per-regime heads — never as hard subsetting.
6. Soft gating dominates hard selection: the prediction path stays continuous exactly where you are least certain.
7. Invariance filtering asks of the regime estimate only what it can deliver, and is the best-value scheme for a model meant to last.
8. Trees cannot extrapolate, so a new regime is a silent failure for them. Regime-stationary features — trailing normalisation or ranking — are the fix, and they are normally mislabelled as feature scaling.
9. The first three steps of the recommended ordering are not regime modelling at all, and they deliver most of the benefit.

11. Research and evaluation

Regime research fails at evaluation more often than at modelling, and it fails in ways that are specific enough to enumerate. This section is a protocol.

11.1 The evaluation ladder

Cheap filters first, so that most ideas die before you have spent a week on them.

```
0.  Persistence check          |lambda_2| half-life vs your horizon    minutes
1.  Naive-null check           beat "yesterday's volatility"?          minutes
2.  In-sample conditional fit  necessary, nowhere near sufficient      an hour
3.  Purged walk-forward        rung-5 states only                      a day
4.  Ablation ladder            each increment measured separately      days
5.  Shuffled-regime control    is the gain from the regime at all?     hours
6.  Regime-stratified report   with EPISODE counts, not day counts     hours
7.  Formal tests               fluctuation test, DSR, conditional PA   hours
8.  Shadow / paper trading     the only test that cannot be gamed      months
```

Steps 0 and 1 kill most ideas in under an hour, and both are skipped almost universally. Step 0 is §1.3: compute the half-life implied by the fitted transition matrix and compare it to your rebalancing horizon. Step 1 is: does your elaborate state estimate beat the single feature "trailing 20-day realised volatility, ranked"? If not — and it frequently does not — stop.

11.2 The eight leaks

Every one of these produces a better backtest, which is why they survive.

#	Leak	Where it hides	Fix
1	**Smoothed states**	$\xi_{t\|T}$ used as a signal	Use $\xi_{t\|t-1}$ (§5.1)
2	**Full-sample parameters**	$\hat{\theta}$, $\hat{\mathbf{P}}$ fitted once on everything	Expanding-window refit
3	**Full-sample labelling**	states sorted by fitted mean or volatility over the whole sample	Sort inside each window, by a stable quantity
4	**Full-sample scaling**	feature z-scores, percentile thresholds, K, the jump penalty λ, the decay half-life	All chosen inside the training window
5	**Label-window normalisation**	$\hat{\sigma}_t$ computed over $[t, t+h]$	Trailing estimator, explicitly lagged
6	**Overlapping labels in CV**	any $h > 1$ target with k-fold	Purge and embargo (§11.3)
7	**Tuning on the test period**	K, λ, half-life, thresholds	Nested validation, or pre-register
8	**Revised macro vintages**	GDP, payrolls, industrial production as regime inputs	Real-time vintage data

Leak 8 is specific to this field and deserves elaboration, because regime models attract macro inputs. **Macroeconomic series are revised, sometimes by more than their own standard deviation, and the revisions can arrive years later.** A model conditioned on final-vintage GDP growth is conditioned on numbers that did not exist at the decision date. The Philadelphia Fed's real-time data set exists precisely for this, and if your regime is macro-driven you need it. The NBER recession indicator is the extreme case: the committee has announced turning points more than a year after the fact (§1.5). **[Fact]**

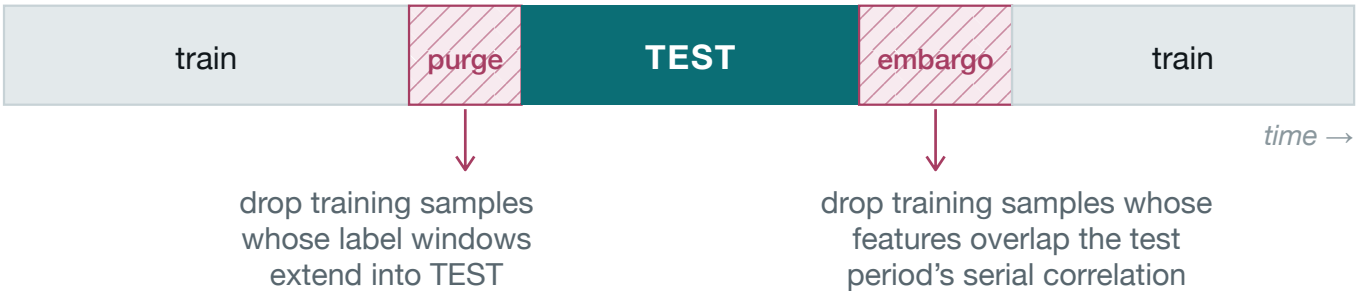
Leak 3 is the subtlest and it does real damage. State labels are exchangeable under the likelihood, so you must impose an identification rule. If you impose it on the full sample — "state 2 is the one with the lower mean over 1990–2024" — you have told the model which state is the bad one using the whole history. The fix is to sort by a quantity that is (a) stable and (b) computed inside the window: fitted volatility, ascending. And as §8.3 shows, when the states do not differ in volatility, no rule works and the pipeline is worse than nothing.

11.3 Cross-validation design

Standard k-fold cross-validation is invalid here for three independent reasons, and each one alone is disqualifying:

1. **Overlapping labels.** A target realised over $[t, t+h]$ shares data with the targets at $t+1, \ldots, t+h-1$. A fold boundary in the middle of that window puts nearly identical observations on both sides.
2. **Serial correlation in features.** Trailing volatility at t and $t+1$ are nearly the same number, so a training point adjacent to a test point is close to a copy of it.
3. **Regime blocks.** This one is specific to regime research and it is the worst. Regimes are persistent, so a random fold split places the *same regime* on both sides of every boundary. The model is then evaluated on regimes it was trained on, which is exactly the question you are trying to answer, answered in advance in the affirmative.

Purging and embargoing fix the first two:



The third needs a different remedy, and it is where the interesting design choice lives:

- **Walk-forward** is the honest default. Train on $[0, T_i]$, test on $(T_i, T_{i+1}]$, roll. Every test period is genuinely future.
- **Combinatorial purged CV** (López de Prado, 2018) constructs many train/test path combinations from the same data with purging applied throughout, giving a *distribution* of backtest outcomes rather than a single path. This is the right answer to "was my walk-forward result luck", and it is worth the implementation cost.
- **Do not stratify folds by regime.** It is a natural instinct — ensure each fold contains both regimes so the estimates are stable — and it destroys the experiment, because it guarantees the model has seen the test regime.
- **Leave-one-episode-out** is the recommended test to run and report. Hold out the entire 2008 crisis, or March 2020, or the 2022 rate shock — one complete regime episode — train on everything else, evaluate there. It is not clean inference from one episode, but it is the closest available approximation to the question you actually care about, which is "how will this behave in a bad period it has not seen".

11.4 The ablation ladder

The specific experiment sequence. Every rung uses the **same** purged walk-forward split, the same test period, and the same evaluation metric, so that the increments are comparable. Report each increment with a standard error.

Rung	Model	What it isolates
A0	Unconditional constant	the null
A1	Base model, raw target, raw features	your actual baseline
A2	A1 + target volatility normalisation	conditional-scale drift (§10.2)
A3	A2 + regime-stationary features (trailing ranks)	feature-range shift (§10.8)
A4	A3 + continuous conditioning variables	observable state, no latent model
A5	A4 + regime posterior as a feature	what the latent model adds
A6	A5 + regime $\times$ signal interactions	explicit concept-drift conditioning
A7	A6 + partial pooling or gating	separate functions per regime
C1	**A5 with a *shuffled* regime**	**does the regime matter at all?**
C2	**A5 with a regime fitted on a different asset**	is it asset-specific or generic?
C3	A5 with the regime replaced by trailing volatility rank	does the latent model beat the obvious proxy?

The two controls in bold are the ones that decide whether you have anything.

C1, the shuffled-regime control, is the most important experiment in this document and almost nobody runs it. Replace the fitted state path with a synthetic one that has the *same marginal distribution and the same persistence* but no relationship to the data — a block bootstrap of the state sequence, or a simulated Markov chain with the fitted $\hat{\mathbf{P}}$. Refit everything. If the shuffled version performs as well as the real one, your improvement came from the extra model capacity and the extra column, not from the regime, and you have learned something valuable for the cost of one rerun.

C3 is the humility check. If the latent-state posterior does not beat "the percentile rank of trailing 20-day realised volatility", use the volatility rank. It has no identification problem, no label switching, no filtering, and no K.

11.5 Report by regime — and count episodes, not days

Any regime-conditional claim must be reported stratified by regime, with:

- performance within each regime,
- the number of **days** in each regime,
- and, separately and much more importantly, the number of **episodes**.

The effective sample size for a regime-conditional claim is the number of independent regime episodes, not the number of observations. Twenty years of daily data with a 25-day turbulent state entered roughly twice a year contains about 40 episodes and about 1,000 turbulent days. Your standard errors are governed by the 40.

The arithmetic is unforgiving. The standard error of an annualised Sharpe ratio estimated over y years is approximately $\sqrt{(1 + S^2/2)/y}$. If the turbulent regime occupies 20% of a 20-year sample — four years of turbulent time — then a within-regime Sharpe of 0.5 carries a standard error of **0.53**.

> You cannot distinguish "this strategy has a Sharpe of 0 in crises" from "this strategy has a Sharpe of 1 in crises" with twenty years of daily data. Any claim about crisis-regime performance that does not acknowledge this is reporting noise.

And that 0.53 is the *optimistic* figure, because the formula treats the four years of turbulent days as independent observations. It answers "how precisely have I measured performance in the crises I saw". The question you actually care about — "will this work in the next crisis" — is governed by the forty episodes, not the thousand days, and its standard error is correspondingly worse. The two numbers are not in conflict; they bound different quantities, and the episode count bounds the one that matters.

This is the same identification problem as §8.1, arriving from a different direction, and it applies just as forcefully to the *evaluation* of a regime strategy as to the *estimation* of a regime model.

11.6 Formal tests worth running

Four, in order of value per hour spent:

Giacomini-Rossi fluctuation test (2010). Compute the rolling relative out-of-sample loss of model A versus model B and compare its path against critical values that account for the fact that you are looking at the maximum over time. This directly answers "does my model's advantage come and go", which is the regime question stated as a forecasting question. **If your regime-aware model is genuinely better, its advantage should be concentrated in the regime you designed it for, and the fluctuation test will show you that.** If the advantage wanders around with no relation to the regime, you have found something else.

Giacomini-White conditional predictive ability (2006). Tests whether A beats B *conditional on the current state* rather than on average. This is precisely the object of interest and it should be standard in this literature. It is not.

Deflated Sharpe ratio (Bailey & López de Prado, 2014). Adjusts an observed Sharpe for the number of trials and for skew and kurtosis. Necessary here because **regime slicing multiplies trials fast**: K regimes $\times$ M specifications $\times$ W window choices, and each combination you looked at is a trial whether or not you reported it. The arithmetic: the expected maximum of 20 independent standard normal draws is about 1.87 and of 100 about 2.51, so with a within-regime Sharpe standard error of 0.53 (§11.5), **twenty honest trials produce an expected best-of-sample crisis Sharpe of about 1.0 from pure noise.** That is a more impressive number than most published regime results.

Stationary bootstrap (Politis & Romano, 1994) for confidence intervals on path-dependent statistics. Block length should exceed your regime persistence, or the bootstrap destroys the very structure you are testing — a subtle and common error: resampling in blocks shorter than the regime half-life produces a null in which regimes do not exist, which flatters any regime strategy.

11.7 Synthetic data: two tests, both essential

You have a generative model. Use it.

Test 1 — recovery. Simulate from a known regime process, run your entire pipeline end to end, and check that it recovers the truth and that your reported confidence intervals have the coverage they claim. Everything in §8 came from this exercise, and every surprise in it was informative.

Test 2 — the null. Simulate from a process with **no regimes at all** — a GARCH or stochastic-volatility model calibrated to the same unconditional moments — and run the identical pipeline. **Your pipeline should report no edge.** If it reports an edge, you have a bug or a leak, and you have found it before it cost you money.

Test 2 is the one nobody runs and it catches more problems than any other single check. It is the direct operationalisation of §2.6: if a no-regime process yields a regime strategy that appears to work, then your positive result on real data means nothing.

11.8 The protocol, condensed

1. Fit the state model on the **training window only**. Refit on an expanding schedule. Never touch the test period.
2. Identify states by a rule computed inside the window (volatility, ascending).
3. Emit $\xi_{t|t-1}$ only. Store the rung with the number.
4. Normalise the target by a trailing volatility estimate; verify by construction that the denominator uses no data after t.
5. Split with purging and embargo sized to your label horizon. Prefer walk-forward; add combinatorial purged CV for a distribution.
6. Run the ablation ladder A0–A7 with identical splits and metrics.
7. Run controls C1 (shuffled regime), C2 (foreign regime), C3 (volatility rank). **If C1 matches A5, stop.**
8. Report stratified by regime, with episode counts and Sharpe standard errors.
9. Run the fluctuation test and the deflated Sharpe with an honest trial count.
10. Run the synthetic null. Confirm no edge.
11. Shadow-trade before sizing.

§11 Key takeaways

1. Two checks costing under an hour — the transition-matrix half-life against your horizon, and beating trailing volatility rank — kill most regime ideas. Run them first.
2. There are eight distinct regime-specific leaks. Full-sample labelling and revised macro vintages are the two that survive otherwise-careful pipelines.
3. k-fold CV fails here for three independent reasons; the regime-block reason is the one purging does not fix.
4. Never stratify folds by regime. It guarantees the model has seen the test regime and destroys the experiment.
5. The shuffled-regime control — same marginal, same persistence, no relationship to the data — is the single most informative experiment available, and it is almost never run.
6. The effective sample size for a regime claim is the number of episodes, not days. Four years of crisis time gives a Sharpe standard error above 0.5.
7. Twenty honest trials produce an expected best crisis Sharpe near 1.0 under the null. Deflate accordingly.
8. Bootstrap block length must exceed regime persistence, or your null has no regimes in it and every regime strategy looks good.
9. Simulate from a no-regime process and confirm your pipeline finds nothing. This catches more than any other single check.

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12. Trading a regime-aware model

You have a state posterior, and perhaps a forecasting model that uses it. This section is about the last mile, where most of the theoretical gain is lost.

12.1 Do not gate. Size.

The instinctive use of a regime model is a switch: risk-on, be invested; risk-off, go flat. The better use is a dial. And the *correct* use is neither — it is to feed the mixture's moments into the position rule and let the arithmetic decide.

The correct object. Your model gives a predictive mixture (§1.2). Its first two moments are

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \sum_k \xi^{(k)}_{t|t-1}\mu_k \;\equiv\; \bar{\mu}_t,$$

$$\mathrm{Var}[r_t \mid \mathcal{F}_{t-1}] = \underbrace{\sum_k \xi^{(k)}_{t|t-1}\sigma_k^2}_{\text{within-state risk}} + \underbrace{\sum_k \xi^{(k)}_{t|t-1}(\mu_k - \bar{\mu}_t)^2}_{\text{state uncertainty}},$$

and the mean-variance position is $\pi_t = \bar{\mu}_t/(\gamma\,\mathrm{Var}[r_t \mid \mathcal{F}_{t-1}])$.

The position shrinks by itself, through the first term. As the posterior shifts toward the turbulent state the within-state term rises from σ_1^2 to σ_2^2, so with the §8 calibration the predictive volatility runs 12.0%, 19.1%, 24.2%, 28.4%, 32.0% as $\xi^{(2)}_{t|t-1}$ goes 0, 0.25, 0.5, 0.75, 1 — and since position scales as $1/\,\mathrm{Var}$, not $1/\sigma$, the mean-variance position at full turbulence (mean held fixed, so only this term is acting) is $(12.0/32.0)^2 = $ **14.1% of its calm-state size**. No threshold, no hysteresis rule, no special case: the de-risking is a consequence of taking the mixture seriously instead of collapsing it to a label.

The second term is real but small, and it is worth knowing why. Both μ_k and σ_k^2 here are *per bar*, and a mean difference enters squared while a variance enters linearly, so the between-state term is smaller by a factor of order A. Concretely, at a 50/50 posterior between the $+10\%$ and -15% states, the between-state term contributes 0.79% of annualised volatility against the within-state term's 24.2% — about one part in a thousand of the variance. It is tempting to write the dispersion of *annualised* means, $(0.125)^2 = 1.6\%$, and conclude that state uncertainty dominates; that is wrong by the factor $A = 252$. Suppose the state stayed put for h bars, which is the case that favours the between-state term most. The squared gap between cumulative means grows as h^2 while the cumulative variance grows only as h, so the between-state term overtakes the within-state term only at a horizon of roughly $4\sigma^2/(\Delta\mu)^2 \approx 940$ bars — about **3.7 years** — where σ^2 and $\Delta\mu$ are per bar and the 4 is $1/[\xi(1-\xi)]$ at a 50/50 posterior. A state that mean-reverts, as these do, only pushes the crossover later. Which is the identification asymmetry (§8.1) arriving once more: uncertainty about *which mean* you face is negligible next to uncertainty about *how volatile* things are, until you are holding for years.

This is the answer to §1.2's warning about argmax labels, stated operationally. **Feed the moments, not the label.**

The measurement. From the same simulation as §8.4, comparing a hard gate (flat when the predicted state is turbulent) against volatility sizing (scale by $1/\sqrt{\mathrm{Var}}$) and the combination:

[Simulated]

States separated by	Buy & hold	Gate	Size	Both
Volatility and mean	0.26	0.54	0.45	**0.57**
Volatility only	0.26	0.29	0.31	**0.31**
Mild volatility	0.42	0.50	0.45	**0.51**
Mean only	0.55	0.55	0.55	0.55

Gating beats sizing when the states genuinely differ in mean; sizing beats gating when they differ only in volatility; the combination wins or ties everywhere. But the important reading is not which column is larger — it is **what each column's advantage depends on**. Gating's edge depends entirely on $\mu_1 - \mu_2$, the quantity §8.1 says you cannot estimate. Sizing's edge depends on σ_2/σ_1, the quantity you can estimate to within a few percent. **Under parameter uncertainty — which is the real world — sizing is the robust choice, because its benefit does not depend on the number you cannot measure.** The simulation above hands both quantities to the strategy for free; your backtest will not.

On real data the worked example of §6.11 runs the same comparison on US equities from 1946 to 2025, with walk-forward parameters. The gate earns a Sharpe ratio of 0.60 and the HMM volatility target 0.62, against 0.52 for buy and hold and a standard error of about 0.12, so the sample cannot rank the two. They differ where the arithmetic above says they should: the gate cut the maximum drawdown further (38% against 44% for the volatility target) but gave up 2.4 points of annual return against buy and hold, because the days it sat out had the same average return as the days it held. It was paid entirely through volatility, which sizing captures without betting on the mean.

12.2 Hysteresis, buffering, and the cube-root rule

A hard threshold at $\xi = 0.5$ produces chatter whenever the posterior loiters near a half, which is exactly when the evidence is weakest and trading is least justified. Three fixes, in increasing order of quality:

1. **Two thresholds (a Schmitt trigger** — the electronics term for a switch whose turn-on and turn-off points differ, so that a noisy input near the boundary cannot make it chatter**).** Enter the defensive state at $\xi > 0.7$, leave it at $\xi < 0.3$. Crude, effective, one extra parameter, and it introduces path dependence you must then model in the backtest.
2. **Continuous sizing** (§12.1). No chatter by construction, because the position is a continuous function of a continuous posterior. The cost moves from a few large trades to many small ones — and in the §8.4 calibration that is a large saving rather than a wash: total annual turnover falls from 8.8 units under the hard gate to 4.0 under continuous sizing. **[Simulated]**
3. **A no-trade band.** Trade only when the target position differs from the current one by more than a band δ. Under proportional transaction costs ε, the classical result (Constantinides, 1986; and the continuous-time treatments that followed) is that the optimal no-trade half-width scales as

$$\delta \propto \varepsilon^{1/3}.$$

The cube root is the practically important part. A ten-fold increase in trading costs widens the optimal band by only $10^{1/3} \approx 2.15$. People consistently over-widen bands in response to cost estimates; the correct response is much milder than intuition suggests. Under *quadratic* costs the answer is different and equally clean: Gârleanu and Pedersen (2013) show the optimal policy is to trade a constant fraction of the way toward an aim portfolio that is tilted toward slower-decaying signals — "aim in front of the target, and trade partially towards the aim". Regime signals decay fast (§1.3), so under this framework they get *less* weight in the aim than their raw strength suggests.

12.3 Does the overlay pay for itself?

Do this arithmetic before building anything, with numbers from §8.4 and the turnover from the same simulation.

[Simulated] The mild-volatility calibration — the realistic one — gives a gated strategy with mean absolute daily position change of **0.0350**, so annual position turnover is $0.0350 \times 252 = 8.8$ units of $|\Delta\pi|$. Each such unit is one unit traded in one direction, so the annual cost is $8.8 \times c$ for a one-way cost c. The strategy runs at 15.0% annualised volatility and beats buy-and-hold by 0.50 − 0.42 = 0.08 Sharpe, which is $0.08 \times 0.150 \approx$ **118 basis points a year** gross — 118 rather than the 120 the rounded factors suggest, because the underlying Sharpe gap is 0.0785, not exactly 0.08. All four numbers are printed by `figures/regime_filtration.py`.

One-way cost	Annual cost	Net of 118 bp gross
2 bp (index future)	18 bp	**+100 bp**
5 bp (liquid ETF)	44 bp	**+74 bp**
10 bp (single stock, in size)	88 bp	+30 bp
20 bp (less liquid)	176 bp	**−58 bp**

The break-even one-way cost is $118/8.8 \approx 13$ basis points. Comfortable on futures and liquid ETFs, marginal on single stocks traded in size, and negative beyond that — and all of it before you account for the 118 bp having come from a simulation that handed you the true parameters, the right K, and the right emission family. This single table should determine whether you start the project.

Three implications. First, regime overlays belong on the most liquid instruments you have — index futures, major FX, rates — which is exactly where the CTA industry applies them, and that is not a coincidence. Second, if you must apply one to a less liquid book, apply it at the *portfolio* level as an overlay on aggregate exposure rather than instrument by instrument, so the turnover is paid once on the cheapest hedging instrument. Third — and this is the cost argument for §12.1's recommendation — **continuous sizing turns over 4.0 units a year against the hard gate's 8.8**, so its trading cost is 45% of the gate's. Its gross Sharpe in this calibration is lower (0.45 against 0.50, §12.1), but it does not depend on the mean spread. The saving is not automatic: on the real data of §6.11 the order reverses, with the HMM volatility target turning over 14.2 units a year against the gate's 9.5. Measure the turnover of your own sizing rule before costing it.

12.4 Where regimes genuinely earn their keep: the covariance matrix

For a multi-asset book, the regime-conditional *covariance* matters more than the regime-conditional mean, for a reason that is obvious once stated: diversification is the thing that stops working when you need it, and a covariance matrix estimated on a full sample understates crisis co-movement.

Practical approaches, in ascending order of ambition:

- **Two-state covariance blending.** Estimate Σ_{calm} and Σ_{stress} on turbulence-sorted subsamples (Chow, Jacquier, Kritzman & Lowry, 1999) and use $\Sigma_t = \xi_t^{(1)}\Sigma_{\text{calm}} + \xi_t^{(2)}\Sigma_{\text{stress}}$. Cheap and it captures most of the effect. This is §9.5's "subset the risk model" prescription, made concrete.
- **Stress-weighted shrinkage.** Shrink the sample covariance toward the stress covariance rather than toward the identity or a constant-correlation target. A one-line change to a standard pipeline with a much better prior.
- **Regime-switching DCC** (Pelletier, 2006) for a full dynamic model. More correct, considerably more machinery.

The caveat travels with the recommendation. Part of the measured correlation increase in crises is a mechanical artefact of the volatility increase (Forbes & Rigobon, 2002). Apply their correction before you conclude that correlations genuinely rose, or you will double-count: your volatility model already captured the variance increase, and the "correlation regime" you then add on top is partly the same effect a second time.

12.5 Capacity, crowding, and reflexivity

Regime signals have a specific and uncomfortable property: **they are public and they all say the same thing at the same time.** Realised volatility, the VIX, the 200-day moving average, and every fitted two-state HMM will agree, because they are all measuring the same underlying quantity (§7.2). The de-risking trade is therefore among the most crowded trades in existence, and it arrives as a simultaneous, mechanical, price-insensitive flow.

The reflexive consequence follows directly from §2.1: **your regime overlay is a component of the mechanism that generates the regime.** Volatility rises, volatility-targeting strategies mechanically reduce exposure, the selling pushes prices down and volatility up, and the next round of de-risking follows. This is the fire-sale channel with systematic strategies as the intermediary. **[Hypothesis]** — the mechanism is well specified and the flow estimates are large, but attributing specific episodes to it is contested.

What to do about it, practically:

- Assume your exits are correlated with everyone else's, and cost them at stress levels rather than at average levels. The cost table in §12.3 is optimistic for exactly the days when the overlay fires.
- Prefer signals with idiosyncratic timing where you can find them — a slower or faster horizon than the industry standard is a real, if small, edge.
- Size for the possibility that you cannot exit at all.

12.6 Monitoring: what breaks, and how you would know

A deployed regime model has specific failure modes with specific detectors.

What breaks	Detector	Trigger
The state definition drifts	$\hat{\sigma}_k$, $\hat{\mathbf{P}}$ tracked across refits	any parameter moving more than its bootstrap CI
Label switching between refits	correlation of the new state path with the old on overlapping data	correlation below ~0.8
The model has lost the plot	posterior entropy $-\sum_k \xi^{(k)} \ln \xi^{(k)}$	sustained rise above its historical range
Occupancy drift	realised time in each state vs the stationary π	large or sustained divergence
Turnover blow-out	realised versus backtested turnover	ratio above ~1.5
Feature extrapolation (trees)	share of predictions at a boundary leaf	any material rise (§10.8)
The strategy itself is dying	BOCPD run-length posterior on the **strategy's own returns**	run-length posterior collapsing

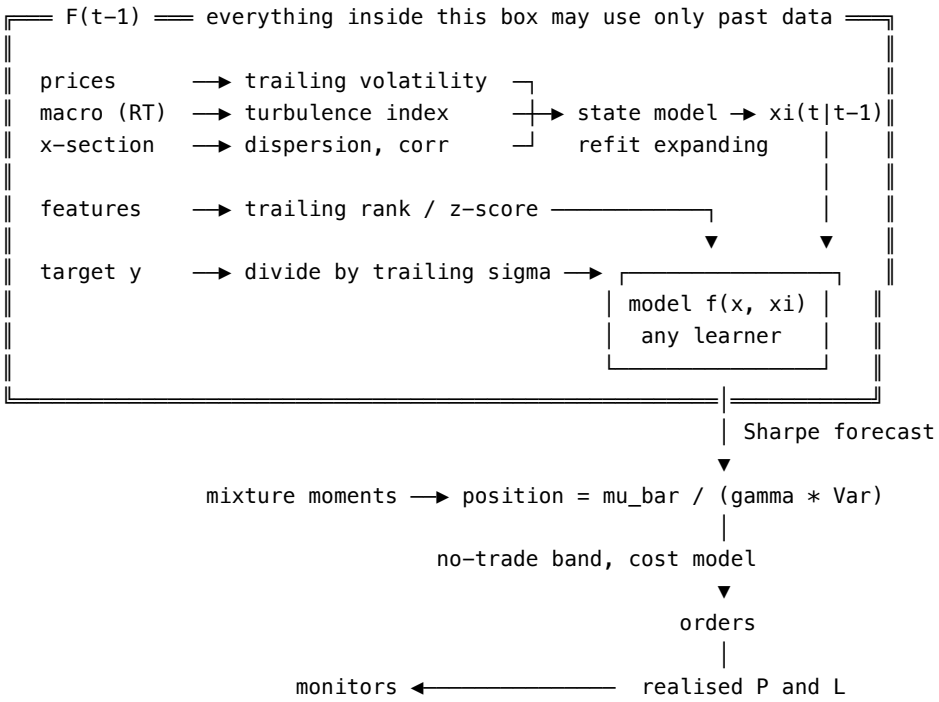
The last row is worth doing regardless of everything else in this document. A change-point detector applied to your own P&L is the cleanest available answer to "has this stopped working", it is model-free, and it is a much better use of change-point

machinery than trying to detect market regimes with it (§5.6).

Set the response in advance, and make it graduated: **reduce size** on entropy or occupancy alarms, **retrain** on parameter drift, **halt and escalate** on turnover blow-out or run-length collapse. Deciding these thresholds during a drawdown is how they get overridden.

12.7 A reference architecture

The pipeline, with the filtration boundary drawn explicitly — everything to the left of it may only use $\mathcal{F}_{t-1}$.



Three properties of this diagram are the point of it. **The state model and the feature scaling sit inside the same causal boundary** — both must be refit on expanding windows, and it is the feature scaling that people forget. **The target normalisation happens before the model, and the position scaling after it**, so the volatility estimate appears twice and both appearances are correct (§10.2). And **the posterior reaches the position rule as moments, not as a label**, so state uncertainty automatically reduces size (§12.1).

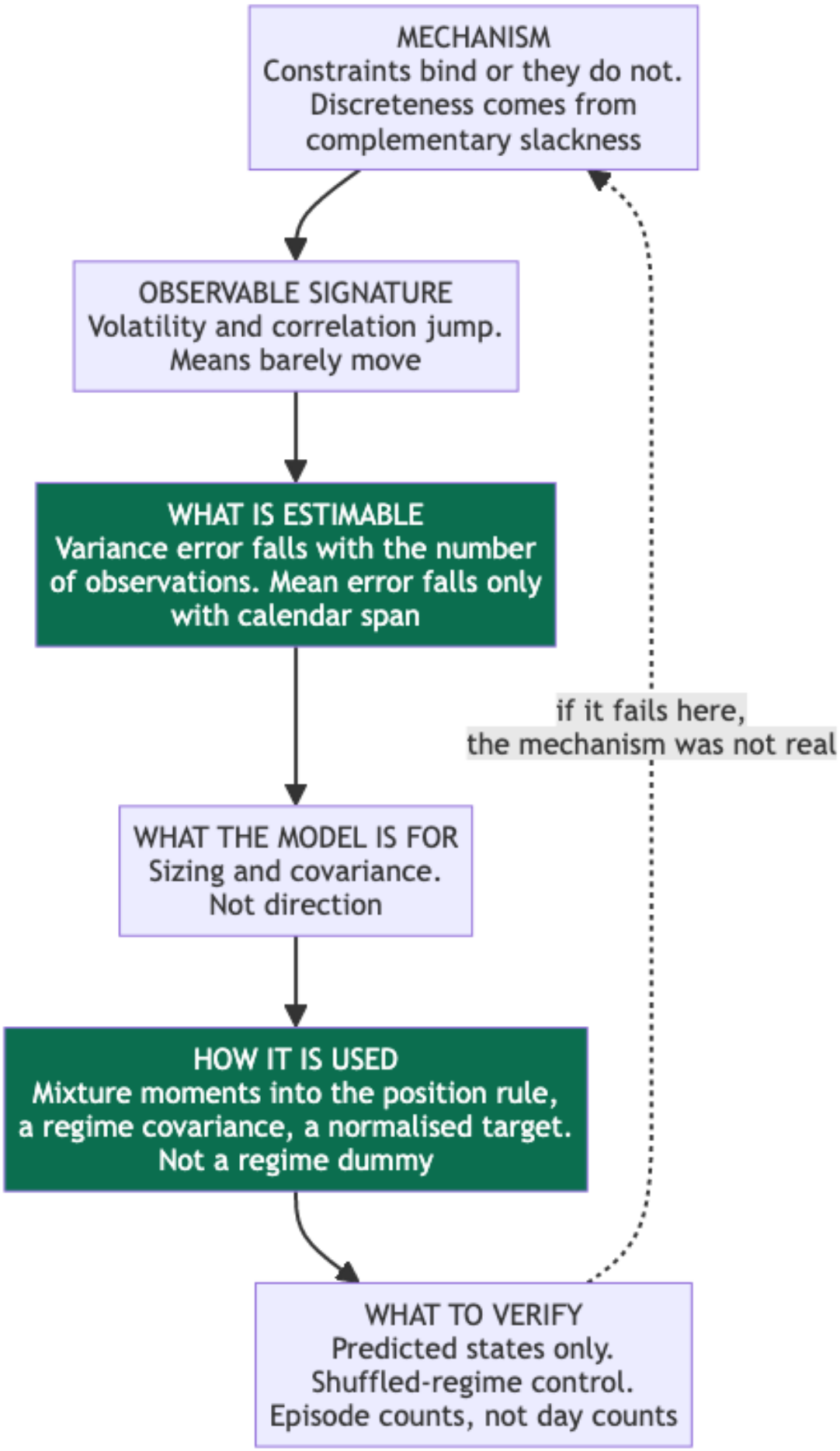
§12 Key takeaways

1. Feed the mixture's moments to the position rule, not the argmax label. State uncertainty then inflates predictive variance and shrinks the position automatically — no threshold, no hysteresis rule.
2. Gating's benefit depends on the mean spread, which you cannot estimate. Sizing's depends on the volatility ratio, which you can. Under parameter uncertainty, size.
3. The optimal no-trade band widens as the *cube root* of transaction cost. A tenfold cost increase roughly doubles the band. Do not over-widen.
4. Regime signals decay fast, so in a Gârleanu-Pedersen aim portfolio they deserve less weight than their raw strength suggests.
5. Run the cost arithmetic first. A regime overlay is worth about 118 bp gross in the realistic calibration, and the break-even one-way cost is 13 bp. That determines whether the project exists.
6. The genuine multi-asset payoff is the regime-conditional covariance, not the regime-conditional mean. Blend a stress covariance in by posterior weight.
7. Correct correlation increases for the volatility artefact before modelling them, or you will count the same effect twice.
8. Everyone's regime signal fires at once, and your overlay is part of the mechanism that produces the regime. Cost your exits at stress levels.
9. Run a change-point detector on your own P&L. It is the best available answer to "has this stopped working", and a better use of the machinery than regime detection.

13. Synthesis

13.1 The framework in one loop

Six links generate everything in this document, and they chain.

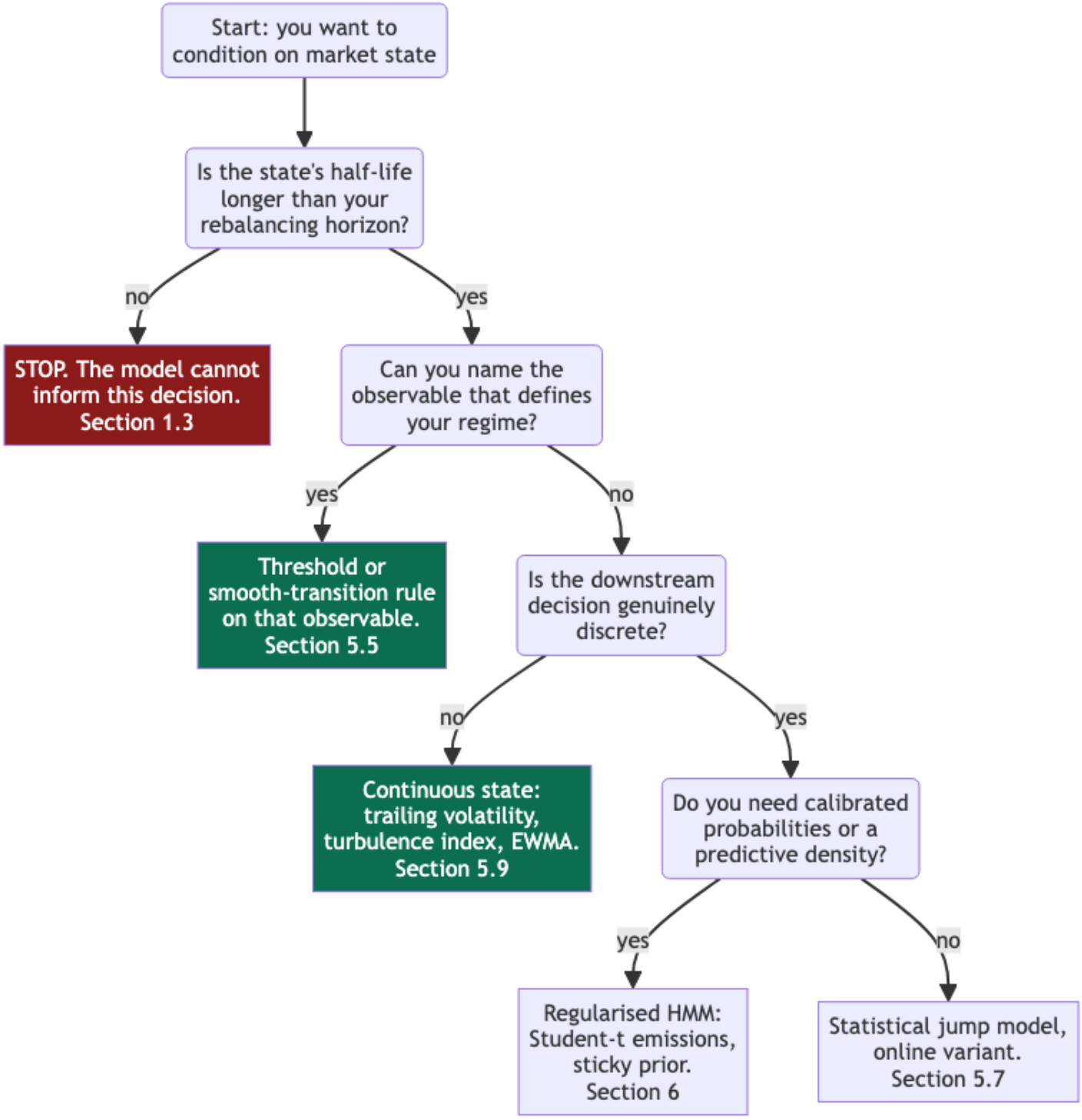

MECHANISM
Constraints bind or they do not.
Discreteness comes from
complementary slackness
OBSERVABLE SIGNATURE
Volatility and correlation jump.
Means barely move
WHAT IS ESTIMABLE
Variance error falls with the number
of observations. Mean error falls only
with calendar span
WHAT THE MODEL IS FOR
Sizing and covariance.
Not direction
HOW IT IS USED
Mixture moments into the position rule,
a regime covariance, a normalised target.
Not a regime dummy
WHAT TO VERIFY
Predicted states only.
Shuffled-regime control.
Episode counts, not day counts
if it fails here,
the mechanism was not real

Stated as prose, and this is the whole document compressed:

> A market regime is not a property of the market; it is a discrete bottleneck you impose on a conditional density, and its only forecasting content is the persistence of the state. Because the standard error of a variance falls with the number of observations while the standard error of a mean falls only with calendar span, a regime model can identify volatility states precisely and return states not at all. Everything downstream follows: the model is worth having for sizing and for covariance, not for direction; it enters decisions through the position rule and the risk model, and a forecasting model through the target rather than through a feature; you pool the means across regimes and let the variances switch; the refinements that pay are the ones that buy persistence from a few dozen transitions, such as heavy-tailed emissions and sticky priors; and the only state estimate that means anything is the one-step-ahead prediction computed with parameters that have not seen the future.

13.2 Decision tree

The tree branches on the four questions that change the answer and ends in a named choice.



Five steps apply on every branch: normalise any forecasting target by trailing volatility; make features regime-stationary; size from mixture moments, never from a label; use predicted states with expanding-window parameters only; and run the shuffled-regime control.

Note where the tree sends most people: to a threshold rule on an observable, or to a continuous state variable. The latent-state branches are reached only when you genuinely cannot name the driver *and* your decision is genuinely discrete. That is a narrower set of situations than the literature's volume would suggest. Of the two latent branches, the HMM is the one to take when the output must be a probability: for a position sized from mixture moments, a covariance blended by state weight, or a decision that weighs the cost of being wrong.

13.3 A staged build, with gates

Each stage has a gate. Failing a gate means stopping, not tuning until it passes.

Stage 0 — Infrastructure (two weeks, and it is not the interesting part). Point-in-time data with real-time macro vintages. A backtester that makes look-ahead structurally impossible rather than merely discouraged — features and targets carrying explicit timestamps, and an assertion that no feature timestamp exceeds its decision timestamp. Purged, embargoed splits. *Gate: reproduce a known-null result — run the whole pipeline on shuffled returns and confirm it finds nothing.*

Stage 1 — The baseline you must beat (one week). Trailing volatility, ranked. Position sized by inverse volatility. No regimes anywhere. *Gate: this is now your benchmark, and every later stage is measured against it, not against buy-and-hold.*

Stage 2 — Target and feature normalisation (one week). Vol-normalise the target; convert every feature to a trailing rank or z-score. *Gate: out-of-sample information coefficient improves, and performance in the most extreme held-out period improves. If not, something is wrong — this almost always helps.*

Stage 3 — Continuous conditioning (two weeks). Add realised volatility, turbulence, dispersion, term-structure slope as features and as explicit interactions with your main signals. *Gate: incremental out-of-sample improvement over Stage 2, with a standard error.*

Stage 4 — A state model (three weeks). A two-state HMM with Student-t emissions, variance-only switching and a sticky prior, fitted by penalised EM (§6.3, §6.7) — or a statistical jump model on volatility features, online variant (§5.7). Expanding-window refit, states identified by fitted volatility. Emit $\xi_{t|t-1}$. Check pseudo-residuals and race the HMM against an EWMA with Student-t innovations (§6.9). *Gate: beats Stage 3, and beats control C3 — the trailing volatility rank. Most projects should end here, having failed this gate, and that is a success: you have saved the three weeks that Stage 5 would have cost.*

Stage 5 — Conditioning on the state (three weeks). Posterior as a feature, time-since-transition, entropy; interactions with signals; regime-conditional covariance for the portfolio. If the HMM is used directly rather than through a forecasting model, this stage is the direct use of §6.10. *Gate: the full ablation ladder of §11.4, including the shuffled-regime control C1. If C1 matches, stop.*

Stage 6 — Sizing and execution (two weeks). Mixture moments into the position rule. No-trade band from the cube-root rule. The cost table of §12.3, computed with your own costs. *Gate: net of realistic stressed costs, the overlay still pays.*

Stage 7 — Monitoring (one week, and never finished). The table in §12.6, with pre-agreed graduated responses.

Stages 0 through 2 are infrastructure and normalisation. They are where the benefit is, and they are the stages people skip in order to get to the modelling.

13.4 Ten rules for someone starting today

1. **Compute the transition matrix's second eigenvalue before anything else.** If its half-life is shorter than your rebalancing horizon, the project is over and you have spent ten minutes.
2. **Normalise your target by trailing volatility.** It is one line, it addresses the shift that actually dominates, and without it a fifth of your sample supplies two thirds of your loss.
3. **Assume the regime is volatility.** It nearly always is. Design for that and you will be right; design for a mean regime and the data will not support you.
4. **Never report a smoothed state probability without labelling it.** And never trade one.
5. **Sort states by fitted volatility, inside the estimation window.** Sorting by mean, or sorting on the full sample, is a look-ahead that hides well.
6. **Run the shuffled-regime control.** One rerun, and it is the difference between knowing and believing.
7. **Count episodes, not days.** Forty crises is your sample size, not five thousand observations, and the standard errors follow from the forty.
8. **Do the cost arithmetic before building.** 118 basis points gross against 176 basis points of costs is a project you should not start.
9. **Pool the means, switch the variances, and use heavy-tailed emissions.** The identification asymmetry says which parameters to let vary, and Student-t emissions stop single outliers from inventing regime changes.
10. **Try to falsify the regime rather than to find it.** Simulate from a no-regime process and confirm your pipeline reports nothing. If it reports something, everything else you have measured is uninterpretable.

13.5 What is known, what is not, and where the evidence points

Known. [Fact] Volatility comes in persistent levels, and models that represent this — discrete or continuous — forecast better than models that do not. Correlations rise in stressed periods, partly mechanically and partly not. Regime-aware position sizing reduces realised volatility and drawdown, reliably, across markets and decades. The likelihood-ratio test for the number of regimes is invalid, and the standard errors on regime-conditional means are far larger than the literature's presentation implies.

Not known. Whether financial "regimes" are genuinely discrete or a discretisation of a continuous latent process. The evidence is compatible with both and Diebold and Inoue (2001) showed the two are analytically confusable, so this may not be resolvable with return data at all. Whether regime-conditional expected returns exist in a form stable enough to trade — the recession- predictability results are real in sample and contested out of it. Whether the crowding of systematic de-risking has changed the dynamics of the regimes themselves: plausible, and not demonstrated. **[Hypothesis]**

Where the evidence points. This is the chapter's assessment, not a settled result. Most of the value practitioners attribute to regime models comes from three things that are not regime models: volatility-normalised targets, volatility-scaled positions, and regime-stationary features. When a fitted state model is justified at all, its correct use is to supply moments to a sizing rule and a covariance to a portfolio optimiser — never a label to a classifier. The worked example of §6.11 is consistent with all of it: on a century of US data the HMM's states separate volatility by a factor of 2.3 out of sample and separate returns not at all, and the gain from using them is a modest Sharpe improvement and a large drawdown reduction. And the highest-value unexploited idea in this space is the inverse one: **using regimes as environments to find the relationships that do not change, rather than as conditions to adapt to.** Invariance is a better match than adaptation to what a forty-episode sample can support, and it is where research time is best spent.

The largest open methodological problem, and it is the same one that afflicts trend-following: the long-history evidence that regime overlays add value comes disproportionately from firms that sell them, and the academic evidence is dominated by in-sample fits and smoothed state probabilities. A study that ran the protocol of §11 — rung-5 states, shuffled-regime controls, episode-counted standard errors, realistic stressed costs — across many markets and a century of data would settle more than any new model could. Nobody has an incentive to publish it, which is precisely why it would be worth reading.

The consolation is that the parts of this field that survive scrutiny are the parts you can verify yourself in an afternoon of simulation, and they point in a consistent direction. Volatility is knowable and worth acting on. Direction, in regimes as everywhere else, is not.

Appendix A. Concepts and prerequisites

Everything the main text leans on without stopping to explain. Entries are ordered by **dependency**, not alphabetically — later entries use earlier ones — so the appendix reads as a build-up rather than a dictionary. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper.

Index. Where each concept is first used:

Concept	First used	Concept	First used
Filtration and measurability	§1.2	Bayesian nonparametrics	§6.4
Predictive density	§1.2	Sharpe ratio and its error	§5.1
PIT and pseudo-residuals	§6.9	Risk premia and priced factors	§2.1
Markov chains and mixing	§1.3	Limits to arbitrage	§2.3
Covariate-dependent transitions	§6.6	Complementary slackness	§2.1
Sojourn times and the constant hazard	§6.5	Dataset shift	§9.1
Mixture distributions	§1.1	Importance weighting and ESS	§9.1
Heavy tails and stable laws	§3.2	Partial pooling and shrinkage	§6.10
Student-t as a scale mixture	§6.3	Invariant prediction	§9.6
Latent variables and HMMs	§1.2	Gradient boosting	§9.3
Filtering, smoothing and Viterbi	§5.1	Neural-network components	§3.7

Concept	First used	Concept	First used
Maximum likelihood and EM	§5.2	Conditioning a network	§10.8
Standard errors for latent models	§6.1	Mixture of experts	§6.6
Online EM and forgetting	§6.8	Variational inference	§5.10
Identification and label switching	§5.1	Posterior collapse	§5.10
Bayesian computation for HMMs	§6.7	Purging, embargo, CPCV	§5.11
Non-standard LR tests	§1.1	Multiple testing	§3.6
ARCH, GARCH, realised vol	§1.2	Block bootstrap	§11.4
Long memory	§2.6	Forecast-comparison tests	§6.9
Unit roots and breaks	§2.6	Conformal prediction	§10.8
Change points and run length	§5.6	Real-time data vintages	§1.5
Clustering and distances	§1.5	Mean-variance sizing	§6.10
Mahalanobis distance and PCA	§3.6	Transaction-cost models	§12.2

A.1 Filtration and measurability

The idea. Time-series modelling needs a formal way to say "the information a decision-maker actually has at time t", so that you can check whether a quantity was computable then. A *filtration* is that bookkeeping device: a family of information sets, one per time point, that only ever grows.

Formally. A filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is an increasing family of σ-algebras, $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$, on the underlying probability space. A random variable X is **$\mathcal{F}_t$-measurable** if its value is determined by the information in $\mathcal{F}_t$ — equivalently, if $\{X \leq x\} \in \mathcal{F}_t$ for every x. A process $\{X_t\}$ is **adapted** if X_t is $\mathcal{F}_t$-measurable for every t, and **predictable** if X_t is $\mathcal{F}_{t-1}$-measurable.

Why it appears here. The entire filtration ladder of §5.1 is a statement about which σ-algebra a state estimate is measurable with respect to: $\xi_{t|t-1}$ is $\mathcal{F}_{t-1}$-measurable and therefore tradable at the open of bar t; $\xi_{t|t}$ is only $\mathcal{F}_t$-measurable; $\xi_{t|T}$ is measurable with respect to information that will not exist for years. "Look-ahead bias" is the informal name for using a quantity that is not measurable with respect to the information set your decision rule claims to use.

Deeper. Williams, *Probability with Martingales* (CUP, 1991), chapters 3 and 9. A companion note, Stochastic Processes, builds the measure-theoretic apparatus from σ-algebras up.

A.2 Conditional distributions and the predictive density

The idea. A forecast is not a number; it is a distribution. The object a probabilistic time-series model produces at each step is the distribution of the next observation given everything known so far.

Formally. The **one-step predictive density** is $p(y_t \mid \mathcal{F}_{t-1})$, the conditional density of y_t given the information available at $t-1$. A point forecast is a functional of it — the conditional mean $\mathbb{E}[y_t \mid \mathcal{F}_{t-1}]$ minimises expected squared error, the conditional median minimises expected absolute error — so choosing a point forecast is choosing a loss function, not a modelling step.

The quality of a density forecast is measured by a **scoring rule**. The **log score** is $\ln p(y_t \mid \mathcal{F}_{t-1})$ evaluated at the outcome that occurred; averaged over an out-of-sample period, higher is better. It is **proper**: in expectation it is maximised by reporting the true predictive density, so a forecaster cannot gain by shading the forecast. Maximum likelihood maximises the in-sample average log score.

Why it appears here. §1.2 defines a regime model *as* a predictive density with a particular structure, and §5.2 shows that the normalising constant of the Hamilton filter's update step is exactly this density, which is why maximising the likelihood is maximising one-step predictive performance and nothing else. §6.4, §6.9 and §6.11 compare models by their out-of-sample average log score.

Deeper. Gneiting & Katzfuss, "Probabilistic Forecasting," *Annual Review of Statistics and Its Application* 1 (2014), 125–151.

A.3 The probability integral transform and pseudo-residuals

The idea. If a forecast distribution is right, then the probability it assigned to "an outcome no larger than what happened" should look like a fair random number between 0 and 1, every day. Checking that is a way to test a whole predictive density, not just its mean.

Formally. For a continuous predictive distribution function $F_t(y) = \Pr(y_t \le y \mid \mathcal{F}_{t-1})$, the **probability integral transform** is $\text{PIT}_t = F_t(y_t)$. If F_t is the true conditional distribution, the PIT_t are independent and uniform on $[0, 1]$ (Diebold, Gunther & Tay, 1998). Applying the inverse standard normal distribution function gives **pseudo-residuals** $e_t = \Phi^{-1}(\text{PIT}_t)$, which are then independent standard normal; normal quantile plots and autocorrelation functions of e_t, e_t^2 and $|e_t|$ test the tails and the dynamics separately. For an HMM, F_t is the mixture $\sum_k \xi_{t|t-1}^{(k)} F(\cdot \mid \theta_k)$.

Why it appears here. §6.9 uses pseudo-residuals as the main model check for an HMM, computed from predicted probabilities and walk-forward parameters so the check is itself out of sample.

Deeper. Zucchini, MacDonald & Langrock (2016), chapter 6, in §4.2.

A.4 Markov chains, transition matrices, and the second eigenvalue

The idea. A Markov chain is the simplest model of a system that moves between a finite set of states with memory of only where it is now. Its long-run behaviour is governed by the eigenvalues of its transition matrix: the largest is always 1 and describes the equilibrium, and the second largest tells you how fast you forget where you started.

Formally. A time-homogeneous Markov chain on $\{1, \dots, K\}$ has transition matrix $\mathbf{P}$ with $p_{jk} = \Pr(s_{t+1} = k \mid s_t = j)$, rows summing to one. A **stationary distribution** π satisfies $\pi^\top \mathbf{P} = \pi^\top$; for an irreducible aperiodic (**ergodic**) chain it is unique and $\mathbf{P}^h \to \mathbf{1}\pi^\top$. Ordering the eigenvalues $1 = \lambda_1 > |\lambda_2| \ge \dots \ge |\lambda_K|$, the convergence is geometric at rate $|\lambda_2|$, which is called the **spectral gap** $(1 - |\lambda_2|)$ or the chain's **mixing rate**. For $K = 2$, $\lambda_2 = p_{11} + p_{22} - 1$.

A state k is **absorbing** if $p_{kk} = 1$: once entered, it is never left. A fitted $\hat{p}_{kk}$ very close to 1 behaves almost the same way within any realistic sample.

Why it appears here. §1.3 uses $|\lambda_2|$ to put a hard ceiling on the horizon at which a regime model can inform a decision: beyond a few multiples of $\ln 2/(-\ln|\lambda_2|)$ bars, the state forecast has reverted to the unconditional distribution and carries no information. §5.3 warns that near-absorbing fitted states mean the model has found a break rather than a recurring regime.

Deeper. Levin & Peres, *Markov Chains and Mixing Times*, 2nd ed. (AMS, 2017), chapter 12 for the eigenvalue connection.

A.5 Covariate-dependent transitions and the multinomial logit

The idea. A transition matrix whose rows change with the weather. Each row is still a probability distribution over where to go next, but the probabilities are now a smooth function of something observed.

Formally. The **multinomial logit** maps K real scores $\zeta_1, \dots, \zeta_K$ to probabilities $\exp(\zeta_k)/\sum_l \exp(\zeta_l)$, which are positive and sum to one. Adding the same constant to every score changes nothing, so one score per row is fixed at zero for identification. A **time-varying transition probability** model sets the scores of row j to $z_{t-1}^\top \psi_{jk}$, with $\psi_{jj} = 0$, so $p_{jk,t} = \exp(z_{t-1}^\top \psi_{jk})/\sum_l \exp(z_{t-1}^\top \psi_{jl})$. For two states each row reduces to the logistic function $1/(1+e^{-u})$ of one linear score. The chain is no longer homogeneous, so the stationary distribution and the second eigenvalue of §1.3 become functions of z.

Why it appears here. §6.6 builds covariate-driven transitions this way, and §7.2 records the limit in which they become a mixture of experts.

Deeper. Diebold, Lee & Weinbach (1994) and Filardo (1994), both in §4.1.

A.6 Sojourn times and the constant hazard

The idea. How long a Markov chain stays in a state before leaving is determined entirely by its self-transition probability, and the resulting distribution has a peculiar property: however long you have already been in a state, the chance of leaving in the next step is unchanged.

Formally. The **sojourn time** in state k is $D_k = \min\{d \ge 1 : s_{t+d} \ne k \mid s_t = \dots = s_{t+d-1} = k\}$, and for a Markov chain $\Pr(D_k = d) = p_{kk}^{d-1}(1 - p_{kk})$ — a geometric distribution with mean $1/(1 - p_{kk})$ and modal value 1. Its **hazard**

rate, $\Pr(D_k = d \mid D_k \geq d) = 1 - p_{kk}$, is constant in d: the distribution is memoryless. A **hidden semi-Markov model** replaces the geometric with an arbitrary duration distribution, restoring duration dependence at the cost of substantially more machinery. The usual choice is the shifted **negative binomial**, $\Pr(D_k = d) = \binom{d+m-2}{d-1} q^m (1-q)^{d-1}$, which is the geometric at $m = 1$, has a rising hazard for $m > 1$ and a falling hazard for $m < 1$. Any duration distribution can be approximated by an ordinary HMM in which each regime is a block of sub-states visited in sequence; the approximation is exact up to the block length.

In real time the current spell is **right-censored**: you know it has lasted at least d periods, not how long it will last. The relevant quantity is then the conditional distribution $\Pr(D_k = d+u \mid D_k > d)$ of the remaining duration, which under the geometric is the same for every d and under a semi-Markov model is not.

Why it appears here. §6.5 identifies the constant hazard as the reason fitted regime paths flicker, and §12.2 shows that flicker is what destroys a strategy through turnover. §10.1 recommends time-since-transition as a feature precisely because it restores the duration information the Markov assumption discards.

Deeper. Yu, "Hidden Semi-Markov Models," *Artificial Intelligence* 174 (2010), 215–243.

A.7 Mixture distributions and scale mixtures of normals

The idea. A mixture is what you get when the data come from several different distributions and you do not observe which. Mixing normals with different variances is the standard way to build a fat-tailed, symmetric distribution out of thin-tailed ingredients.

Formally. A finite mixture has density $p(y) = \sum_{k=1}^{K} \pi_k f(y \mid \theta_k)$ with weights $\pi_k \geq 0$ summing to one. A **scale mixture of normals** is the special case $f(y \mid \theta_k) = \mathcal{N}(y; \mu, \sigma_k^2)$ with a common mean: the result is symmetric about μ and has excess kurtosis $3\left(\mathbb{E}[\sigma^4]/(\mathbb{E}[\sigma^2])^2 - 1\right) > 0$ whenever the σ_k differ. Finite mixtures of Gaussians are dense in the space of densities, so with enough components they approximate anything.

Why it appears here. §1.1 uses the universal-approximation property to argue that "the data have regimes" is not on its own a testable claim. §7.2 notes that a two-state model differing only in variance produces an unconditional distribution that is a scale mixture of normals, hence indistinguishable from a fat-tailed i.i.d. law on unconditional moments alone — which is §2.6's null.

Deeper. Frühwirth-Schnatter, *Finite Mixture and Markov Switching Models* (Springer, 2006), chapters 1–3.

A.8 Heavy tails, stable distributions, and infinite variance

The idea. Financial returns have far more extreme observations than a normal distribution allows. The first explanation offered was not that two regimes take turns but that returns follow a single law with genuinely heavier tails — and the most radical version of that claim gives up finite variance altogether.

Formally. A distribution has **heavy tails** if $\Pr(|X| > x) \sim c x^{-\alpha}$ for some tail index $\alpha > 0$; moments of order $\geq \alpha$ do not exist. **Excess kurtosis** $\mathbb{E}[(X-\mu)^4]/\sigma^4 - 3$ is the standard finite-sample summary and is positive for **leptokurtic** (fat-tailed, peaked) distributions. The **stable** (or Lévy-stable) family is the class closed under convolution — sums of independent stable variables are stable — parameterised by $\alpha \in (0, 2]$, with $\alpha = 2$ the Gaussian and $\alpha < 2$ implying $\mathbb{E}[X^2] = \infty$.

Why it appears here. §3.2: Mandelbrot's stable-Paretian proposal was the field's first answer to non-normality, and it is a *single-regime* answer. §2.6 and §7.2 record that the debate was never settled — a two-state scale mixture (mixtures) and a fat-tailed i.i.d. law produce the same unconditional moments, so unconditional evidence cannot separate them.

Deeper. Nolan, *Univariate Stable Distributions* (Springer, 2020), chapter 1; Cont, "Empirical Properties of Asset Returns," *Quantitative Finance* 1(2) (2001), 223–236, for what the stylised facts actually are.

A.9 The Student-t distribution as a scale mixture of normals

The idea. A Student-t variable is a normal variable whose variance was itself drawn at random, from a distribution that occasionally produces a large value. That is why it has heavy tails, and why it can be estimated with the same tools as a normal.

Formally. With location μ, scale σ and degrees of freedom $\nu > 0$, the density is $\frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\nu\pi}\,\sigma}\left(1 + \frac{(y-\mu)^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}$. Its tails decay as $|y|^{-(\nu+1)}$, so moments of order ν and above do not exist; the variance is $\sigma^2\nu/(\nu-2)$ for $\nu > 2$, and $\nu \to \infty$ recovers the normal. The **scale-mixture representation** is $y = \mu + \sigma\varepsilon/\sqrt{\omega}$ with $\varepsilon \sim \mathcal{N}(0,1)$ and $\omega \sim \text{Gamma}(\nu/2, \nu/2)$

(shape and rate) independent. Given y, the latent precision has conditional mean $\mathbb{E}[\omega \mid y] = (\nu+1)/(\nu+(y-\mu)^2/\sigma^2)$, which is small for outlying y. Treating ω as missing data turns maximum likelihood for μ and σ into an EM algorithm of weighted least squares, with the outliers downweighted.

Why it appears here. §6.3 recommends Student-t emissions for daily returns, and the weights ω_{tk} there are this conditional mean, state by state. §7.2 lists the representation among the exact equivalences.

Deeper. Lange, Little & Taylor, "Robust Statistical Modeling Using the t Distribution," *Journal of the American Statistical Association* 84(408) (1989), 881–896; Bulla (2011) in §4.3 for HMMs.

A.10 Latent variables and hidden Markov models

The idea. A latent-variable model explains observed data by positing an unobserved quantity that, if you knew it, would make the observations simple. A hidden Markov model is the version where the unobserved quantity is a Markov chain.

Formally. An HMM specifies a latent chain s_t with transition matrix **P** and **emission** (observation) densities $f(y_t \mid s_t = k; \theta_k)$, with the conditional independence assumptions that s_t depends on the past only through s_{t-1}, and y_t depends on everything only through s_t. The joint density of a path is $p(y_{1:T}, s_{1:T}) = \pi_{s_1} f(y_1 \mid \theta_{s_1}) \prod_{t=2}^{T} p_{s_{t-1}s_t} f(y_t \mid \theta_{s_t})$, and the observed-data likelihood is this summed over all K^T paths — which is why the recursion of §5.2 matters.

Why it appears here. Every latent-state model in §5 is an HMM or a generalisation of one. Hamilton's Markov-switching regression is an HMM whose emissions are regressions.

Deeper. Rabiner (1989) for the clearest exposition; Zucchini, MacDonald & Langrock (2016) for practice; Cappé, Moulines & Rydén (2005) for theory. All in §4.2.

A.11 Bayes filtering, smoothing, and Viterbi decoding

The idea. Three different questions about a latent state, with three different answers: what is the state now, given the data so far (filtering); what was the state in the past, given all the data (smoothing); and what single sequence of states best explains the whole record (decoding).

Formally. Given observations $y_{1:T}$:

- **Filtering** computes $\Pr(s_t \mid y_{1:t})$ by the forward recursion of §5.2 — predict through the chain, then reweight by the new observation's likelihood. Cost $O(TK^2)$.
- **Smoothing** computes $\Pr(s_t \mid y_{1:T})$ by combining the forward pass with a backward pass (Kim's recursion, or the α–β form). Same cost, but requires the whole sample.
- **Viterbi decoding** computes $\arg\max_{s_{1:T}} \Pr(s_{1:T} \mid y_{1:T})$, the single most probable *path*, by dynamic programming with max in place of $\sum$. Note that the Viterbi path is not the sequence of per-time-point marginal modes, and the two can differ substantially.

The Kalman filter is the same three algorithms for a linear-Gaussian continuous state; the discrete and continuous cases are instances of one recursion.

The **backward variable** $b_t^{(k)} \propto p(y_{t+1:T} \mid s_t = k)$ is computed from $b_T^{(k)} = 1$ by $b_t^{(j)} \propto \sum_k p_{jk}\, f(y_{t+1} \mid \theta_k)\, b_{t+1}^{(k)}$, and the smoothed probability is proportional to $\xi_{t|t}^{(k)} b_t^{(k)}$; this is the "$\alpha$–$\beta$" form, with α the unnormalised forward variable. The **pairwise** posterior $\Pr(s_{t-1} = j, s_t = k \mid \mathcal{F}_T)$ multiplies the filtered belief at $t-1$, the transition probability, the emission density at t and the backward variable at t. Viterbi decoding, the jump model's state assignment and offline change-point segmentation are all **dynamic programmes**: each finds the best path by keeping, for every state at every date, only the best way of arriving there.

Why it appears here. §5.1's ladder is precisely the distinction between filtering and smoothing plus one extra step of prediction. §7.2 shows the statistical jump model is a Viterbi decode of a restricted HMM. §6.1 uses the backward variable and the pairwise posteriors in the E-step, and §6.2 compares the four state beliefs, including the Viterbi path; §5.6 and §5.7 rely on the same dynamic-programming idea.

Deeper. Särkkä & Svensson, *Bayesian Filtering and Smoothing*, 2nd ed. (CUP, 2023) — free online, and it presents the discrete and continuous cases under one framework.

A.12 Maximum likelihood and the EM algorithm

The idea. With latent variables, the likelihood is a sum over all unobserved configurations and is awkward to maximise directly. EM sidesteps this by alternating between guessing the latent variables' distribution and maximising as if that guess were data.

Formally. The **E-step** computes $Q(\theta \mid \theta^{(i)}) = \mathbb{E}_{s_{1:T}|y,\theta^{(i)}}[\ln p(y, s_{1:T} \mid \theta)]$; the **M-step** sets $\theta^{(i+1)} = \arg\max_\theta Q(\theta \mid \theta^{(i)})$. Each iteration weakly increases the observed-data likelihood, so the sequence converges — to a *local* optimum or a saddle point, not necessarily the global one. For HMMs the E-step is forward-backward and the M-step is weighted moment matching; the pair is called **Baum-Welch**.

The alternative to EM is **direct numerical maximisation** of the log-likelihood. Constrained parameters are first mapped to unconstrained ones — each transition row through a multinomial logit, each scale through its logarithm — so that a **quasi-Newton** optimiser, which builds up an approximation to the curvature from successive gradients, can search freely.

Why it appears here. §6.1 is the practical failure list: local optima (restart many times), degeneracy as $\sigma_k \to 0$ (the likelihood is unbounded, so constrain or regularise), and the small effective sample for transition probabilities. §6.1 recommends EM to get close to the optimum and quasi-Newton steps to finish.

Deeper. Dempster, Laird & Rubin, "Maximum Likelihood from Incomplete Data via the EM Algorithm," *JRSS-B* 39(1) (1977), 1–38.

A.13 Standard errors for latent-variable models

The idea. A maximum-likelihood estimate comes with a measure of how sharply the likelihood peaks around it. The sharper the peak, the more precisely the data determine the parameter. Two ways to measure it: the curvature at the peak, or the spread of estimates across simulated samples.

Formally. The **observed information** is the negative Hessian of the log-likelihood at the estimate, $\mathcal{J}(\hat\theta) = -\nabla^2 \ell(\hat\theta)$, and $\mathcal{J}(\hat\theta)^{-1}$ approximates the covariance of $\hat\theta$. For an HMM the Hessian is computed numerically from the filter's log-likelihood, or from the EM quantities by Louis's identity. The approximation relies on the estimate being interior and the likelihood being locally quadratic, and both fail for a transition probability near 1; working with logit(p_{kk}) helps. The **parametric bootstrap** avoids the approximation: simulate many samples from the fitted model, refit each, and read the standard error off the spread of the refitted parameters.

Standard errors for a function of the parameters, such as the expected spell length $1/(1 - p_{kk})$, follow from the **delta method**: for a smooth g, $\mathrm{se}(g(\hat\theta)) \approx |g'(\hat\theta)|\ \mathrm{se}(\hat\theta)$, with the gradient in place of the derivative for a vector θ.

Why it appears here. §6.1 recommends the bootstrap for HMM standard errors and shows that a turbulent state's expected spell length is uncertain by about 16% even when the state path is known.

Deeper. Louis, "Finding the Observed Information Matrix When Using the EM Algorithm," *Journal of the Royal Statistical Society: Series B* 44(2) (1982), 226–233; Zucchini, MacDonald & Langrock (2016), chapter 3, in §4.2.

A.14 Online EM and exponential forgetting

The idea. Instead of refitting a model from scratch whenever data arrive, update a running summary of the data and re-derive the parameters from it. Discounting old observations in that summary lets the parameters drift with the market.

Formally. EM's M-step for many models, the HMM included, depends on the data only through expected **sufficient statistics** — weighted counts, sums and sums of squares. **Online EM** updates them recursively, $\hat S_t = (1 - \gamma_t)\hat S_{t-1} + \gamma_t\, \tilde S_t$, where $\tilde S_t$ is the new observation's expected contribution and γ_t a step size, then recomputes the parameters from $\hat S_t$. A step size that shrinks like $1/t$ converges to the batch estimate; a constant step size γ weights the observation from u periods ago by $\gamma(1-\gamma)^u$, which is **exponential forgetting** with factor $\lambda = 1 - \gamma$ and effective memory of about $1/\gamma$ observations — the same weighting as an EWMA.

Why it appears here. §6.8 uses forgetting to let HMM parameters drift, and recommends a shorter memory for volatilities than for transition probabilities.

Deeper. Cappé (2011) in §4.2; Nystrup, Madsen & Lindström (2017) in §4.3 for the financial application.

A.15 Identification, label switching, and likelihood degeneracy

The idea. Three distinct ways a mixture-like model can fail to have a well-defined answer: the parameters may not be recoverable even in principle; the labels attached to components may be arbitrary; and the likelihood may be infinite at degenerate parameter values.

Formally. A model is **identified** if distinct parameter values imply distinct distributions. Finite mixtures are identified only **up to permutation of the components** — the likelihood is invariant under relabelling — which is **label switching**. Resolving it requires an ordering constraint (for example $\sigma_1 < \sigma_2 < \dots$), and *which* constraint you choose is a modelling decision with consequences. Separately, a Gaussian mixture's likelihood is **unbounded**: placing one component's mean on a data point and letting its variance go to zero sends the likelihood to infinity, so the maximum-likelihood estimator does not exist without a constraint or a prior.

Why it appears here. §5.1's label leakage, §6.1 and §11.2's leak 3. Sorting states by fitted mean, or sorting on the full sample, is a look-ahead. §8.3 measures what happens when the identification rule has nothing to grip: with equal volatilities the sort order is arbitrary, and the strategy built on it turns a 0.50 Sharpe into zero.

Deeper. Frühwirth-Schnatter (2006), chapter 3, is the standard treatment of both problems.

A.16 Bayesian computation for HMMs: conjugate priors, Gibbs sampling and FFBS

The idea. Bayesian estimation replaces "the best parameters" with a distribution over plausible parameters. For an HMM that distribution has no closed form, but it can be explored by simulation: alternately guess the hidden state path given the parameters, and the parameters given the path, many times over.

Formally. A prior is **conjugate** when the posterior is in the same family. The **Dirichlet** distribution on a probability vector $(p_1, \dots, p_K)$, with density proportional to $\prod_k p_k^{a_k - 1}$, is conjugate to counts: after observing n_k outcomes of each type the posterior is Dirichlet$(a_1 + n_1, \dots, a_K + n_K)$, so the a_k act as prior pseudo-counts. The normal and inverse-gamma distributions are conjugate for a mean and a variance. **Gibbs sampling** draws each block of unknowns from its distribution conditional on the current values of the others; repeated, the draws converge to the joint posterior. For an HMM the blocks are the state path, the transition rows and the emission parameters. **Forward filtering, backward sampling** draws the whole path at once: run the filter, draw s_T from $\xi_{T|T}$, then draw each s_t from $\Pr(s_t = j \mid s_{t+1}, \mathcal{F}_t) \propto \xi^{(j)}_{t|t} p_{j s_{t+1}}$ going backwards. Label switching (the previous entry) reappears as draws that swap state labels between sweeps.

Two practical notes. The posterior **mode** of a Dirichlet is $(a_k - 1 + n_k)/\sum_l (a_l - 1 + n_l)$, so in penalised EM each prior parameter contributes $a_k - 1$ pseudo-counts, not a_k. And a ridge penalty $\frac{\rho}{2}\|\psi\|^2$ on regression-type coefficients is the log of a Gaussian prior with variance $1/\rho$, so penalised maximum likelihood is a posterior mode. MCMC output must be checked for **mixing**: draws are autocorrelated, and samplers that update one state at a time mix far more slowly than block samplers such as FFBS. Running several chains from dispersed starts and comparing them is the minimum check.

Why it appears here. §6.7 uses this machinery for priors that encode persistence and remove degeneracy, and recommends its posterior-mode shortcut, penalised EM, as the default fitting routine. §6.6 recommends a ridge penalty on the transition coefficients, which is the same idea.

Deeper. Chib (1996) and Scott (2002) in §4.2; Frühwirth-Schnatter (2006), chapters 11–13.

A.17 Likelihood-ratio tests when the null is non-standard

The idea. The usual χ^2 null distribution for a likelihood-ratio test requires the models to be nested in a well-behaved way. Testing "is there an extra regime?" violates the requirements in two ways at once, and the standard critical values are then meaningless.

Formally. Under the null of K regimes, testing against $K + 1$ fails regularity because (i) the parameters θ_{K+1} of the extra component are **unidentified under the null** — any value gives the same likelihood, so the information matrix is singular (the *Davies problem*) — and (ii) the transition probabilities into the extra state lie **on the boundary** of the parameter space, where the asymptotic normality of the MLE fails. Hansen (1992) obtains a valid bound by treating the likelihood ratio as an empirical process indexed by the nuisance parameters and taking a supremum; Cho and White (2007) derive the limiting distribution under weaker conditions.

Two terms this drags in. **Information criteria** — AIC $= -2\ell + 2k$ and BIC $= -2\ell + k \ln n$ for k parameters — penalise fit by parameter count and are the usual fallback for choosing K; both are derived under the same regularity conditions that fail here, which is why they over-select. A **parametric bootstrap** sidesteps the theory entirely: simulate many samples *from*

the fitted K-state model, refit both K and $K+1$ to each, and read the null distribution of the likelihood-ratio statistic off the simulations rather than off an asymptotic formula.

Why it appears here. §1.1 and §3.5 state the problem, and §8.2 works it through. The practical consequences are that likelihood improvement is not evidence, information criteria over-select, and the parametric bootstrap is the only routinely correct route. §6.4 applies them to choosing the number of states.

Deeper. Davies, "Hypothesis Testing When a Nuisance Parameter Is Present Only Under the Alternative," *Biometrika* 74(1) (1987), 33–43; then Hansen (1992) in §4.4.

A.18 ARCH, GARCH, stochastic volatility, and realised volatility

The idea. Several ways to say that volatility varies over time, differing in whether the variance is a deterministic function of the past, a latent random process, or something you measure directly from high-frequency data.

Formally. With $r_t = \sigma_t \varepsilon_t$ and $\varepsilon_t \sim (0,1)$ i.i.d.:

- **ARCH(q)**: $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i r_{t-i}^2$. The variance is $\mathcal{F}_{t-1}$-measurable — there is nothing latent to filter.
- **GARCH(1,1)**: $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$. Persistence is measured by $\alpha + \beta$; values near 1 imply shocks to variance decay very slowly ("IGARCH").
- **Stochastic volatility**: $\ln \sigma_t^2$ follows its own latent autoregression driven by a separate innovation. Now there *is* something to filter, and the state is continuous.
- **DCC** (dynamic conditional correlation; Engle, 2002) extends GARCH to many assets by modelling each series' variance separately and then letting the *correlation* matrix follow its own GARCH-like recursion — the multivariate member of this family, and the base that §12.4's regime-switching DCC extends.
- **Realised volatility**: $\mathrm{RV}_t = \sum_i r_{t,i}^2$ summed over intraday returns, a direct estimate rather than a model. The **HAR** (heterogeneous autoregressive) model regresses future RV on its daily, weekly, and monthly averages — three horizons standing in for three classes of trader.
- **EWMA**: $\hat{\sigma}_t^2 = (1-\lambda) r_{t-1}^2 + \lambda \hat{\sigma}_{t-1}^2$, which is GARCH(1,1) with $\omega = 0$ and $\alpha + \beta = 1$ — one parameter, no estimation, and the baseline that the rest of this list has to beat. Its **half-life** is $\ln 2 / (-\ln \lambda)$ bars.

Why it appears here. §1.2 names GARCH and stochastic volatility as the continuous alternatives to a regime model and §1.5 distinguishes them from regimes; §5.4 notes Hamilton and Susmel's finding that apparent IGARCH persistence is partly unmodelled regime switching; §8.7 argues the continuous models usually win.

Deeper. Andersen, Bollerslev, Christoffersen & Diebold, "Volatility and Correlation Forecasting," *Handbook of Economic Forecasting* vol. 1 (2006).

A.19 Long memory and fractional integration

The idea. A process has long memory when its autocorrelations decay slowly enough that they are not summable — the past keeps mattering far longer than an ARMA model can express. It matters here because a process with occasional regime changes looks exactly like one.

Formally. A stationary process has **long memory** if its autocorrelation satisfies $\rho_k \sim c k^{2d-1}$ as $k \to \infty$ for some $d \in (0, 1/2)$, so that $\sum_k |\rho_k| = \infty$; equivalently its spectral density diverges at the origin. The canonical model is **ARFIMA(p, d, q)**, which applies the fractional differencing operator $(1-L)^d$ with non-integer d.

Why it appears here. §2.6 and §7.2: Diebold and Inoue (2001) prove that stochastic regime switching generates estimated long memory and vice versa, even asymptotically. The two are near-observationally-equivalent, which is why a likelihood improvement from adding regimes is not evidence that regimes exist.

Deeper. Beran, Feng, Ghosh & Kulik, *Long-Memory Processes* (Springer, 2013), chapter 1; and Diebold & Inoue (2001) in §4.4.

A.20 Unit roots, non-stationarity, and the break/persistence confusion

The idea. A series can wander without returning to any fixed level either because shocks to it never die away, or because its level occasionally jumps to a new value. In finite samples these look almost identical, and the confusion is the direct ancestor of §2.6's regime-versus-long-memory problem.

Formally. $y_t = \rho y_{t-1} + \varepsilon_t$ has a **unit root** when $\rho = 1$, in which case shocks are permanent, the variance grows without bound, and the process is **non-stationary** (no time-invariant distribution). Tests such as augmented Dickey-Fuller

test $H_0 : \rho = 1$. A **trend-stationary with breaks** alternative instead has $\rho < 1$ around a deterministic level that shifts at a few dates. Perron (1989) showed that ignoring a genuine break makes the unit-root null very hard to reject, and conversely.

Why it appears here. §2.6 and §4.4: this is the precedent for the document's central identification worry. Occasional breaks masquerade as persistence and persistence masquerades as breaks, in the unit-root literature (Perron), the long-memory literature (long memory; Diebold and Inoue), and the regime literature alike. It is the same confusion three times.

Deeper. Hamilton, *Time Series Analysis* (Princeton, 1994), chapters 15–17.

A.21 Change points, run length, and segmentation

The idea. A different question from "which state are we in": has the data-generating process changed, and when? Offline you segment a completed record; online you maintain a belief about how long it has been since the last change.

Formally. Offline **multiple change-point detection** partitions $\{1, \dots, T\}$ into $m+1$ segments minimising $\sum_i \mathcal{C}(y_{\tau_i+1:\tau_{i+1}}) + \beta m$, where $\mathcal{C}$ is a segment cost (often twice the negative log-likelihood) and β penalises segments. Dynamic programming solves this exactly; **PELT** does so in $O(T)$ amortised time by pruning candidates that can never be optimal. Online, **Bayesian online change-point detection** maintains a posterior over the **run length** ρ_t — the number of periods since the last change — via $p(\rho_t, y_{1:t}) = \sum_{\rho_{t-1}} p(\rho_t \mid \rho_{t-1})\, p(y_t \mid \rho_{t-1}, y^{(\rho)})\, p(\rho_{t-1}, y_{1:t-1})$, and forecasts by mixing over run lengths.

Why it appears here. §5.6 argues change-point methods answer the "has it changed" question better than regime models do, and §12.6 recommends running one on your own strategy returns as the cleanest available death detector.

Deeper. Truong, Oudre & Vayatis (2020) for the offline survey; Adams & MacKay (2007) for BOCPD. Both in §4.5.

A.22 Clustering: k-means, density-based methods, and distances between distributions

The idea. Unsupervised grouping of observations, and the question of what "similar" means. When the objects being grouped are themselves distributions or covariance matrices rather than points, you need a distance that respects that.

Formally. k**-means** minimises $\sum_t \|u_t - c_{z_t}\|^2$ over assignments z and centroids c, alternating between assigning each point to its nearest centroid and recomputing centroids as means. It assumes roughly spherical, equal-sized clusters and requires k in advance. **Density-based** methods (DBSCAN, HDBSCAN) instead define clusters as connected regions of high density, need no k, produce arbitrary shapes, and label sparse points as noise. **Agglomerative** methods merge the closest pair repeatedly, producing a dendrogram. For distributions, the **Wasserstein** (optimal-transport, "earth mover's") distance $W_p(P,Q) = \left(\inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{(x,y)\sim\gamma} \|x - y\|^p\right)^{1/p}$ measures the minimum cost of transporting one distribution onto the other, and unlike a KL divergence it is finite and well-behaved for distributions with different supports.

Why it appears here. §1.5, where regimes (partitions of time) are told apart from cross-sectional clusters, and §3.7 and §5.8, where clustering is one route to regime definition — and where the document's objection is that plain clustering has no persistence term, which is exactly what §5.7's jump penalty adds.

Deeper. Campello, Moulavi & Sander, "Density-Based Clustering Based on Hierarchical Density Estimates," PAKDD (2013), for HDBSCAN; Peyré & Cuturi, "Computational Optimal Transport", arXiv:1803.00567, for Wasserstein distances.

A.23 Mahalanobis distance, PCA, and the absorption ratio

The idea. Two ways to summarise a cross-section of returns in one number: how unusual today's pattern is relative to its historical shape, and how concentrated the market's risk currently is in a few common factors.

Formally. The **Mahalanobis distance** of an observation $r \in \mathbb{R}^N$ from a distribution with mean μ and covariance Σ is $d = (r-\mu)^\top \Sigma^{-1} (r-\mu)$. It is scale-invariant and accounts for correlation, so it is large both when individual moves are extreme and when the *relative pattern* of moves is unusual. Under multivariate normality, $d \sim \chi^2_N$. **Principal component analysis** diagonalises $\Sigma = V \Lambda V^\top$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$; the **absorption ratio** is $\sum_{i=1}^{n} \lambda_i / \sum_{i=1}^{N} \lambda_i$ for a chosen n, the share of total variance explained by the leading components.

Why it appears here. §3.6 introduces the turbulence index by name. §5.9 presents both as scalar regime indices needing no latent state and no K, and says that if such an index gets you most of the benefit, the latent-state model is a research project rather than an edge; §8.7 lists a turbulence index among the continuous alternatives to fit first. Note that Σ^{-1} is unstable when N approaches the estimation window length; shrink it.

Deeper. Kritzman & Li (2010) and Kritzman, Li, Page & Rigobon (2011), both in §4.6; Ledoit & Wolf, "Honey, I Shrunk the Sample Covariance Matrix," *Journal of Portfolio Management* 30(4) (2004), 110–119, for the shrinkage.

A.24 Bayesian nonparametrics: Dirichlet processes and the sticky HDP-HMM

The idea. Rather than choosing the number of clusters or states in advance, put a prior over models with *any* number of them and let the posterior decide, with the number allowed to grow as data accumulate.

Formally. A **Dirichlet process** $\mathrm{DP}(\alpha, H)$ is a distribution over distributions: draws are discrete with probability one, and the stick-breaking construction generates weights $\beta_k = \beta'_k \prod_{j<k}(1-\beta'_j)$ with $\beta'_k \sim \mathrm{Beta}(1, \alpha)$. Using it as the prior on mixture weights gives an infinite mixture whose effective number of occupied components grows like $\alpha \ln n$. A **hierarchical** DP shares one set of atoms across several DPs, which is what lets every row of an HMM's transition matrix draw from the same countable state set — the **HDP-HMM**. The **sticky** variant adds κ to the self-transition weight, encoding a prior belief in persistence and preventing the rapid state-switching that plain HDP-HMMs are prone to.

Why it appears here. §6.4 offers this as the principled answer to "how many regimes", and §4.2 recommends Fox et al. (2011). The document's reservation is practical rather than theoretical: it returns more states than a trading decision can use.

Deeper. Teh & Jordan, "Hierarchical Bayesian Nonparametric Models with Applications," in *Bayesian Nonparametrics* (CUP, 2010); Fox et al. (2011) in §4.2.

A.25 The Sharpe ratio, the information coefficient, and their standard errors

The idea. The standard performance measures, and — more importantly here — how badly they are estimated. Almost every disagreement about whether a regime strategy works is really a disagreement about sampling error.

Formally. The **Sharpe ratio** is $S = \mathbb{E}[r - r_f] / \mathrm{sd}(r)$, conventionally annualised by multiplying a per-bar estimate by $\sqrt{A}$. For i.i.d. normal returns over y years its estimator has approximate standard error

$$\mathrm{se}(\widehat{S}) \approx \sqrt{\frac{1 + S^2/2}{y}},$$

which depends on the **calendar span**, not the sampling frequency — the same asymmetry as §8.1. Serial correlation and non-normality inflate it further. The **information coefficient** is the cross-sectional correlation between forecast and realised return; the **fundamental law of active management** relates it to the achievable information ratio as $\mathrm{IR} \approx \mathrm{IC}\sqrt{N}$ for N independent bets.

Why it appears here. §5.1 and §6.11 quote Sharpe ratios and their error first. §11.5: four years of turbulent time gives $\mathrm{se}(\widehat{S}) \approx 0.53$, so within-regime performance claims are mostly noise. §8.3 and §8.4 report Sharpe ratios throughout. §6.11 quotes the standard error of a Sharpe ratio over 81 years of data.

Deeper. Lo, "The Statistics of Sharpe Ratios," *Financial Analysts Journal* 58(4) (2002), 36–52 — including the correction for serial correlation.

A.26 Risk premia, priced factors, and discount-rate variation

The idea. Asset-pricing vocabulary that the economic-mechanism sections assume. An asset's expected return above the risk-free rate compensates for bearing some risk; a variable is "priced" when exposure to it earns compensation; and prices move either because expected cash flows changed or because the rate at which they are discounted changed.

Formally. With stochastic discount factor M_{t+1}, no-arbitrage implies $\mathbb{E}_t[M_{t+1} R_{t+1}] = 1$ for every gross return, which rearranges to $\mathbb{E}_t[R^e_{t+1}] = -\mathrm{Cov}_t(M_{t+1}, R^e_{t+1}) / \mathbb{E}_t[M_{t+1}]$ for excess returns: **risk premia are compensation for covariance with bad times**. A **factor** f is **priced** if M loads on it, so that assets with higher exposure β_f earn higher average returns. Campbell's return decomposition splits unexpected returns into news about future cash flows and news about future discount rates; the empirical finding that most price variation is **discount-rate variation** means expected returns move a great deal over time — which is what makes conditional and regime-based models worth considering at all. A **no-arbitrage objective** in a machine-learning model (§5.10) means training the model so that its implied M prices the assets, rather than fitting returns directly.

Why it appears here. §2.1 (intermediary net worth as a priced state variable), §2.5 (Fama-French and Cochrane on cyclical expected returns), and §5.10 (Chen, Pelger and Zhu's end-to-end objective).

Deeper. Cochrane, *Asset Pricing*, rev. ed. (Princeton, 2005), chapters 1 and 20; Cochrane, "Presidential Address: Discount Rates," *Journal of Finance* 66(4) (2011), 1047–1108.

A.27 Limits to arbitrage and multiple equilibria

The idea. Textbook arbitrage assumes that mispricings are corrected by traders with unlimited capital and unlimited patience. Real arbitrageurs manage other people's money, face margin calls, and can be forced to liquidate exactly when the opportunity is best — which can make mispricings self-reinforcing rather than self-correcting.

Formally. Shleifer and Vishny (1997) model an arbitrageur whose capital depends on past performance: an adverse price move causes withdrawals, forcing liquidation, which moves the price further adversely. The feedback loop means the price impact of a shock is amplified rather than damped, and — when the feedback is strong enough — the system admits **multiple equilibria**: two or more self-consistent price levels for the same fundamentals, with which one obtains determined by beliefs or by history rather than by fundamentals. Formally these arise when the excess-demand function crosses zero more than once, which requires the positive feedback to dominate locally.

Why it appears here. §2.3. Multiple equilibria are the cleanest theoretical route to genuine *discreteness* other than binding constraints (complementary slackness), because distinct equilibria are distinct by construction. The document tags the family **[Hypothesis]** because distinguishing "the system jumped between equilibria" from "a large shock hit a continuous system" using return data alone has not been done convincingly.

Deeper. Shleifer & Vishny, "The Limits of Arbitrage," *Journal of Finance* 52(1) (1997), 35–55; Gromb & Vayanos, "Limits of Arbitrage," *Annual Review of Financial Economics* 2 (2010), 251–275.

A.28 Complementary slackness and occasionally binding constraints

The idea. When an optimiser faces an inequality constraint, exactly one of two things is true at the optimum: either the constraint is slack and its shadow price is zero, or it binds and its shadow price is positive. That "exactly one of two" is a genuine discreteness, produced by optimisation rather than assumed.

Formally. For $\min f(x)$ subject to $g(x) \leq 0$, the Karush-Kuhn-Tucker conditions include **complementary slackness**: $\lambda\, g(x^\star) = 0$ with $\lambda \geq 0$. So either $g(x^\star) < 0$ and $\lambda = 0$, or $g(x^\star) = 0$ and $\lambda \geq 0$. The multiplier λ — the shadow price of the constraint — jumps from zero to positive as the constraint starts to bind, and the solution's local behaviour changes qualitatively at that point.

Why it appears here. §2.1 argues this is the only mechanism family that predicts genuine discreteness in markets rather than continuous variation: intermediaries' leverage limits, margin requirements, and VaR budgets are inequality constraints, so "constrained" and "unconstrained" really are two states.

Deeper. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), section 5.5; then Brunnermeier & Pedersen (2009) for the financial mechanism.

A.29 Dataset shift: covariate shift, concept drift, prior shift

The idea. "The distribution changed" is too coarse to act on. There are several distinct things that can change, they have different consequences, and they call for different remedies.

Formally. Write training and deployment joints as $p_{\text{tr}}(x, y)$ and $p_{\text{te}}(x, y)$, and factor $p(x, y) = p(x)\, p(y \mid x)$. Then:

- **Covariate shift**: $p_{\text{tr}}(x) \neq p_{\text{te}}(x)$ but $p(y \mid x)$ unchanged.
- **Concept drift** (*real* drift): $p(y \mid x)$ changes. This is the only one that changes the function you are trying to learn.
- **Prior / label shift**: $p(y)$ changes with $p(x \mid y)$ fixed — the reverse factorisation, and the natural model for classification when class frequencies move.
- **Conditional-scale drift** is not standard terminology; §9.1 coins it for the case where $\mathbb{E}[y \mid x]$ is fixed but $\text{Var}(y \mid x)$ changes. It is formally a species of concept drift under a distributional loss, but it leaves the conditional *mean* function untouched, and its remedy is entirely different, so the document separates it.

Why it appears here. §9.1 and §9.2 use this taxonomy as the decision rule for which integration scheme to reach for, and §9.3 explains that a regime feature addresses only concept drift — which is why it so often does nothing.

Deeper. Quiñonero-Candela et al. (2009) and Gama et al. (2014), both in §4.7.

A.30 Importance weighting and effective sample size

The idea. If your training distribution is wrong, you can reweight training points to look like the distribution you care about. It is unbiased and it is expensive: reweighting concentrates the fit on fewer effective observations.

Formally. Under covariate shift with $p(y \mid x)$ unchanged, weighting each training point by $w(x) = p_{\text{te}}(x)/p_{\text{tr}}(x)$ makes the weighted training risk an unbiased estimate of the test risk (Shimodaira, 2000). But the variance of a weighted estimator grows with the dispersion of the weights, and the standard summary is the **effective sample size**

$$n_{\text{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \leq n,$$

with equality only for uniform weights. Critically, if the model is **correctly specified**, the unweighted MLE is already consistent under covariate shift, so weighting buys nothing and costs variance; weighting helps only under misspecification or in finite samples.

Why it appears here. §9.1's counterintuitive row, and §10.3, where n_{eff} is the number you should quote in place of T after any weighting scheme — an exponential-decay half-life of 5 years on 20 years of daily data leaves roughly 3,200 effective observations out of 5,040.

Deeper. Shimodaira (2000) in §4.7; Sugiyama, Krauledat & Müller, "Covariate Shift Adaptation by Importance Weighted Cross Validation," *JMLR* 8 (2007), 985–1005.

A.31 Partial pooling, shrinkage, and hierarchical models

The idea. When you have several related groups, you can estimate each separately (noisy), pool them all together (biased), or do something in between. The optimal in-between is determined by how much the groups genuinely differ relative to how noisily you measure each one.

Formally. In the canonical normal hierarchical model, group k has true parameter $\beta_k = \beta + \delta_k$ with $\delta_k \sim (0, \tau^2)$, and the per-group estimate $\hat{\beta}_k$ has sampling variance v_k. The posterior mean (and the minimum-MSE linear combination) is

$$\hat{\beta}_k^{\text{opt}} = w_k \hat{\beta}_k + (1 - w_k)\hat{\beta}_{\text{pooled}}, \qquad w_k = \frac{\tau^2}{\tau^2 + v_k},$$

so $w_k \to 1$ (no pooling) when between-group variation dominates and $w_k \to 0$ (complete pooling) when estimation noise dominates. This is the James-Stein phenomenon in its interpretable form: shrinkage toward a common mean strictly improves total MSE when groups are numerous and individually noisy.

Why it appears here. §6.10 first applies a pooling rule (equal state means, switching covariances). §9.5 turns the identification asymmetry into this decision rule — pool the mean model across regimes (τ^2 small against v), subset the risk model ($\tau^2 \gg v$) — and §10.4 gives recipes (linear, boosting, network) that implement partial pooling in practice.

Deeper. Gelman & Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models* (CUP, 2007), chapter 12.

A.32 Invariant prediction and invariant risk minimization

The idea. Instead of adapting a model to each environment, look for the relationship that is the *same* in every environment. Under assumptions, the predictors that survive are causal (not necessarily all of the causal ones), and a model built on them generalises to environments you have never seen.

Formally. Given data from environments $e \in \mathcal{E}$, **invariant causal prediction** seeks the subsets $\mathcal{S}$ of predictors for which $Y^e \mid X_S^e = x$ has the same distribution for all e. Peters, Bühlmann and Meinshausen (2016) test this hypothesis for each candidate subset and take the intersection of the accepted ones, giving conservative confidence sets for the causal predictors. **Invariant risk minimization** is the gradient-based relaxation: find a representation Φ such that a single classifier w is simultaneously optimal on top of Φ in every environment, enforced by penalising $\sum_e \|\nabla_{w|w=1} R^e(w \cdot \Phi)\|^2$.

Why it appears here. §9.6 and §10.6 argue this is the best-value underused idea available: it uses regimes as *environments* to filter features for robustness rather than as conditions to adapt to, which is a far better match to what a forty-episode sample can support.

Deeper. Peters, Bühlmann & Meinshausen (2016) and Arjovsky et al. (2019), both in §4.7 — and Rosenfeld, Ravikumar & Risteski (2021) for IRM's failure modes.

A.33 Gradient boosting: what the model class can and cannot represent

The idea. A boosted tree ensemble builds a prediction as a sum of small decision trees, each fitted to the errors of the ones before it. Two structural properties of that class drive everything in §10.8: trees split on thresholds, so they are invariant to monotone rescaling of features; and trees are piecewise constant, so they cannot extrapolate.

Formally. Gradient boosting (Friedman, 2001) fits $F_M(x) = F_0(x) + \nu \sum_{m=1}^{M} h_m(x)$, where each h_m is a regression tree fitted to the negative gradient of the loss at the current prediction and ν is the learning rate. A tree of depth d represents interactions of order at most d. Because every split is a threshold test $\mathbb{1}[x_j > c]$, the model is invariant to any strictly monotone transformation of x_j — which means an indicator $\mathbb{1}[x_j > c_0]$ you supply is exactly representable from x_j alone and adds no function to the class. And because leaves hold constants, for x_j outside the training range the prediction equals the boundary leaf's value, constant forever.

Why it appears here. §9.3 uses the tree's freedom to place its own threshold to explain why a regime dummy shows little importance. §10.8 makes the redundancy of a threshold-derived regime feature precise, and identifies the inability to extrapolate as the most serious regime-related weakness of GBTs and prescribes trailing normalisation or ranking as the fix. §10.4's `init_score` recipe uses the fact that boosting starts from an arbitrary offset F_0.

Deeper. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *Annals of Statistics* 29(5) (2001), 1189–1232; Grinsztajn et al. (2022) in §4.7 for why trees keep winning on tabular data.

A.34 Neural-network components: recurrence, switching dynamics, and ensembles

The idea. The pieces of deep-learning machinery that §5.10 and §10.7 name in passing. None is specific to finance.

Formally.

- **Recurrent networks and the LSTM.** A recurrent network carries a hidden state $h_t = \phi(h_{t-1}, u_t)$ along the sequence, so h_t is a learned summary of the history — a continuous, learned regime variable. The **long short-term memory** cell adds gated paths and an additive cell state, so gradients flow across long lags instead of vanishing.
- **Recurrent switching linear dynamical systems (rSLDS).** An HMM (latent variables and HMMs) whose discrete state indexes a *linear dynamical system* over a continuous latent state, and whose discrete transitions depend on that continuous state: $\Pr(s_t \mid s_{t-1}, h_{t-1})$. Hamilton's model is the special case with no continuous state and state-independent transitions; the TVTP model of §6.6 is the case where the continuous driver is observed.
- **Deep ensembles and MC dropout.** Train several networks from different random initialisations and take the spread of their predictions as an uncertainty estimate, or keep dropout active at inference and take the spread over stochastic forward passes. Both widen in regions the training data did not cover.
- **Adversarial heads and meta-learning**, in one line each. A gradient-reversal layer trains a shared representation to make an auxiliary regime classifier *fail*, so the representation carries no regime information. Model-agnostic meta-learning trains parameters so that a few gradient steps on new data perform well, at the cost of a nested optimisation.

Why it appears here. §3.7 and §5.10 and §10.7 (learned and switching latent states, adversarial invariance, meta-learning) and §10.8 (cheap uncertainty for sizing). The chapter's verdict on all of them is the same: correct machinery aimed at a sample size that rarely supports it, with the partial exception of ensembling, which is nearly free.

Deeper. Hochreiter & Schmidhuber, "Long Short-Term Memory," *Neural Computation* 9(8) (1997), 1735–1780; Linderman et al. (2017) in §4.7; Lakshminarayanan, Pritzel & Blundell, "Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles", NeurIPS 2017.

A.35 Conditioning a neural network: FiLM, and batch versus layer norm

The idea. There is a weak way and a strong way to tell a network about a side variable. The weak way is to append it to the input. The strong way is to let it *modulate* the network's internal activations multiplicatively.

Formally. Feature-wise linear modulation (FiLM) computes, from conditioning information c, a per-channel scale and shift and applies them to a hidden layer: $a^{(\ell)} \leftarrow \gamma^{(\ell)}(c) \odot a^{(\ell)} + \beta^{(\ell)}(c)$, where $a^{(\ell)}$ is the activation vector of layer ℓ. Because the scale is multiplicative, a single conditioning value can switch whole pathways on or off, which a concatenated input can only achieve through learned interactions. **Batch normalisation** standardises activations using statistics computed across the mini-batch during training and a running average at inference; **layer normalisation** standardises across the features of a single example, so it uses no cross-sample statistics and is unaffected by a shift in the input distribution at inference.

Why it appears here. §10.8: FiLM is the right injection point for a regime posterior, and batch norm's running statistics are calibrated to the training regime, which makes it a hidden source of distribution-shift failure.

Deeper. Perez et al. (2018) in §4.7; Dumoulin et al., "Feature-wise Transformations", *Distill* (2018), for the unifying view.

A.36 Mixture of experts and gating

The idea. Rather than one model that must handle every situation, train several specialists and a router that decides how much to trust each one. When the router's input is a regime posterior, this is regime switching in machine-learning clothing.

Formally. A mixture of experts predicts $\widehat{y}(x) = \sum_{k=1}^{K} g_k(z)\, f_k(x)$ with gate outputs $g_k(z) \geq 0$ summing to one, typically a softmax of a learned function of gating inputs z. Trained jointly, the gradient of the loss with respect to expert k is scaled by g_k, so experts specialise on the regions the gate assigns them. **Sparse gating** keeps only the top few g_k nonzero for efficiency, and then needs an auxiliary load-balancing loss to prevent **expert collapse**, where the gate routes everything to one expert.

Why it appears here. §10.5. The key structural point is that soft gating dominates hard selection: the prediction path stays continuous through a transition, so turnover is bounded exactly where confidence is lowest. §7.2 records that a smooth-transition model is exactly (for the conditional mean) a two-expert MoE with a logistic gate on one observable, and that a covariate-driven HMM whose transitions ignore the current state is one too (§6.6).

Deeper. Jacobs, Jordan, Nowlan & Hinton (1991) and Shazeer et al. (2017), both in §4.7.

A.37 Variational inference and the evidence lower bound

The idea. Exact Bayesian inference over a latent variable requires an integral that is usually intractable. Variational inference replaces the integral with an optimisation: pick a tractable family of approximate posteriors and find the member closest to the true one.

Formally. For latent z and data y, the marginal likelihood $\ln p(y) = \ln \int p(y \mid z) p(z)\, dz$ is intractable. Introducing any distribution $q(z)$ gives the decomposition

$$\ln p(y) = \underbrace{\mathbb{E}_q[\ln p(y \mid z)] - \mathrm{KL}(q(z) \,\|\, p(z))}_{\text{ELBO } \mathcal{L}(q)} + \mathrm{KL}(q(z) \,\|\, p(z \mid y)),$$

and since the final KL is non-negative, $\mathcal{L}(q)$ is an **evidence lower bound**: maximising it over q simultaneously tightens the bound and drives q toward the true posterior. **Amortised** inference replaces a per-datapoint optimisation with a neural network $q_\phi(z \mid y)$ that outputs the approximate posterior directly — the "encoder" of a variational autoencoder.

Why it appears here. §5.10's deep state-space models are trained this way, and §5.10 warns that variational approximation error is one more source of uncertainty stacked on top of every identification problem in §8. It is also the setting in which posterior collapse occurs.

Deeper. Blei, Kucukelbir & McAuliffe, "Variational Inference: A Review for Statisticians", *JASA* 112(518) (2017), 859–877.

A.38 Posterior collapse

The idea. A latent-variable model trained by variational inference can learn to ignore its own latent variable entirely, becoming a plain conditional model with an unused component — and the training loss will not tell you.

Formally. Take the ELBO of variational inference, $\mathbb{E}_{q(z|y)}[\ln p(y \mid z)] - \mathrm{KL}(q(z \mid y) \,\|\, p(z))$. If the decoder $p(y \mid z)$ is expressive enough to model y without z — a powerful autoregressive decoder, for instance — the optimiser can drive the KL term to zero by setting $q(z \mid y) = p(z)$, at which point the latent carries no information about the data and the encoder has been trained to ignore its input. Nothing in the loss signals this: the ELBO looks healthy throughout, because the decoder has absorbed the work.

Why it appears here. §5.10 lists it as the first failure mode of learned latent-state models, and §10.7 gives the diagnostic: ablate the latent and confirm that performance actually degrades. In the financial setting the second diagnostic matters just as much — regress the learned state on realised volatility and check whether you have discovered anything beyond a noisy volatility proxy.

Deeper. Bowman et al., "Generating Sentences from a Continuous Space", CoNLL 2016, where the phenomenon was first characterised.

A.39 Purging, embargo, and combinatorial purged cross-validation

The idea. Financial ML targets are realised over a window, so nearby training and test examples share outcome data. Standard cross-validation then measures how well the model memorises overlapping labels. Purging and embargoing remove the contaminated observations.

Formally. Let example i have feature time t_i and label window $[t_i, t_i + h_i]$. **Purging** removes from the training set every i whose label window overlaps any test example's label window. **Embargo** additionally removes training examples in a short interval immediately following the test block, to break the serial correlation that survives purging via features rather than labels. **Combinatorial purged cross-validation** forms many train/test splits by choosing k of N contiguous groups as test blocks, applying purging and embargo throughout, and thereby yields a *distribution* of backtest paths rather than a single one.

Why it appears here. §11.3, where the third failure of k-fold — persistent regimes appearing on both sides of every fold boundary — is the one purging does *not* fix, and requires walk-forward or leave-one-episode-out instead. §5.11 raises the overlapping-label problem first, for supervised regime labels.

Deeper. López de Prado (2018), chapters 7 and 12, in §4.8.

A.40 Multiple testing, selection bias, and the deflated Sharpe ratio

The idea. If you try many strategies and report the best, the best one looks good partly because it is the best of many, not because it is good. Correcting for that requires knowing how many things you tried.

Formally. The expected maximum of n independent standard normal draws grows roughly like $\sqrt{2 \ln n}$ — concretely about 1.87 for $n = 20$ and 2.51 for $n = 100$. So if each trial's Sharpe estimate has standard error se, the *expected* best-of-n estimate under a true Sharpe of zero is about 1.87 se at $n = 20$. The **deflated Sharpe ratio** computes the probability that an observed Sharpe exceeds what selection alone would produce, adjusting for the number of trials, the variance across trials, the sample length, and the skewness and kurtosis of returns.

Why it appears here. §3.6 and §5.5 flag unadjusted specification search; §11.6 corrects for it. Regime work multiplies trials fast — K regimes times M specifications times W windows — and §11.5's standard error of 0.53 for a four-year crisis sample means twenty honest trials produce an expected best crisis Sharpe near 1.0 from pure noise.

Deeper. Bailey & López de Prado (2014) and Harvey, Liu & Zhu (2016), both in §4.8.

A.41 The block and stationary bootstrap

The idea. Resampling observations independently destroys serial dependence, which is exactly the structure a time-series null needs to preserve. Block methods resample contiguous stretches instead.

Formally. The **moving block bootstrap** resamples blocks of fixed length ℓ with replacement and concatenates them. The **stationary bootstrap** (Politis & Romano, 1994) uses geometrically distributed block lengths with mean $1/q$, which makes the resampled series stationary. Consistency requires $\ell \to \infty$ with $\ell/T \to 0$; in practice ℓ must exceed the dependence horizon you wish to preserve.

Why it appears here. §11.4's shuffled-regime control resamples the state path in blocks, and §11.6's warning: if your block length is shorter than the regime half-life, the bootstrap null contains no regimes, and any regime strategy will look significant against it. The block length is not a nuisance parameter here — it defines the hypothesis being tested.

Deeper. Politis & Romano, "The Stationary Bootstrap," *JASA* 89(428) (1994), 1303–1313; Lahiri, *Resampling Methods for Dependent Data* (Springer, 2003).

A.42 Forecast-comparison tests under instability

The idea. Comparing two forecasting models means comparing two sequences of losses. The standard test asks whether one is better *on average*; the tests that matter here ask whether one is better *right now*, and whether the answer changes over time.

Formally. Let $d_t = L(y_t, \hat{y}_t^A) - L(y_t, \hat{y}_t^B)$ be the loss differential.

- **Diebold-Mariano** tests $H_0 : \mathbb{E}[d_t] = 0$ using $\bar{d} / \sqrt{\widehat{\text{avar}}(\bar{d})/T}$ with a HAC variance estimate.
- **Giacomini-White conditional predictive ability** tests $H_0 : \mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = 0$ — whether the models are equally good *given the current state* — by testing whether d_{t+1} is orthogonal to a chosen set of $\mathcal{F}_t$-measurable instruments.

- **The Giacomini-Rossi fluctuation test** computes a rolling Diebold-Mariano statistic over windows and compares its **maximum over time** against critical values that account for taking that maximum, so that "model A was better in the 2000s and worse in the 2010s" becomes a testable statement rather than an eyeball impression.

A **HAC** (heteroskedasticity and autocorrelation consistent) variance estimate, of which the **Newey–West** estimator is the standard example, adds down-weighted autocovariances of d_t up to some lag to its variance, so that the standard error of $\bar{d}$ remains valid when the loss differentials are serially correlated, as daily ones are.

Why it appears here. §11.6 ranks the fluctuation test as the most useful diagnostic in the document, because a genuinely regime-aware model should have its advantage *concentrated in the regime it was designed for* — and the test shows you whether it does. §6.9 and §6.11 compare density forecasts this way, with Newey–West standard errors.

Deeper. Giacomini & White (2006) and Giacomini & Rossi (2010), both in §4.8; Rossi, "Forecasting in the Presence of Instabilities," *Journal of Economic Literature* 59(4) (2021), 1135–1190, for the survey.

A.43 Conformal prediction under distribution shift

The idea. A way to produce prediction intervals with a guaranteed coverage rate that does not depend on the model being correct — and an adaptive version whose guarantee survives the distribution changing underneath it.

Formally. Split conformal prediction holds out a calibration set, computes non-conformity scores $s_i = |y_i - \hat{f}(x_i)|$ on it, and forms the interval $\hat{f}(x) \pm \hat{q}_{1-\alpha}$ where $\hat{q}_{1-\alpha}$ is the empirical $(1-\alpha)$ quantile of the scores. Under exchangeability this has marginal coverage at least $1-\alpha$ regardless of the model. Exchangeability fails under distribution shift, so **adaptive conformal inference** (Gibbs & Candès, 2021) updates the nominal level online, $\alpha_{t+1} = \alpha_t + \eta(\alpha - \mathbb{1}[y_t \notin C_t])$, which recovers the long-run coverage rate without any distributional assumption.

Why it appears here. §10.8. It is the only tool in this document that offers a guarantee holding *under* shift rather than a method that hopes to adapt to it, and its output — an interval that widens in unfamiliar conditions — is directly usable as a sizing input.

Deeper. Angelopoulos & Bates, "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification", arXiv:2107.07511; Gibbs & Candès (2021) in §4.7.

A.44 Real-time data vintages

The idea. Macroeconomic statistics are estimates, published with a lag and revised repeatedly afterwards. The number you can download today for a date in 1995 is not the number anyone had in 1995.

Formally. A **vintage** is the complete set of values for a series as published at one point in time. A real-time data set is therefore two-dimensional — indexed by observation date *and* vintage date — and a point-in-time query asks for the value of period s as known at date $v \geq s$. The Federal Reserve Bank of Philadelphia maintains such a database for U.S. macro series, and ALFRED provides vintage-aware access to FRED. Revisions are frequently larger than the series' own standard deviation, and initial releases of quarterly GDP are revised for years.

Why it appears here. §1.5's note on NBER recession dates, which are announced 6–18 months after the turning points they label; §6.6, which requires real-time vintages for macro covariates; and §11.2's leak 8. Any regime model that conditions on macro data and does not use vintages is conditioning on information that did not exist.

Deeper. Croushore & Stark, "A Real-Time Data Set for Macroeconomists," *Journal of Econometrics* 105(1) (2001), 111–130.

A.45 Mean-variance sizing, Kelly, and volatility targeting

The idea. Given a forecast of return and a forecast of risk, how large should the position be? All the standard answers have the same shape: expected return over variance, scaled by risk appetite.

Formally. Maximising $\mathbb{E}[\pi r] - \frac{\gamma}{2}\mathrm{Var}[\pi r]$ over the position π gives

$$\pi^\star = \frac{\mathbb{E}[r \mid \mathcal{F}_{t-1}]}{\gamma\,\mathrm{Var}[r \mid \mathcal{F}_{t-1}]},$$

with the **Kelly** criterion the special case $\gamma = 1$ (log utility) and **volatility targeting** the special case where the expected return is treated as proportional to volatility, giving $\pi^\star \propto \sigma^\star/\hat{\sigma}_t$. For a mixture predictive distribution the two moments are

$\bar{\mu} = \sum_k \xi^{(k)} \mu_k$ and Var $= \sum_k \xi^{(k)} \sigma_k^2 + \sum_k \xi^{(k)} (\mu_k - \bar{\mu})^2$ — a within-state term plus a between-state term reflecting uncertainty about which state you are in.

Why it appears here. §12.1 is the argument that these moments, not an argmax label, are what a regime posterior should feed. §7.2 records the corollary that every volatility-targeted strategy is already regime-conditional with a continuum of states. §6.10 extends the two-moment decomposition, which is the law of total variance, to cumulative returns over h bars.

Deeper. MacLean, Thorp & Ziemba, eds., *The Kelly Capital Growth Investment Criterion* (World Scientific, 2011), part I.

A.46 Transaction-cost models and no-trade bands

The idea. How costs are shaped determines the shape of the optimal trading rule. Costs proportional to the amount traded produce a region where you do nothing; costs quadratic in the trading *rate* produce partial adjustment toward a target.

Formally. Under **proportional** costs $\varepsilon|\Delta\pi|$, the optimal policy for a Merton-style problem is a **no-trade band**: do nothing while the position lies inside an interval around the frictionless target, and trade only to its nearest edge when you leave. The classical result is that the band's half-width scales as $\varepsilon^{1/3}$ for small ε — so a tenfold cost increase widens it only about $2.15\times$. Under **quadratic** costs $\frac{\lambda}{2}\Delta\pi^\top \Sigma\, \Delta\pi$, the optimal policy is linear: trade a constant fraction of the way toward an **aim portfolio**, itself a weighted average of current and expected future frictionless targets that overweights slowly-decaying signals.

Why it appears here. §12.2. The cube-root scaling matters because people consistently over-widen bands in response to cost estimates; and the aim-portfolio result matters because regime signals decay fast (§1.3) and therefore deserve *less* weight in the aim than their raw strength suggests.

Deeper. Constantinides (1986) and Gârleanu & Pedersen (2013), both cited in §12; Davis & Norman, "Portfolio Selection with Transaction Costs," *Mathematics of Operations Research* 15(4) (1990), 676–713, for the continuous-time band.

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